GARRISON ENGINEER (I) CAMPBELL BAY
ARMY LAND, CAMPBELL BAY, GREAT NICOBAR-744302

NAME OF WORK	:	**SPECIAL REPAIRS OF ROAD FROM PERIPHERY ROAD TO HRDF SITE, PASSAGE OF AGE E/M OFFICE, SECURITY FENCE AROUND BAAZ HOUSE AT INS BAAZ, CAMPBELL BAY UNDER GE(I) CAMPBELL BAY**

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Signature of Contractor

AGE (Contracts)
for Accepting Officer

Tele : 03192- 291898
Email: garrisonengineercampbellbay@gmail.com

SPEED POST

Garrison Engineer (I), Campbell Bay
Military Engineer Services
Army Land, Campbell Bay
Great Nicobar-744302

8146/ 34 /E8

30 Aug 2025

M/s _____
_____
_____
_____

SPECIAL REPAIRS OF ROAD FROM PERIPHERY ROAD TO HRDF SITE, PASSAGE OF AGE E/M OFFICE, SECURITY FENCE AROUND BAAZ HOUSE AT INS BAAZ, CAMPBELL BAY UNDER GE(I) CAMPBELL BAY

Dear Sir(s),

1. Tender documents in respect of above work are uploaded on the site www.defproc.gov.in The tender is based on single stage two cover/two stage three cover e-tendering system. The contents of Cover I & Cover II are specified in NOTICE OF TENDER.

2. Bids will be received online by ACCEPTING OFFICER upto the date and time mentioned in the **NOTICE INVITING TENDER (NIT)**. No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as non bona fide.

3. Bid will be opened on line on a due date fixed for opening as per critical dates given in the portal. Cover 1 will be opened first. Tenderers may see the result of the opening of Cover 1 on the tender portal. Date of opening of Cover 2 shall be decided after technical evaluation of Cover 1 keeping in view the various eligibility criteria given in the NIT and the same will be intimated to the tenderers while uploading the technical evaluation on the tender portal.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents through e-mail and/or depute your technical representative for discussion on tender/drawings and to clarify doubts, if any, before the bid submission start date.

5. Unenlisted contractors are required to submit the scanned copies (in pdf file) of documents required as per eligibility criteria mentioned in instructions for filling the tender documents and Appendix ‘A’ to NIT along with **EARNEST MONEY DEPOSIT (EMD)** and tender fee on procurement portal mentioned above and submit the physical documents in the office of **GE (I) Campbell Bay** within time limit specified in NIT. Inadequacy /deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.

6. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appx ‘A’ to NIT on e-procurement portal and submit physical documents in the office of **GE (I) Campbell Bay** before date & time fixed for this purpose.

7. The contractor must ensure that the tender/bid is uploaded on the tender portal using the authorised DSC in time as the Accepting Officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand/through post from tenderer/bidder.

8. Keeping in view delays due to system failure or other communication related failures, it is suggested that tender/bid may be uploaded sufficiently in advance of the last due date and time fixed on the portal.

9. General Conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, Schedule of Minimum Fair Wages and MES SSR (Part-I and Part-II) are not enclosed with these documents. These are available for perusal in this office and other MES offices.

10. PERFORMANCE SECURITY DEPOSIT : The successful Contractor shall be required to lodge **Performance Security** for an amount equal to **5%** of the Contract Sum within 28 Days of receipt of Letter of Acceptance in the form of Bank Guarantee in the prescribed format or Government Securities or FDR or any other Government Instruments stipulated by the Accepting Officer in terms of Condition – 19 of IAFW – 2249, General Conditions of Contracts, **failing which action as stipulated in Condition 19 of GCC shall be taken.** Work order will be placed after receipt of the said Performance security from the contractor

11. ANY TENDERER, WHICH PROPOSES ALTERATIONS TO ANY OF THE CONDITIONS, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR ANY NEW CONDITION, WHATSOEVER, IS LIABLE TO BE REJECTED

12. Your particular attention is drawn to consequences of newly introduced amendments No 41 to 49 to IAFW-2249 General Conditions of Contracts. You are hereby requested to go through the same in detail. Accordingly your special attention is drawn to the following:-
(a) Inclusion of Performance Security in prescribed form at 5% of Contract Sum by successful bidder and removal of Risk & Cost provision in contracts.
(b) Defect liability period for this work in terms of Condition 46 of IAFW-2249 General Condition of Contracts shall be Twenty Four months after the works have been handed over to Government.
(c) Removal of furnishing Additional Security Deposit and Individual Security Deposit.
(d) Forfeiture of Performance Security, Retention money upto last paid RAR, confiscation of entire tools and plants and materials lying at site without any compensation whatsoever. Debarring Proprietor/Partners /Directors of failed company from participating in tender for balance works either in his/her individual capacity or as a partner of any other company/firm
(e) Effecting recovery from contractor in terms of condition 67 of IAFW-2249, General Conditions of Contract from Performance Security amount (or from Standing Security Deposit amount or Security Bond) if Performance Security amount is not adequate.

Yours faithfully,

AGE (Contracts)
for Accepting Officer

Signature of Contractor

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S)

1. EARNEST MONEY DEPOSIT

1.1 Contractor(s) who are not enlisted with MES/who are enlisted but have not executed the Standing Security Bond shall submit Earnest Money Deposit as detailed in Notice of Tender in one of the following forms, along with their tender/bid:-

(a) Deposit at Call Receipt from a Scheduled Bank in favour of GE (I) Campbell Bay
(b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of GE (I) Campbell Bay

1.2 It is advisable that Earnest Money is deposited in the form of deposit call receipt from an approved Schedule Bank for easy refund. In case the tenderer/bidder wants to lodge 'Earnest Money Deposit' in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of concerned GE(I).

NOTES:- Earnest Money Deposit (EMD) in the form of cheque/Bank Guarantee etc will not be accepted. **Non-Submission of Earnest Money Deposit (EMD) (scanned copy alongwith Technical Bid & hard copy before the date & time fixed for opening of BOQ) will render the bid disqualified for opening of Cover -II (finance bid).**

2. GENERAL INSTRUCTIONS FOR COMPLIANCE

2.1 The bids received only in the electronic form will be considered. All bids shall be submitted on www.defproc.gov.in portal. Documents should be scanned and forwarded in 'pdf' format and 'xls' form as indicated. Bids shall be uploaded on www.defproc.gov.in portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like date email/fax/by hand/through post will be considered.

2.2 The bid shall be DIGITALLY signed using authorised DSC. All pages of tender documents, drawings, corrections/alterations shall be signed/initialed by the lowest bidder after acceptance of tender for making original and CTC of the contract.

2.3 Digital signatory of the bid/tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Article of Association of the Company) in all the matters pertaining to the contract with Union of India including arbitration clause. A scanned copy of the documents in confirmation of such authority shall be attached with the tender/bid in 'pdf' form in Cover 1. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Article of Association.

2.4 Drawings, if issued in physical form, must be returned duly initialled by the tenderer/bidder in separate envelope indicating his name and address.

2.5 The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialled. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (CONTD)

3. **BOQ :-**

3.1 The tenderer shall quote his rates on the BOQ EXCEL FILE only as per guide line of e-procure web portal. No alteration to the format will be accepted and such bid will be disqualified.

3.2 In case any tenderer wishes to revise/modify the rates quoted in the BOQ file, he can do so only in the BOQ files before uploading the tender through **https://defproc.gov.in** site only before closing date & time.

3.3 After the uploading of tender, Department may upload the errata/ amendment through corrigendum. The tenderer/bidder should submit their offer considering the errata/amendment carried out through corrigendum issued from time to time.

3.4 While uploading the bid, the tenderers/bidders should specifically check whether any **revised BOQ** has been uploaded by department through corrigendum prior to **Bid submission start date**. Tenderers/bidders attention is specifically drawn to the fact that they should submit their offer on revised BOQ only. In case any tenderer/bidder submits offer on pre-revised BOQ in lieu of Revised BOQ, **it will be considered as a willful negligence by the tenderer/bidder and quotation shall be considered non-bonafide**.

4. **PERFORMANCE SECURITY**

4.1 Within 28 days of receipt of the letter of Acceptance, the successful contractor shall deliver to the Accepting Officer, a performance security in any of the forms given below for an amount equivalent to 5% of the contract sum.

(a) A Bank Guarantee in the prescribed form.

(b) Government Securities, FDR or any other Government instruments stipulated by the Accepting Officer.

4.2 If the performance security is provided by the successful contractor in the form of a Bank Guarantee, it shall be issued by Nationalized / Scheduled Indian Bank but its confirmation shall be done only from the Head Office of the Bank.

4.3 Failure of the successful contractor to comply with the requirement of clause 4.1 here-in-before shall constitute sufficient grounds for cancellation of the award of work and forfeiture of the Earnest Money. In case of MES enlisted contractor, amount equal to the Earnest money stipulated in the Notice Inviting Tender (NIT), shall be notified to the tenderer for depositing the amount through MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest money is deposited in Government Treasury.

4.4 All compensation or other sums of money payable by the contractor to the Government under the terms of this contract or under any other contract with Government may be deducted from, or paid by the sale of a sufficient part of the Performance Security or from the interest arising there from or from any sums which may be due or become due to the contractor by the Government on any account whatsoever and in the event of his Performance Security being reduced by reason of any such deduction, or sale as aforesaid, the contractor shall within ten days thereafter make good in cash or securities, endorsed as aforesaid, any sum of sums which may have been deducted from or realized by the sale of his Performance Security or any part thereof.

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (CONTD)

4.5 Government shall not be responsible for any loss of securities or any depreciation in the value of securities while in their charge nor for loss of interest theron.

4.6 In the event of contract being cancelled, under condition 52, 53 & 54 of General Conditions of Contract, the Performance Security shall be forfeited in full and shall be credited into Consolidated Fund of India.

4.7 **REFUND OF PERFORMANCE SECURITY DEPOSIT :**

4.7.1 The Performance Security Deposit mentioned in condition 19 of IAFW-2249 shall be refunded to the contractor after the expiration of the defects liability period (See Condition 46 of IAFW-2249) by the GE, provided always that the contractor shall first have been paid the final bill and have rendered a No-Demand Certificate (I.A.F.W. - 451). (See Condition 68 of IAFW-2249)

5. **POWER OF ATTORNEY (PoA)**

5.1 In the technical bid, a scanned copy of Power of Attorney in favour of the person uploading the bid using his/ her DSC shall be uploaded. In case the digital signatory himself is the sole proprietor, scanned copy of an affidavit on stamp paper of appropriate value to this effect stating that he has authority to bind the firm in all matters pertaining to contract including the Arbitration Clause, shall be attached in 'pdf' form. In case of partnership concern or a limited company, digital signatory of the bid/ tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Articles of Association of the Company) in all the matters pertaining to the contracts with Union of India including arbitration clause. A scanned copy of the documents confirming of such authority shall be attached with the tender/ bid in 'pdf' form, if not submitted earlier. The person uploading the bid on behalf of another partner(s) or on behalf of a firm or company using his DSC shall upload with the tender/ bid a scanned copy (in 'pdf' form) of Power of Attorney duly executed in his favour by such other or all of the Partner(s) or in accordance with constitution of the company in case of company, stating that he has authority to bind such other person of the firm or the Company, as the case may be , in all matters pertaining to the contract including the Arbitration clause.

5.2 Even in case of Firms or Companies which have already given Power of Attorney to an individual authorizing him to sign tender in pursuance of which bids are being uploaded by such person as a routine, fresh Power of Attorney duly executed in his favour stating specifically that the said person has authority to bind such partners of the Firm, or the Company as the case may be, including the condition relating to Arbitration Clause, should be uploaded in 'pdf' form with the tender/bid unless such authority has already been given to him by the Firm or the Company. It shall be ensured that Power of Attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Article of Association.

6. **REVOKATION/ REVISION OF OFFER/ OFFERING VOLUNTARY REDUCTION, AFTER OPENING OF FINANCIAL BIDS BY LOWEST BIDDER**

6.1 In the event of lowest tenderer/ bidder revoking his offer or revising his rates upward/ offering voluntary reduction, after closing of bid submission date & time, his offer will be treated as revoked and the Earnest Money deposited by him shall be forfeited. In case of MES enlisted contractors, the amount equal to the Earnest Money stipulated in the Notice of tender, shall be notified to the tenderer/ bidder for depositing the amount through MRO.

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (CONTD)

6.2 Bids of such contractors/ bidders shall not be opened till the aforesaid amount equal to the Earnest Money is deposited by him in Govt Treasury. In addition, bids of such tenderer/ bidder and his related firm shall not be opened in second call or subsequent calls. Reduction offered by the tenderer/ bidder on the freak high rates referred for review shall not be treated as voluntary reduction.

7. **CPM (CRITICAL PATH METHOD)**

7.1 The tender is based on CPM.

7.2 The tenderer/ bidder is expected to be fully conversant with the CPM technique and employ technical staff, who can use the technique in sufficient details. Sufficient books and other literatures on the subject are widely available which the tenderer / bidder may make use of.

7.3 The tenderer's attention is drawn to special conditions of the tender regarding preparation of the detailed network analysis and time schedules for the work and his liability for employing sufficient resources to adhere to this schedule. Any inability on the part of the tenderer/ bidder in using the technique will be taken as his technical insufficiency and will affect his class of enlistment and future prospect/ invitation to tenders for future works.

7.4 Department may issue amendments/ errata on the form of CORRIGENDUM to tender/ revised BOQ to the tender documents. The tenderer/ bidder is requested to read the tender documents in conjunction with all the errata/ amendments/ corrigendum, if any issued by the department and no claim whatsoever on account of any lapse or misunderstanding shall be entertained

8. These instructions shall form part of the tender documents.

9. The tender for the works shall remain open for acceptance for a period of Sixty (60) days from due date of submission of the bid excluding the date of submission of the bid.

10. In case any tampering/alteration is noticed between the documents as uploaded at the time of the bid online and the original one as uploaded by the office of GE (I) Campbell Bay, then the bid submitted shall become invalid and the Government shall, without prejudice to any other right or remedy, be at liberty to forfeit the said earnest money as aforesaid.

11. **ELIGIBILTY CRITERIA FOR**

(A) **MES ENLISTED CONTRACTORS**:- The contractor shall be enlisted in the requisite class and category as mentioned in Appendix 'A' to Notice Inviting Tender (NIT). Scanned copy of enlistment letter along with its enclosures shall be uploaded along with the bid.

(B) **UNENLISTED CONTRACTORS**:-

(B-1) **EXPERIENCE CRITERIA**

(a) Past Experience of "similar nature of works" completed in last 5 years.

(a) Two works costing not less than Rs 25.00 lakhs each.

or

(b) One work costing not less than Rs 40.00 lakhs.

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (CONTD)

Note:- Contractor has to submit details of "similar nature of works" executed by you financial year wise in tabular form giving name of work, Accepting Officer's details, viz, address, telephone, email etc, date of acceptance of tender and actual date of completion. It should indicate whether extension was granted or compensation was levied. Attested copy of acceptance letter and completion certificate shall be enclosed for each work. In case performance report has been given by the bidder same shall also be submitted duly attested. Format for affidavit of containing list of works(last three years) in hand period of completion, date of commencement and completion of work and remaining works to be completed as on date of application duly signed on an affidavit on non-judicial stamp paper of ₹ 100.00 duly notarized.

(i) **YEAR WISE LIST OF COMPLETED WORK DURING LAST THREE YEAR (SCANNED COPY OF SUPPORTING DOCUMENTS .PDF DOCUMENTS TO BE UPLOADED)**

Ser No	CA No. & Name of work	CA Amount (in Rs)	Completion Amount (in Rs)	Period of Completion in months	Date of completion

(ii) **LIST OF BALANCE WORK IN HAND:- (SCANNED COPY OF SUPPORTING DOCUMENTS .PDF DOCUMENTS TO BE UPLOADED)**

Ser No	CA No. & Name of work	CA Amount (in Rs)	Period of completion in months	Balance Amount of work (in Rs)

(b) Average annual turnover for last two consecutive financial years shall not be less than Rs 50.00 lakhs.

Note : Please submit turnover certificate issued by Chartered Accountant along with relevant pages of audited balance sheets in support thereof.

(i) **ANNUAL TURN OVER OF LAST TWO YEAR (SCANNED COPY OF SUPPORTING DOCUMENTS .PDF DOCUMENTS TO BE UPLOADED)**

Ser No	Year	Annul Turn Over Amount

(B-2) **FINANCIAL CRITERIA**

(a) Solvent upto Rs 20.00 lakhs.

or

Financially sound for engagement upto Rs 50.00 lakhs

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (CONTD)

(b) Working Capital not less than Rs 7.50 lakhs.

Note:-

1. Proforma of above documents is given below. Please upload above documents issued by your Banker in Original before date and time for bid submission end date. Above certificates shall not be older than 06 months as on date for opening of technical bid. Signatory shall endorse his rubber seal, indicate his name and bank employee number, bank address particulars like full address, telephone & fax number.
2. Alternatively Contractor may submit certificate obtained from the Deputy Commissioner/ Collector/ First Class Magistrate for assessment of wealth tax.

FORM OF SOLVENCY CERTIFICATE FROM A NATIONALISED/SCHEDULED BANK

This is certified that to the best of our knowledge and information M/s/ Shri/Smt_____ _____having proprietor / partners (Name of proprietor/partners) and address_____ _____, a customer of our bank having A/C No …….…… is / are respectable and can be considered solvent / financially sound for engagement upto Rs……..... (Rupees……..……only) issued without any guarantee or responsibility on the bank or any of the officers.

Place Signature

Name, Designation, personal code

Date No of signatory & Seal of Bank

NOTE : In case of partnership firm, certificate to include names of all partners as recorded with the Bank.

FORM OF WORKING CAPITAL CERTIFICATE FROM NATIONALISED/ SCHEDULED BANK

This is certified that M/s Shri/Smt _____ having proprietor/partners (Name of proprietor/partners) and address has/have been maintaining a saving Bank account/current account/fixed deposit account having A/C Nos. _____ with this branch of Bank since _____ and an amount not less than Rs_____(Rupees _____) has available to credit in his /her/their accounts for last three months. The firm is enjoying over draft/credit facility up to limit of Rs_____. This certificate is issued without any guarantee of responsibility on the bank or any of the officers.

Place Signature

Name, Designation, personal code

Date No of signatory & Seal of Bank

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (CONTD)

NOTE : In case of partnership firm, certificate to include names of all partners as recorded with the Bank.

(B-3) **FIXED ASSETS** : Limit of minimum reserves/total of movable or immovable property or fixed assets (in case of companies) (80% of minimum reserve/assets shall be in shape of immovable property) shall be Rs12.50 lakhs

Note:-

(i) Please submit affidavits on non Judicial stamp paper of ₹ 100/- each duly notarized and supported by copies of present valuation certificates for both immovable and movable property separately. It should also indicate that immovable property is free from mortgages, hypothecation or any other disputes and encumbrances and clearly belongs to your firm and it should indicate that immovable property will not be sold, transferred, gifted or otherwise disposed off without prior intimation to this office.

(ii) Immovable asset value indicated should be supported by Valuation Certificate from a registered valuer and last tax paid receipt. Report of valuation of immovable property (Other than agricultural lands, plantations, forests, mines and quarries)

(iii) Movable assets like Scooter/ Car etc or NSCs/ fixed deposits etc should be supported by certified photocopies of updated RCC books, valid current Insurance papers, NSCs/ fixed deposits.

(B-4) **ENGINEERING ESTABLISHMENT**

(a) One graduate Engineer from a Government recognized institution with minimum experience of 2 years.

or

(b) One diploma Engineer from Government recognized institution with minimum experience of 4 years

NOTE:- Please submit affidavits on non judicial stamp paper of ₹ 100/ each duly notarized from your side having employed each engineer and from engineer side of taking your employment duly supported by attested copy of engineering degree / diploma as applicable to the individual.

(B-5) **T &P / MACHINERY**:

(a)	One bag capacity concrete mixer (diesel)	--	01 No
(b)	Vibrators (Needle & plate type)	--	02 Nos
(c)	Steel shuttering with soans, props etc (Sqm)	--	500 Sqm

Affidavit containing details of possession and having ownership of T& P with present value of each T& P along with ownership details like make and year of manufacturer, certified copies of updated RC books and valid current insurance papers.

(B-6) (a) **CONSTITUTION OF FIRM** : Affidavit of Constitution of firm indicating status of firm i.e. whether it is proprietorship or partnership or limited firm.

Note: Please also upload attested copy of partnership deed in case of partnership firm. Memorandum of Article of constitution in case of limited firm.

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (CONTD)

(b) **CERTIFICATE OF RELATIVE** :- Affidavit that no near relative(s) of the contractor or their employees/agent is/are working as Gazetted/Commissioned Officer in MES/Corps of Engineer/Ministry of Defence. If his/their near relative(s) is/are working in such capacity, he/they shall furnish details. The contractor shall not be entitled to tender for the works in entire area of CE Zone/project under whose control his/their near relatives are working as Gazetted/Commissioned Officers.

(c) **CERTIFICATE OF RELATIVES** :- Affidavit that no near relative(s) of the contractor is/are working as Junior Engineer in MES/Corps of Engineer. If his near relative(s) is/are working in such capacity, the contractor shall furnish details. The contractor shall not be entitled to tender for the works in entire area of GE(I)/GE in which his near relative(s) is/are working as Junior Engineer.

(d) **POLICE VERIFICATION** :- Copy of police verification certificate from police authority of the area where registered office is located. Alternatively notarized copy of valid passport of proprietor/each partner/each director can be submitted.

(e) **CERTIFICATE REGARDING NO RECOVERY OUTSTANDING IN ANY GOVT DEPARTMENT** :- Affidavit to the effect that no recovery of any kind is outstanding in any Government department to be furnished.

(f) **CERTIFICATE REGARDING NO ARBITRATION/LITIGATION CASES** :- Affidavit to the effect that no Arbitration/litigation cases is outstanding in any Government department to be furnished.

(g) Affidavit by firm/bidder containing details of each employed engineer in firm.

(h) Affidavit from engineer side of taking firm/bidder employment duly supported by attested copy of degree/diploma certificate and experience certificate as applicable.

(j) Certified true copies of PAN/TIN of proprietor, partners/firm.

(k) Certified true copies of enlistment letters with various Govt Deptt/PSUs.

(l) Copy of last three years Income Tax return filed.

(m) **INDEMNITY BOND:** Indemnity bond on non judicial stamp paper duly attested by Notary (on format given below)

INDEMNITY BOND

This bond of indemnity is executed……..on this…….day by M/s/ Shri …….… having registered office at……to indemnify the Govt of India of the following :-

(a) We undertake to pay the Govt of India any damages that may be found to be recoverable on order of our contracts.

(b) We undertake that in case Govt is put to pay any loss or disadvantage in monetary of contract by the firm which shall indemnify the Govt for each loss or disadvantage.

This bond of indemnity is executed by has signed before me. Whatever stated above is true and correct to the best knowledge and belief.

Signature of Contractor Notary

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (CONTD)

(n) Contractor shall get themselves registered with the Employees Provident Fund Organisation and deposit the necessary contributions with the EPFO. Further, it is a mandatory requirement that all construction workers should be provided Universal Account Number (UAN) by the contractor by appropriately registering them on the EPFO Portal. The contractor shall also ensure compliance of the EPF & MP Act, 1952 by the sub-contractors, if any engaged by the contractor for the subject work.

(o) All the Workers deployed by contractors are enrolled as members of Provident Fund and should be given the Universal Account Number (UAN).

(p) Certificate to the effect that all the workers employed directly or indirectly are registered for EPF and the due consideration have been credited into their account shall be rendered by the contractor along with each and every RAR/Final bill.

Note : Uploading a Scanned copy of Temporary or Permanent Provident Fund Code Number issued by Concerned authority, scanned copy of GST Registration Number in addition to other documents required shall be mandatory for Technical Evaluation. Contractor not in possession of this number will be disqualified in 'T' bid evaluation and his Finance bid will not be opened.

(q) Un-enlisted contractors will render an affidavit categorically stating that they have not been banned by any Govt Department/ Semi Govt/ PSU/ State Govt agency for participating in tendering.

(C) **FOR BOTH ENLISTED AND UNENLISTED CONTRACTORS**

(C-1) Scanned copy of DD / Banker's Cheque towards cost of tender.

(C-2) Scanned copy of application of firm requesting issue of tender.

(C-3) Scanned copy of Temporary or Permanent Provident Fund Code Number obtained from concerned Authorities (Employees Provident Fund Organization).

(C-4) Scanned copy of GST Registration Number shall be a prerequisite for qualifying in 'T' bid. Contractors not in possession of this number will be disqualified in 'T' bid evaluation and his Finance bid will not be opened

12. Scanned copies of the documents as mentioned hereinbefore, as applicable, shall be uploaded by the tenderers in .pdf or .rar form in one folder alongwith their bids. The original documents shall be submitted by the bidder on or before the date and time stipulated for technical opening of bid in the sealed cover marked on outside the Name of work as per the tender documents. Failure to adhere to the above requirement will result in rejection of their bids. In case any of the tenderer does not meet the eligibility criteria or submits incomplete documents or provides false information, their technical bids will be summarily rejected and financial bid will not be opened. No claim for refund of the tender fee submitted by the tenderers in such cases shall be entertained.

INSTRUCTIONS FOR COMPLETION OF TENDER DOCUMENTS TO BE COMPLIED WITH BY THE TENDERER(S) (CONTD)

13. Any firm which is banned due to slow progress in WLR or against which a Technical BOO has been ordered for defective work shall be considered automatically ineligible to quote for the tender. The capability of contractor as indicated in quarterly work load return of the Department shall be also verified/checked.

14. DSC used by the tenderer shall be issued on behalf of the applicant firm.

15. Contractor shall upload copies of all the above mentioned documents (as applicable) prior to closing date of bid submission.

16. Hard copies of all above documents shall be sent by the contractor to the Tender issuing authority well in advance to be received before the date & time fixed for the same.

17. The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to relevant provisions in Special Condition referred here-in-after and also Conditions 24 & 25 of IAFW-2249 (General Conditions of Contract).

Signature of Contractor AGE (Contracts)

Dated: for Accepting Officer

[In lieu of IAFW-1779A (Revised 1955)]

TO BE USED IN CONJUNCTION WITH GENERAL CONDITIONS OF CONTRACT IAFW-2249 (1989 PRINT)

MILITARY ENGINEER SERVICES

Telephone: 03193-291898

Garrison Engineer (I)
Army Land, Great Nicobar Islands
Campbell Bay – 744 302

8146/34 /E8

30 Aug 2025

WORK REQUIRED IN THE EXECUTION OF ITEM RATE TENDER AND CONTRACT FOR "SPECIAL REPAIRS OF ROAD FROM PERIPHERY ROAD TO HRDF SITE, PASSAGE OF AGE E/M OFFICE, SECURITY FENCE AROUND BAAZ HOUSE AT INS BAAZ, CAMPBELL BAY UNDER GE(I) CAMPBELL BAY."

1. A tenderer/ bidder who is qualified as per the eligibility criteria mentioned in the tender documents and has interalia sound past record is hereby authorized to tender for the above work. The tender/ bid [both COVER-1 & COVER-2] shall be submitted/uploaded on the MES website portal as per time and date mentioned in tender documents.

2. The tender shall be published by this office on the MES portal of www.defproc.gov.in dates can be referred in the web portal.

3. All correspondence must be returned to this office whether or not the tender has been online submitted.

4. Any Correspondence concerning this tender should be addressed as indicated at the top of this sheet, quoting the reference to the tender id mentioned in the www.defproc.gov.in portal.

THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY TENDER

Signature of officer issuing the tender documents
AE (QS&C)
AGE (Contracts)
for Accepting Officer

(Signature of Contractor)
Dated: _____

SCHEDULE 'A'

NAME OF WORK : SPECIAL REPAIRS OF ROAD FROM PERIPHERY ROAD TO HRDF SITE, PASSAGE OF AGE E/M OFFICE, SECURITY FENCE AROUND BAAZ HOUSE AT INS BAAZ, CAMPBELL BAY UNDER GE(I) CAMPBELL BAY.

NOTES:-

1. The scope of work covered under this contract is given in Schedule 'A' (BoQ) in a separate Excel sheet.

2. The quantities shown in Col 3 of the BoQ are approximate and are inserted as guide line only. These shall however not be varied beyond the limits laid down in condition 7 of IAFW-2249 (General Conditions of contracts).

3. Tenderers are required to insert unit rates in figures under Col 5 of the BoQ against each item of schedule `A' and upload after due validation. In the event of failure to quote for any item in BOQ, the same is liable to be rejected.

4. The rate quoted for each items under Col 5 of Sch 'A' (BoQ) shall include for all necessary works required for the entire completion all as specified.

5. The entire works covered under this contract shall be completed within **180 (One hundred and Eighty)days** from the date of commencement as indicated in Work Order No 1. Work Order shall be placed by the GE after receipt of Performance Security from the contractor. Before commencement of work joint inventory shall be prepared and shall be signed by the Engineer-in-Charge and contractor, indicating the details, condition of materials and the quantities existing at site. The site/ buildings for carrying out repairs will be handed over to the Contractor as available at the site and as per user's requirement. The Contractor will proceed and complete the items of the work within the time period on priority all as directed by Engr-in-Charge.

6. The unit rate quoted by the tenderer under Col 5 of Sch 'A' (BoQ) shall be deemed to inclusive for all incidental and additional labour and materials to cover all the work enumerated and described under the description of items and in the preamble to similar items in SSR Part II 2020. It is an express condition of this contract that the rates quoted in this tender for various items of work shall be deemed to be for the full and entire execution and completion of this contract. The Government will not entertain any claim whatsoever on account of in accuracies, misunderstanding, if any in the aforesaid rates. Unless otherwise specifically mentioned in schedule 'A' the rate quoted for schedule 'A' items shall be deemed to include for the provisions of all materials, labour, processor, operations including requirements as detailed in the particulars specifications.

7. Unit rate quoted by the tenderers against various items of Schedule 'A'(BoQ) shall interalia, be deemed to include minor details of construction which are obviously and fairly intended but which may not have been specifically stated in tender documents but are essential for execution and completion of work. In case of any difference of opinion between the contractor and the GE as to whether or not any items of work constitutes "minor construction details" which is deemed to have been included in the contractor's quoted prices, the decision of the Accepting Officer shall be final, conclusive and binding.

SCHEDULE 'A' NOTES (CONTD..)

8. Description of items given in Schedule 'A' (BoQ) are in brief. These are deemed to be amplified and read in conjunction with latest ISS, Special Conditions, Particular Specifications, Specification of materials of workmanship and condition given in the relevant sections of the SSR Part I 2009 (Specifications) and SSR Part II 2020 (Rates).

9. The subject work to be executed in Restricted Area. Hence the tenderer shall satisfy himself by visiting the site with prior appointment of GE regarding working conditions, local conditions, nature of site/ground, climatic conditions, nature of work and means to access to site and other issues/circumstances which may affect the execution of work at a later stage. No extra payment with regard to aforesaid reasons shall be admissible under any circumstances.

10. The contractor shall make all precautionary arrangements not to damage the existing assets to the extent possible. It is becomes inevitable to execute the subject contract including cutting etc, the disturbed/ damaged surfaces/assets shall be made good to the original specifications without any extra cost to the govt.

11. As per Government of India, Ministry of Labour and Employment New Delhi Circular No S.35025/02/2014 – SS II dated 15 July 2015, it is mandatory that , Contractor shall get themselves registered with the Employees Provident Fund Organisation and deposit the necessary contributions with the EPFO. Further, it is a mandatory requirement that all construction workers should be provided Universal Account Number (UAN) by the contractor by appropriately registering them on the EPFO Portal. The Contractor has to upload Scanned Copy of Permanent or Temporary Provident Fund Code Number issued by Concerned Authority. The contractor shall also ensure compliance of the EPF & MP Act, 1952 by the sub-contractors, if any engaged by the contractor for the subject work.

12. It is mandatory on the part of the GE that while clearing the bills/ payments (RAR) of Contractors, a certificate be obtained from the contractor that all workers employed by him directly or indirectly are registered for EPF and the due contributions have been credited into their accounts.

13. The rate quoted by the tenderer shall be inclusive of all types of taxes / cesses viz GST and any other tax imposed / modified by the central / state govt during the currency of the contract. Nothing extra shall be admissible on this account. Tenderers are advised to independently ascertain the rate / percentage of goods & service tax, surcharge etc., and shall quote their rates accordingly. No claim on account of any kind on misunderstanding on this issue shall be entertained at a later date.

14. In case the BOQ is revised by the deptt and the bidder has failed to quote his rates in revised BOQ (i.e. the contractor has quoted in unrevised BOQ), such Bid shall be treated as wilful negligence on the part of contractor and such Bid/Quotation shall be considered Non-Bonafide. Also necessary action against such contractor(s) shall be taken by the Accepting Officer. Further it is clarified that in such cases, lowest tender shall be determined by the lowest amount amongst the valid/bonafide bids only.

SCHEDULE 'A' NOTES (CONTD..)

15. **LOCATION :** The tenderer's particular attention is also drawn to the fact that the Andaman & Nicobar Island are remote. Port Blair is situated approximately 1200 KM away from Mainland connected by airlines from Kolkata and Chennai and by Ship from Kolkata, Chennai and Vishakhapatnam. The work is required to be carried out at CampBell bay which is located further 550 KM away from Port Blair. Therefore the tenderer's are advised to well acquaint themselves about the working conditions and availability of local materials in these Islands and all other matters affecting the execution and completion of works before quoting their rates and no extra claims shall be entertained at a later stage.

16. The Coarse aggregate i.e, Crushed stone aggregate shall be conforming to the requirement as laid down in relevant IS from any approved quarry either from Main Land India or Port Blair as approved by GE.

17. The fine aggregate i.e, river sand to be incorporated in the work shall be obtained from Main land India conforming to the requirements as laid down in relevant IS as approved by the GE.

18. **PAYMENT TO LABOUR DURING EXECUTION OF WORK:** The payment to labour engaged by the contractor shall be made as per Wages Act. No payments to the contractor /firms will be made unless proof of payment of wages is submitted by the contractor/firm. In case payments are made through current coins or currency notes, certificate will be submitted by the contractor along with undertaking by the concern workmen asking for payments other than by cheques or by crediting in the bank account of the employee/workmen. Adherence to these instructions will be checked during various inspection. Being Principal Employer, the GE shall ensure that the contractor fully comply with above provisions. The documents held by the contractor shall be subject to scrutiny by law enforcement agencies. The contractor shall also provide proof of monthly payment details under EPF & ESI with account numbers alongwith details of regular payments which shall be verified by GE.

19. **MAKES/BRAND OF MATERIAL**: A list of makes is enclosed at Appx 'A' to Particular Specification and few makes are mentioned in the PS itself, however in case specific make is not mentioned in the tender documents, the material shall be procured from authorised dealer of manufacturers for which samples are kept in GE's office. The material so procured shall be ISI marked.

20. **ELECTRICAL LICENSE**

20.1 The exemption referred in Sec 184 of Electricity Act, 2003 is not applicable to MES Contracts. Valid Electrical License issued by concerned State Government / Union Territory shall only be applicable.

20.2 For execution of electrical works / electrical component of works requiring Electrical License in accordance with Rule 29, the contractor shall have valid Electrical License issued by the concerned State / Union Territory in the name of the Firm or get the work executed through an agency having valid Electrical License.. Prior to the commencement of Electrical works, copy of valid Electrical License in the name of the contractor or copy of agreement with agency having valid Electrical License alongwith agency's Electrical License shall be submitted by the contractor to the Accepting officer through GE with copy to CWE.

SCHEDULE 'A' NOTES (CONTD..)

20.3 In case contractor does not possess valid Electrical License and he intends to get such work executed through agency having valid Electrical License and with whom he has agreement, prior approval of such agency shall be got done from competent authority. Copy of License shall be kept on record by GE/ AGE as well as CE/CWE for future reference.

20.4 GE and concerned AGE/ Engr-in-Charge shall ensure that electrical work/ electrical component of works and other works required to be executed in accordance with Rule 29 are executed by contractor/ agency having valid Electrical License. In addition, Supervisor for execution of electrical works/ electrical components of works employed by the contractor shall possess Supervisor Certificate of Competency issued by concerned State Govt/ Union Territory and the worker/ tradesman for execution of electrical works/ electrical components of works shall hold necessary permit issued by concerned state Govt/ Union Territory. Reference of Electrical License and details of Supervisor's Certificate of Competency and Permit of tradesman employed for execution of electrical work/ electrical component of works shall be duly incorporated in the Works Diary and relevant site documents respectively for the whole period during which such work is executed.

21. **ENGINEER & T&P, MACHINERY & TRANSPORT:** Please note that you will be required to strictly ensure engagement of engineers and deployment of 'T&P, Machinery & Transport' as stipulated in the contracts. Inadequate engagement of engineers/deployment 'T&P, Machinery & Transport' as per contract conditions shall be considered as serious lapse attracting ban/removal/downgrading/debarment of the Firm/Company. In this connection General Condition of Contract (IAFW-2249) Condition 15, 16 & 25 shall be referred and strictly followed.

22. **GST (GOODS AND SERVICE TAX ACT)**

22.1 GST (Goods and Service Tax) as per GST Act- 2017 is applicable for this work. Tenderer shall be quote the unit rates/ amount accordingly.

22.2 The unit rate quoted by the tenderer for various items of Schedule 'A' shall deemed to include for new market materials, transportation all materials to site of works and labour including all types of taxes, levies including GST, labour welfare cess etc., as prevailing on the date of submission of tender. It may be noted that any tender imposing any condition in this regard or on any other account shall be treated as a conditional tender and the same shall be liable to be rejected.

22.3 Certain taxes such as central excise duty, service tax, additional custom duty, state level value added tax, octroi and other levies which were applicable on interstate transportation of goods are subsumed by GST, thus special condition here-in-after for Reimbursement / refund on variation in prices in taxes directly related to contract value of tender shall be deemed to be amended taxes directly related to contract value of tender shall be deemed to be amended incorporating GST in lieu of taxes mentioned in the condition but subsumed by GST.

22.4 The Contractor is required to submit proof for payment of Service Tax and cess to the Department as and when demanded by GE for verification. In addition to that it is mandatory that contractor shall upload the scanned copy of GST Registration Number while submitting their Technical bid. Contractor not in possession of this number will be disqualified in 'T' bid evaluation and his Finance bid will not be opened. Nothing extra whatsoever shall be admissible on this account. The clauses pertaining to tax liabilities wherever mentioned in the tender shall be deemed to be amended accordingly.

23. In case there are two identical items contractor shall quote same rates for such similar items, if there is an variation in rates, the rate quoted on lower side shall be considered for payment and contractor shall have no claim whatsoever on this account.

24. **DEFECTS LIABILITY PERIOD**

24.1 Defects liability period for the subject work shall be twenty four calendar months from the certified date of completion issued by GE.

SIGNATURE OF CONTRACTOR AGE(Contracts)
DATED: For ACCEPTING OFFICER

SCHEDULE OF CREDIT

1. The following old materials which are expected to be obtained under this contract shall become the property of the contractor subject to recovery of their salvage value indicated against each.

2. The quantities shown in Column – 4 are approximate and may vary.

3. The tenderer shall inspect the site with the prior approval of GE to ascertain the condition of the materials at site before submission of the tender. Irrespective of their actual visit to site, the tenderers are deemed to have visited the site. No claim whatsoever will be entertained after submission of the tender.

4. The rates inserted by the department in Column – 5 shall be made applicable for deducting the credit amount. The total amount of Schedule of Credit indicated in this Schedule of Credit will be considered as firm and will be deducted from the total sum quoted for **BOQ / Schedule "A"** to arrive Contract Sum for the purpose of acceptance.

5. The materials obtained from schedule of credit shall be taken away by the tenderer only after payment of credit/deducting of credit amount and after obtaining permission for its removal from GE.

6. The Contractor shall not re-use the material obtained from the Schedule of Credit.

7. The Garrison Engineer reserves the right to take back any or the full quantity of demolished/dismantled materials which in his opinion are serviceable. However, the recovery shall be affected only for the quantities issued to the contractor.

8. Serviceable materials / items other than shown herein below obtained from demolition / taking down if any, shall remain the Government property and the contractor shall deposit the same with Engineer-in- Charge as directed and debris / unserviceable material shall be disposed off the site as directed by the Engineer-in-Charge without any extra cost to Government.

S.No	Brief Description	A/U	Qty	Rate	Amount
1	**2**	**3**	**4**	**5**	**6**
1.00	Unserviceable fencing post	RM	142.80	10.00	1428.00
2.00	Unserviceable scrap iron	Qtl	14.55	150.00	2182.50
3.00	Old unserviceable block walling any type and size	Cum	9.00	300.00	2700.00
4.00	Unserviceable wall paneling any type including fillets	Sqm	38.00	20.00	760.00
				Total	**7070.50**

SIGNATURE OF CONTRACTOR
DATED :

AGE (CONTRACTS)
FOR ACCEPTING OFFICER

TO

BOQ

SUMMARY OF SCH 'A'

1.	Total amount of Sch 'A' quoted on Sl Page No 36 of (BOQ)	:	₹
2.	Deduct amount of Schedule of Credit brought forward from Sl Page No 32	:	₹ (-) 7,070.50
			₹

SIGNATURE OF CONTRACTOR
DATED :

AGE (CONTRACTS)
FOR ACCEPTING OFFICER

SCHEDULE 'B'
ISSUE OF MATERIALS TO THE CONRACTOR
(SEE CONDITION 10 OF IAFW-2249)

Sl No	Particulars	Rate at which materials will be issued to contractor		Place of issue (By name)	Remarks
		Unit	Rs Ps		
1	2	3	4	5	6

_____ Nil _____

SIGNATURE OF CONTRACTOR
DATED :

AGE (Contracts)
For ACCEPTING OFFICER

SCHEDULE 'C'
ISSUE OF TOOLS AND PLANTS (OTHER TRANSPORT) WHICH WILL BE HIRED TO THE CONTRACTOR (REFER CONDITION 15&34 OF IAFW – 2249)

Sl No	Qty	Particulars	Details of MES crew supplied	Hire charges per unit working day Rs. Ps.	Stand by Charges per unit per off Day Rs. Ps.	Place of Issue by name
1	2	3	4	5	6	7

_____ Nil _____

SIGNATURE OF CONTRACTOR
DATED :

AGE (Contracts)
For ACCEPTING OFFICER

SCHEDULE 'D'
TRANSPORT TO BE HIRED TO THE CONTRACTORS
(See condition 16 & 35 of IAFW-2249)

Sl No	Qty	Particulars	Rate per unit per working day	Place of issue (by name)	Remarks
1	2	3	4	5	6

____ Nil _____

SIGNATURE OF CONTRACTOR
DATED :

AGE (Contracts)
For ACCEPTING OFFICER

(In lieu of IAFW-1779A (Revised 1955))

TENDER

To

THE PRESIDENT OF INDIA

Having examined and perused the following documents.

1 Specifications signed by AGE (Contracts)

2. Drawings detailed in the specifications.

3. Schedule 'A', 'B', 'C' and 'D' attached hereto.

4. MES Standard Schedule of Rates 2009 (Part I specification) and 2020 -Part II rates hereinafter referred to as the MES Schedule together Errata/amendments 1 to 3 for Part I and together with amendment Nos 1 to 122 for Part II (Rates) and errata Nil as applicable to the above said schedule.

5. General conditions of contracts, IAFW-2249, (1989 print) together with amendments 1 to 49 and errata Nos. 1 to 20.

6. WATER CONDITION 31 OF IAFW-2249, General Conditions of Contracts.

Water will not be supplied by MES to the contractor and the contractor shall make his own arrangement for water required if any for the work as well as for his labourers.

7. Should this tender be accepted I/We agree: -

*(a) I/WE agree that the sum of **₹ 57,000.00 (Rupees Fifty Seven thousand only)** forwarded as earnest money shall either be retained as part of performance security deposit or be refunded by the Government of receipt of appropriate amount of performance security deposit all as per condition 19 of IAFW- 2249.

(b) To execute all the works referred to in the said documents upon the terms and conditions contained or referred to therein into at the item rates contained in the aforesaid BOQ or at such other rates as may be fixed under provision of condition 62 and to carry out such deviations as may be ordered vide Condition 7 of IAFW-2249 upto a **maximum of 10 (TEN) percent** and further agree to refer all disputes as required by Condition 70 of IAFW-2249 to the Sole Arbitration of a serving officer having degree in Engineering or equivalent or having passed final/direct final examination of sub division II of Institution of Surveyor(India) recognised by Government of India to be appointed by the Chief Engineer (A&N) Zone, Port Blair or in his absence, the officer officiating as Chief Engineer (A&N) Zone, Port Blair or if specifically delegated in writing by Chief Engineer (A&N) Zone, Port Blair whose decision shall be final, conclusive and binding.

**Delete whichever is not applicable.

TENDER (CONTD)

Signature ---------- in the capacity of ---------- duly authorised to sign the tender for and on behalf of M/s ----------(Name in Block Capitals)

Witness _____ Postal Address _____

Address_____ Telegraphic Address_____

_____ Telephone No._____

ACCEPTANCE

_____ alterations have been made in these documents and as evidence that these alterations were made before the execution of the of the contract agreement they have been initialed by the contractor and Shri _____ The said Officer / (s) is/ are hereby authorized to sign and initial on my behalf the documents forming part of this contract. The above tender was accepted by me on behalf of the President of India for an amended/revised contract sum of Rs_____ (Rupees_____ _____ _____only) on _____ 2025.

Signature:_____ Dated this _____ day of _____ 2025.

(FOR AND ON BEHALF OF THE PRESIDENT OF INDIA)

Appointment: Garrison Engineers (I)
GE (I) Campbell Bay
Army Land, Great Nicobar Islands
Campbell Bay – 744 302

GENERAL CONDITIONS OF CONTRACTS

[IAFW- 2249- 1989 PRINT]

FOR ITEM RATE CONTRACTS (IAFW-1779A)

A copy of the GENERAL CONDITIONS OF CONTRACTS (IAFW-2249 1989 Print) with Errata 1 to 20 and Amendments No. 1 to 49 has been supplied to me/us and is in my/our possession. I/we have read and understood the provisions contained in the aforesaid GENERAL CONDITIONS OF CONTRACTS before submission of this tender and I/We agree that I/we shall abide by the terms and conditions thereof as modified, if any, elsewhere in these tender documents.

It is hereby further agreed and declared by me/us, that the General Conditions of Contracts (IAFW-2249-1989 Print) including Condition 70 & 71 thereof pertaining to settlement of disputes by Arbitration and concilation with Errata 1 to 20 and Amendment Nos. 1 to 49 form part of these tender documents.

Note : In case of difference in interpretation due to wordings of English and Hindi versions of the General Conditions of contracts (IAFW-2249) (1989 print), the English version will prevail.

* Delete which is not applicable

AGE (Contracts)
For Accepting Officer

SIGNATURE OF CONTRACTOR
DATED:

SCHEDULE OF MINIMUM FAIR WAGES

(PAYABLE BY THE CONTRACTOR UNDER MINIMUM FAIR WAGES ACT)

(See Condition 58 of IAFW-2249)

1. It is hereby agreed by me/ us that the schedule of Minimum Fair Wages as published vide Government of India Ministry of Labour & Employment shall form Part of these tender documents state Govt/ Union Territory Notification forms part of these tender documents.

2. It is also agreed by me/ us that although latest notification, as available with the Accepting Officer, has been formed part of the contract, in case the Government has further revised the wages, the latest revised wages of labours shall only be applicable.

3. My/ our signature (s) here under is deemed to my/ our having read and understood the provisions contained therein and having signed the aforesaid Schedule of Minimum Wages forming part of this tender and in agreement of provisions of Para 2 above * I/ we agree that * I/ we shall abide by the same and that aforesaid documents form part of the tender.

4. The latest revised wages of labour as fixed by State Govt/ Govt of India/ Union Territory as applicable to the work whichever is higher will only be applicable in the contract.

NOTE : Schedule of Minimum fair wages is available for reference in the office of the GE (I) C'BAY or any other Office of CE/CWE/GE of MES.

AGE (Contracts)
For Accepting Officer

SIGNATURE OF CONTRACTOR
DATED:

SPECIAL CONDITIONS

1. **GENERAL**

1.1 The following special conditions shall be read in conjunction with the General Conditions of Contracts. IAFW-2249 [1989 print] and IAFW-1779 'A' [Revised 1955] / IAFW 2159 [Revised 1947] including Errata/Amendments thereto. If any provisions in these special conditions are at variance with that of the aforesaid documents, the former shall be deemed to take precedence there over.

2. **ADMISSIONS TO SITE BY CONTRACTORS TO ASCERTAIN HIS OWN INFORMATION**

2.1 The tenderers shall contact the Garrison Engineer for the purpose of the inspection of site[s] and relevant documents other than those sent herewith who will give reasonable facilities for this purpose. The tenderers shall also make themselves with working conditions, accessibility of site [s] availability of materials and other cogent conditions etc which may affect the entire completion of work under this contract.

2.2 The tenderers shall be deemed to have visited the site[s] and made themselves familiar with the working conditions, accessibility of site [s], availability of materials and other cogent conditions, etc which may affect the entire completion of work under this contract, whether they actually inspect the site[s] or not. No extra payment consequent on any mistake or misunderstanding otherwise on this account will be allowed.

3 **SECURITY AND PASSES**

3.1 Contractors attention is invited to condition 25 of IAFW-2249. He shall employ only Indian Nationals after verifying their antecedents and loyalty. The contractor shall on demand by the Engineer-in-Charges, submit list of his agents, employees and working people concerned and shall satisfy the Engineer-in-charge as to the bonafides of such people.

3.2 The Engineer-in-charge shall at his discretion have the right to issue passes as per rules and regulations of the installation/area in force to control the admission of the contractor, his agents, and employees and work people to the site of the work or any part thereof. Passes should be returned at any time on demand by the Engineer-in-Charge of the authorities concerned and in any case on completion of works.

3.3 The contractor and his agents, employees and work people shall observe all the rules promulgated by the authority controlling installation/area in which the work is to be carried out e.g., prohibition of smoking and lighting, fire precautions, search of persons on entry and exit, keeping to specific routes, observing specified timing etc. Nothing extra shall be admissible for any man hour etc lost on this account.

4. **MATERIALS AND SAMPLES**

4.1 Refer condition 10 of IAFW-2249 and clause 1.6 & 1.7 of MES Standard Schedule of Rates-2009 Part-I.

SPECIAL CONDITIONS

4.2 Materials provided by the contractor for incorporation in the works shall, unless otherwise specified in the particular specifications be ISI marked. IS means Indian Standards as issued by the Bureau of Indian Standards. Wherever in the specifications 'IS' is referred to, it means the edition with all amendments, current on the due date of receipt of the tender documents.

4.3 The tender is advised to inspect other materials, which are displayed in the office of GE, before submitting his tender. The tenderer shall be deemed to have inspected the samples and satisfied himself as to the nature and quality of materials, he is required to incorporate in the work irrespective of whether he has actually inspected them or not. The materials to be incorporated in the work by the contractor shall be ISI marked or shall be equal or superior in quality to sample displayed and shall comply with the specifications given hereinafter.

4.4 The contractor shall produce samples of all materials, articles, fittings, accessories etc that he proposes to use and get these approved in writing by the Garrison Engineer within reasonable time from the date of commencement of works as per work order. The materials, articles, etc as approved shall be labeled as such and shall be signed by the GE and the contractor's representative. These samples shall be kept in the custody of the Garrison Engineer/Engineer-in-Charge.

4.5 The contractor shall not procure materials unless the samples are first got approved by the Garrison Engineer.

4.6 The brand of all materials, articles, fittings, etc, approved together with the names of the manufacturers and firms from which supplies have been arranged shall be recorded in the works passing register.

4.7 The Govt. reserves the right to get the items/articles to be incorporated in the work tested in approved laboratories. The cost of sample for testing shall be borne by the contractor and the remaining expenses such as cost of transportation of sample to laboratory and testing fee shall be borne by the Govt, if the test result is found to be satisfactory. However, in the event of the test result being found unsatisfactory, the entire cost of testing including cost of sample shall be borne by the contractor. Government may get more than one sample of the same material tested and the cost of such testing shall be borne by the Government.

5. PROPRIETORY/BRANDED MATERIAL

5.1 Proprietary/branded materials such as paints, liquid chemicals, waterproofing compound etc. like quality of which cannot be checked after incorporation in the work, shall when collected at site, be recorded in measurement book and signed both by the Engineer-in-Charge and the contractor as a check to ensure that the required quality has been brought at site for incorporation in the work.

5.2 Materials brought to site shall be stored as directed by the Engineer-in-Charge and those already recorded in measurement book shall be suitably marked for identification.

SPECIAL CONDITIONS

5.3 The contractor shall obtain proprietary/branded materials from manufacturers or from manufactures authorised stockists where such authorised stockist have been appointed. The contractor shall, on demand, produce original receipted vouchers/ invoices of suppliers to the Garrison Engineer, to ensure that the contractor has actually brought the required quality of the materials from the authorised dealers/manufactures and also to find out the rates thereof.

5.4 The original vouchers, invoices shall be defaced and stamped by Engineer-in-Charge, indicating contract number, name of work, under his dated signature. The contractor shall ensure that the materials are brought to site, in original sealed containers/packing bearing manufacturer's marking except in the case of the requirement of material(s) being less than the smallest packing.

6. CRITICAL PATH METHOD IN LIEU OF TIME AND PROGRESS CHART

6.1 The CPM chart to be prepared as per condition 11 of IAFW-2249 (General Conditions of Contracts) shall consist of detailed network analysis and a time Schedule. The critical path network will be drawn jointly by the Garrison Engineer and the contractor soon after acceptance of the tender. The time scheduling of the activities will be done by the contractor so as to finish the work within the stipulated time. On completion of the time schedule, firm calendar date schedule will be prepared and submitted by the contractor to the Garrison Engineer who will approve it after due scrutiny. The schedule will be submitted in four copies within two weeks from the date of handing over the site.

6.2 During the currency of the contract, the contractor is expected to adhere to the time schedule and this adherence will be a part of the contractor's performance under this contract. During the execution of work, the contractor is expected to participate in the reviews and updating of the network undertaken by the GE. These reviews may be undertaken at the discretion of the Garrison Engineer either as a periodic appraisal measure or when the quantum of work ordered on the contractor is substantially changed through deviation orders or amendments. Any revision of the time schedule as a result of the review will be submitted by the contractor to the Garrison Engineer within a week for his approval after due scrutiny. The contractor will adhere to the revised schedule thereafter. In case of contractor disagreeing with the revised schedule, the same will be referred to the Accepting Officer, whose decision will be final, conclusive and binding. Garrison Engineer's approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to grant of extension of time.

6.3 Extension of time shall be considered and decided by the appropriate authority mentioned in condition 11 of IAFW-2249 and separately regulated.

6.4 The contractor shall mobilise and employ sufficient resources to achieve the detailed schedule within the broad framework of the accepted methods of working and safety.

6.5 No additional payment will be made to the contractor for any multiple shift work or other intensive methods contemplated by him in his work schedule, even though the time schedule is approved by the department.

SPECIAL CONDITIONS

6.6 Contractor shall plan for execution of completion of work with foresight to ensure timely execution with the quality of work desired.

6.7 Contractor shall employ qualified site engineers to supervise the work so as plan and got executed contract work as per standard engineering practice.

6.8 Contractor shall deploy requisite tools/plants and machineries to complete the job in a schedule time.

7. **SECURITY OF CLASSIFIED DOCUMENTS**

7.1 Contractor's special attention is drawn to condition 2-A and 3 of General Conditions of Contracts (IAFW-2249). The contractor shall not communicate any classified information regarding the works either to sub-contractors or others without prior approval of the Engineer-in-Charge. The contractor shall also not make copies of the design/drawings and other documents furnished to him in respect of the works and he should return all documents furnished to him in respect of the works on completion of the work or earlier termination of the contract. The contractor shall along with the final bill attach a receipt from the Engineer-in-Charge in respect of his having returned the classified documents as per condition 3 of General Conditions of Contracts (IAFW-2249).

8. **FAIR WAGES:** The contractor shall have no claims, whatsoever if on account of any rules and regulation or otherwise he is required to pay wages in excess of the fair wages shown in the schedule of wages under condition 58 of General Conditions of Contracts IAFW-2249.

9. **RECORD OF CONSUMPTION OF CEMENT**

9.1 For purpose of keeping record of cement brought by and consumed in works, the contractor shall maintain a pucca bound register with serially numbered pages with all pages initialed by Engineer-in-Charge against numbering in the form approved by the Engineer-in-Charge showing daily receipts of cement, quantity used in works and balance in hand. The register shall be signed daily by representative of MES and contractor in token of their verification of its correctness and will be checked by Engineer-in-Charge at least once a week and on the days cement is brought by the contractor.

9.2 The aforesaid provision will not however, absolve the contractor from his responsibility to justify the consumption of cement at the time of finalisation of his accounts.

9.3 The register shall be kept at site in the safe custody of the contractor during progress of the work and he shall on demand produce the same for verification by inspecting officers. On completion of the works cement register shall be handed over to the Engineer-in-Charge for record with MES.

10. **QUARRIES AND ROYALTIES** :

10.1 Delete the existing condition 14 of IAFW-2249 and in lieu add-'Neither quarries are available nor the quarrying shall be permitted on Defence Lands.

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11. LAND AND LABOUR ACCOMMODATION, AND STORES AND WORKSHOP ETC.

11.1 Delete lines 5 to 9 of Para 1 of condition 24 of General Conditions of Contracts IAFW-2249 i.e., from "In the event of area of land' to 'land allotted to him' and insert as under. 'Separate land may be allotted by the GE to the contractor for the storage of materials, temporary workshop and offices, for which he shall pay nominal rent of Re 1/- per year. No MD land is available for accommodation of labour and canteen etc in restricted area for which contractor has to make own arrangement outside the premises.

12. CO-OPERATION WITH OTHER AGENCIES

12.1 The contractor shall permit free access and generally afford reasonable facilities to other agencies or departmental workmen engaged by Government to carry out their part of the work, if any, under separate arrangements.

12.2 The contractor's prices shall be deemed to cater for all the above contingencies and nothing extra shall be admissible on these accounts.

13. LABOUR (REGULATION & ABOLITION) ACT

13.1 Contract labour (Regulation & Abolition) Act 1970 is applicable to MES contractors. Rates quoted by the tenderer shall be deemed to take into account the cost, etc, required to comply with the provisions contained in the said act and the rules framed under the said act.

13.2 Refer condition 58 of IAFW-2249. The 'Schedule of Minimum Wages' as published vide Govt. of India Notification as available on date of receipt of tender shall forms part of these tender documents. However, the contractor shall not pay wages lower than minimum wages for labour as fixed by the Govt of India/State Govt/Union territory under Minimum Wages Act or contract Labour (Abolition and Regulation Act), whichever is higher.

13.3 The fair wages referred to in condition 58 of IAFW-2249 will be deemed to be the same as the minimum wages payable as referred to above.

13.4 The contractor shall have no claim whatsoever, if on account of local factors and/or regulations, he is required to pay the wages in excess of minimum wages as described above during the execution of the works.

14. WATER SUPPLY

14.1 Water will not be supplied by the MES. Contractor shall make his own arrangement for Water.

15. ELECTRIC SUPPLY

15.1 No electricity for lighting and power shall be supplied to the contractor by the MES. The contractor shall have to make his own arrangements to generate electricity or to get electric connection from the electric supply agency of their own arrangements.

15.2 In case Electricity is available and supplied at any point by Engr-in-Charge, recovery shall be effected as per local electricity departmental charges/ All-in-Cost rates as applicable.

16. RECORD OF MATERIALS

16.1 The quantity of materials such as paints, bitumen, water proofing compound, chemical for anti-termite treatment and the like, as directed by the Engineer-in-Charge (The quantity of which cannot be checked after incorporation in the work) shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work. Batch No./ Lot No. mentioned on packing of the items shall also be recorded in the MB.

16.2 Materials brought to site shall be stored as directed by the Engineer-in-Charge and those already recorded in measurement book shall be suitably marked for identification.

16.3 The contractor shall produce to the GE, original receipted vouchers/invoices in respect of the supplies. The vouchers / invoices so produced and verified shall be stamped by Engineer-in-Charge indicating contract number, name of work under his dated signature. The contractor shall ensure that the materials are brought to site in original sealed containers / packing, bearing manufacturer marking except in the case of the requirement of material being less than smallest packing.

16.4 Contractor shall produce vouchers / invoices in original from the manufacturers and / or their authorised distributors only for the full quantity of the following materials **as applicable** and any other item as asked by Accepting Officer / GE as pre-requisite before submitting claims for payment of advances on account of the work done and / or materials collected in accordance with Condition 64 of General Conditions of Contracts IAFW-2249:-

SPECIAL CONDITIONS

Factory made panel/ flush shutters	Paint	Steel
Factory made steel windows/ ventilator	Aluminium door/ window/ ventilator/ shutter/ frame	Floor Spring
Iron Mongery	Chequered tiles, Ceramic, Vitrified Tiles	CGI/PGI Sheet
Pressed steel door frame	Mild Steel Sheet	Seamless Ceiling
Sheet glass	PVC Insulation Panels	Kota Stone
Galvalume Plain/ Corrugated Sheet	GI pipe/ tubes	Rolling Shutter
Tubular Truss	Cement	MCB/ MCB DB/ MCCB and cost of electrical items more than Rs25000
Electric Cables/ Wire/ Fittings/ Fixtures	PVC (SWR) Pipe	Steel Wire Cloth
Plywood	Water Proofing Compound	HDPE water tank
Steel window/ ventilator	Builders Hardware	Kitchen Sink & Rack
PTMT fittings	Sanitary, Bathroom Fittings	All other non-perishable materials as decided by GE.

17. **RETENTION MONEY/ COMPENSATION FOR DELAY**

17.1 Refer condition 64 and 50 of IAFW-2249. For the purpose of calculating retention money under condition 64 of IAFW-2249, and compensation for delay under condition 50 of IAFW-2249, the value of contract as revised by price variation shall be taken into account.

18. **OFFICIAL SECRET ACT**

18.1 Refer Condition 2-A of IAFW-2249 (General Conditions of contracts). The contractor shall be bound by the Official Secret Act and particularly section 5 thereof.

19. **SCOPE OF CONCILIATION**

19.1 The scope of conciliation shall be restricted to the following types of disputes with financial limits as indicated therein:-

(a) Disputes relating to levy of compensation for delay in completion- Actual amount of compensation.

(b) Disputes relating to technical examination of works.

(c) Disputes relating to interpretation of the provisions of the contract with reference to their application to parties.

(d) Disputes relating to non-return of Schedule 'B' stores over-issued to contractor.

(e) Any other disputes having fair chances of being resolved by conciliation and considered fit to be referred to conciliation by the parties.

(f) For (b), (c), (d) and (e) each as stated above the financial limit shall be Rupees two lakh or one percent of the contract amount whichever is less.

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19.2 **COMMENCEMENT OF CONCILIATION PROCEEDINGS**

(a) The party initiating conciliation shall send to the other party a written communication to conciliate briefly identifying the subjects of disputes.

(b) On receipt of such request, the Accepting Officer shall within fifteen days refer the matter to CE (A & N) Zone or in his absence the Officer Officiating as Chief Engineer.

(c) Chief Engineer within one month of receipt of request for appointment of Sole Conciliator shall appoint Sole Conciliator who shall be serving officer having degree in Engineering or equivalent or having passed final/direct final examination of sub division II of Institution of Surveyors (India) recognized by Govt and refer the disputes for settlement/agreement between the parties to the contract.

(d) The Sole Conciliator so appointed within fifteen days will enter into reference and fix a hearing to represent the matter by both the parties and finalize the findings within two months after entering into reference and shall forward his proposed settlement / agreement to both the parties to contract.

(e) After receipt of settlement/agreement, the Accepting Officer and the other party within one month shall communicate their consent or rejection of settlement/agreement to each other in writing and if the settlement/agreement as suggested by Sole Conciliator is agreed to by both the parties, shall have the same status and effect as it is on arbitral award on agreed terms within the provisions of Arbitration and Conciliation Act 1996. The silence of both the parties after receipt of settlement/agreement as arrived by Sole Conciliator after one month shall be considered as finality of the settlement/agreement and shall be binding on both the parties not subject to challenge again. However, if any party rejects the settlement/agreement within one month from the date of receipt of settlement/agreement, the dispute may be referred by the aggrieved party for adjudication by arbitrator as per agreed terms of contract.

(f) The amount of settlement shall be treated as an agreed amount payable under the contract and such payment shall be processed through hand receipt.

19.3 **NUMBER OF CONCILIATORS**

19.3.1 There shall be a sole conciliator.

19.4 **APPOINTMENT OF CONCILIATOR**

19.4.1 All disputes brought out in para 23.1 (a) to (e) above shall be referred to the sole conciliator viz serving officer not below the rank of Superintending Engineer / Superintending Engineer (QS & C) having degree in Engineering or equivalent or having passed final / direct final examination of Sub division II of Institution of Surveyors (India) to be appointed by the officer officiating as Engineer-in-Chief or Director General of works specifically delegated by the Engineer-in-Chief, Army Headquarters, New Delhi or in his absence the officer officiating as Engineer-in-Chief or Director General of works specifically delegated by the Engineer-in-Chief in writing.

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19.5 STATUS OF EFFECT OF SETTLEMENT AGREEMENT

19.5.1 The settlement agreement signed by the parties as a result of conciliation proceedings shall have the same status and effect as it is an arbitral award on agreed terms.

20. REIMBURSEMENT / REFUND ON VARIATION IN "TAXES DIRECTLY RELATED TO CONTRACT VALUE "

20.1 The rates quoted by the contractor shall be deemed to be inclusive of all taxes / cesses viz GST, duties, Royalties, Octroi & Other levies payable under the respective statutes. No reimbursement/refund for variation in rates of taxes, duties, Royalties, Octroi & other levies, and /or imposition/abolition of any new/existing taxes/ cesses, duties, Royalties, Octroi & other levies shall be made except as provided in Condition 20.2 herein below.

20.2 The taxes/ cesses which are levied by Government at certain percentage rates of contract sum/ amount shall be termed as " taxes directly related to contract value" such as GST, Labour welfare cess/ tax and like as applicable but excluding Income Tax. The tendered rates shall be deemed to be inclusive of all "taxes directly related to Contract value" with the last due date for receipt of tenders. Any increase in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on last due date for receipt of tenders shall be reimbursed to the Contractor and any decrease in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on last due date of receipt of tenders shall be refunded by the Contractor to the Govt/deducted by the Government from any payments due to the Contractor. Similarly imposition of any new "taxes directly related to Contract value" after the last due date for receipt of tenders shall be reimbursed to the Contractor and of any "taxes directly related to Contract value" prevailing on last due date for receipt of tenders shall be refunded by the Contractor to the Govt/deducted by the Government from the payments due to the Contractor.

20.3 The contractors shall within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further "taxes directly related to Contract value" give written notice thereof to the GE stating that the same is given pursuant to this Special Condition, together with all information relating thereto which he may be in a position to supply. The Contractors shall also submit the other documentary proof/ information's as the GE may require.

20.4 The contractor shall, for the purpose of this connection keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorized representative of Government, and shall further at the request of the GE furnish, verified in such a manner as the GE may require, any documents so kept and such other informations as the GE may require.

20.5 Reimbursement for increase in percentage rates/ imposition of "taxes directly related to Contract value" shall be made only if Contractors necessarily and properly pays additional "taxes directly related to Contract value" to the Government, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Government Authority and submits documentary proof for the same as the GE may require".

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21. **ADVANCES ON ACCOUNT OF NON PERISHABLE - MATERIAL/ITEMS WHICH DO NOT LOOSE IDENTITY AS DECIDED BY THE GE:** Refer Condition 64 of IAFW 2249.

22. **CLEANING DOWN (REFER CONDITION 49 OF IAFW-2249**)

22.1 The contractor shall clean all floors, walls, remove cement/lime/paint marks/drops, the joinery glass panes etc, touch up all painters work and carry out all other necessary items of work in connection therewith and leave the whole premises clean and tidy before handing over the building. No extra payment shall be claimed by the contractor for this operation.

23. **LABEL OF ARTICLES** : Contractor shall provide aluminimum LABEL of not less than 75 mm x 25 mm and of adequate gauge with brass screws on articles like transformer, panel board, generating set etc indicating there on the names of the firm, the contract No, and year as directed by GE. The cost of such aluminium labels shall be deemed to be included in the quoted rates against respective item of Schedule 'A'.

24. **DAMAGE TO STRUCTURE**

24.1 Any damage done to the structure built or being built by other agency during execution of work shall be made good by the contractor at his own cost and the site of work left clean and tidy on completion. Rectification, reinstatement, making good etc. shall conform to the standard of materials originally used in the work and finished work shall match with existing work in all respect to the entire satisfaction of the GE. In case of any dispute on this account the matter shall be referred to the CWE whose decision in writing shall be final, conclusive and binding.

25. **CONDITIONS OF WORKING**

25.1 **CONDITIONS OF WORKING IN RESTRICTED AREA** : The entire work under this contract lies in "RESTRICTED AREA". Restriction for entry to work site and working conditions in Restricted Area shall be as under:-

(a) **DEFINITION** : Restricted area(s) for the purpose of this contract means the area(s) declared as such by the units as shown in site plan/described in Special Conditions.

(b) **VISIT TO SITE WITHIN THE RESTRICTED AREA:** Permission to enter the restricted area(s) at time of submission of tenders can be obtained, through the Garrison Engineer. Tenderers are advised to send prior intimation to the Garrison Engineer about the particulars of the agents, representative etc., if any, the date and the time of their proposed visits so that necessary arrangement may be made by the GE, to secure admission. Whether tenderers visit the site or not they shall be deemed to have full knowledge of the restrictions on entering in, exit from and working within the restricted area.

(c) **ENTRY AND EXIT:** The contractor/his agent(s)/ representative(s)/ workmen etc., and his materials carts, trucks or other means of transport, etc., will be allowed to enter through and leave from only such gate or gates and at such times as the GE or authorities in charge of the restricted area may at their sole discretion permit to be used. The contractor's authorized representative is required to be present at the places of entry and exit for the purpose of identifying his carts, trucks, etc, to the personnel-in-charge of the security of the restricted area.

SPECIAL CONDITIONS

(d) **IDENTITY CARDS OR PASSES**

(i) The contractor, his agents and representatives are required individually to be in possession of an identity card or pass duly verified by the GE. The identity card or pass will be examined by the security staff at the time of entry into or exit from the restricted area and also any time or number of times inside restricted area.

(ii) **IDENTITY OF WORKMEN** :Every workman shall be in possession of an Identity Card. The identity cards shall be issued after a thorough investigation of the antecedents of the labourers, by contractor and attested by the officer-in-Charge of the unit concerned in accordance with the standing rules and regulations of the unit.

(iii) The contractor shall be responsible for conduct of his workmen, agents or representatives.

(e) **SEARCH** : Thorough search of all persons and transport shall be carried out at each gate and for as many times as gate is used for entry or exit and may also be carried out any time or any number of times at the work site within the restricted area.

(f) **FEMALE SEARCHER** : If the contractor desires to employ female labour on works to be carried-out inside the area of factory, depot, park, unit, etc and a female searcher is not borne on the authorised strength of the factory, depot, park, unit, etc at the time of submission of tender, he shall be deemed to have allowed in his tender for pay and allowances etc, for a female searcher (Class IV servant/GP 'D' servant) calculated for the period, female labours are employed by him inside the area. If more than one contractor employs female labour during any month and female searcher(s) has/have to be employed in addition to the authorised strength of the factory, depot, park, unit, etc, the salary and allowances paid to the additional female searcher(s) shall be distributed on equitable basis between the contractors employing female labour taking into consideration the values and periods of completion of their contracts. The GE's decision in regard to the amount recoverable on this account from any contractor shall be final and binding.

(g) **WORKING HOURS** : The units controlling restricted area, usually work during six days in a week and remain closed on the 7th day. The working hours available to contractor's labour/staff shall however accordingly get reduced because of the time taken in security checks observed at the time of entry and exit and during working hours. The exact working hours, working days and non working days observed for these restricted area(s), where works are to be carried out shall be deemed to have been ascertained by contractor before submitting his tender. The tenderers attention is invited to the fact that total number of working hours for a unit are prescribed in regulations and that they cannot be increased by the Garrison Engineer. The definition of "Working Day" as given under condition 1 (t) of IAFW-2249 does not apply in case where the works are carried out in restricted area.

SPECIAL CONDITIONS

(h) **WORK ON HOLIDAYS** : The contractor shall not carry out any work on gazetted holidays, weekly holidays and other non-working days except when he is expressly authorised in writing to do so by the GE. The GE may at his sole discretion declare any as holiday or non-working day without assigning any reason for such declaration.

(j) **ACCESS TO RESTRICTED AREA AFTER COMPLETION** : After the works are completed and surplus stores etc. removed, the contractor, his agents, representatives or workmen etc., will not be allowed any access to the restricted area except for attending to any rectification of defects pointed out to him by the GE.

(k) **FIRE PRECAUTIONS**

(i) The contractor, his agents, representatives, workmen etc., shall strictly observe the orders pertaining to fire precautions prevailing within the restricted area.

(ii) Motor transport vehicle, if allowed by the authorities to enter the restricted area must be fitted with serviceable fire extinguishers.

26. **CONSTRUCTION WORKER WELFARE TAX**

26.1 Consequent upon promulgation of ordnance by the President of India, contractor shall be liable to pay an element for construction Labour Welfare Tax , if any and quoted rates shall be deemed to be inclusive of this element and no extra payment shall be admissible to the contractor

27. **BUILDING AND OTHER CONSTRUCTION WORKERS WELFARE CESS ACT**

27.1 The quoted rates shall deem to be inclusive of the building and other construction workers welfare cess. The cess at the rate as prescribed by the state Govt/Building and other construction workers welfare Board shall be deducted at source and no extra payment shall be admissible to the contractor on this account.

28. **COVERAGE OF BUILDING AND CONSTRUCTION WORKERS(BACW) IN THE SCHEME FRAMED UNDER EMPLOYEES PROVIDENT FUNDS & MISCELLANEOUS PROVISIONS (EPF & MP) ACT**

28.1 Tenderer's attention is invited to employee's provident fund and miscellaneous provision Act 1952. As per this Act, the contractor shall have provident fund code number. The contractor who is not in possession of provident fund code number shall be disqualified in 'T' bid evaluation and his financial bid shall not be opened. It is mandatory to upload copy of EPF registration certificate along with application for tender.

28.2 It is the responsibility of the contractor to ensure that all the workers engaged by him/ them are having universal account number (UAN) duly allotted by Provident fund dept and their contributions are properly credited into their account. A certificate to this effect shall be furnished by the contractor while claiming Final bill. **The contractor shall also ensure compliance of the EPF & MP Act, 1952 by the sub-contractors, if any engaged by the contractor for the subject work.**

29. **PERIOD OF KEEPING THE TENDER OPEN**

29.1 The tender shall remain open for acceptance for a **period of SIXTY(60) days from due date of submission of the bid excluding the date of submission of the bid.**

30. **INTER STATE MIGRANT WORKMAN ACT 1979 :** The tenderer shall be fully conversant with the above Act and take necessary steps to implement the same effectively. The tenderer shall intimate the number of workmen(Skilled/ Unskilled) to be inducted/ imported and total number of labour employed locally at site immediately after issue of work order. Immediately after completion of work,

SPECIAL CONDITIONS

workmen inducted from mainland shall be repatriated. Prior permission shall be obtained if the imported/ inducted workmen are required to stay back for rectification of work if any after the date of completion and handing over the work to the Department. The tenderer shall produce necessary documents as necessary for verification and authentication of having repatriated the imported labour. Suitable amount shall be retained from the Final Bill and will be released only after the Accepting Officer is fully satisfied for the return of the imported labour to main land. Decision of the GE shall be final and binding on this account.

31. **TESTING OF MATERIALS**

(a) **'A' LEVEL TESTS FOR WORKS COSTING 100 LAKHS AND ABOVE :**

The contractor shall set up a site laboratory for testing of materials (except Schedule 'B' materials) for 'A' level tests as listed in Annexure-I hereto. The contractor shall arrange all equipment/machines for the tests specified in Annexure-I as 'A' level tests at his own cost with prior approval of GE. This cost shall be included in the lump-sum cost quoted by the contractor.
The contractor shall employ a competent technical representative as approved by the GE for the purpose of testing and all such tests shall be carried out in the presence of Engineer-in-Charge. The successful tests results thereof shall be recorded and signed jointly by the contractor and the Engineer-in-Charge. The charges for these tests ie 'A' level tests carried out in the site laboratory of the contractor shall not be recovered. In case, the contractor has not set up the site laboratory and the tests are carried out in Zonal lab or any other laboratory setup/approved by GE, the recovery shall be made at rates applicable ie as given in Annexure I.

(b) **'A' LEVEL TESTS FOR WORKS COSTING UPTO 100 LAKHS**: The contractor may set up a site laboratory at his option for works costing upto 100 lakhs. Other stipulations will be the same as specified in preceding para (a). However in case the contractor has not set up the site laboratory and tests are carried out in Zonal or any other laboratory approved/setup by GE, the recovery shall be made at rates applicable ie as given in Annexure –I.

(c) **'B' & 'C' LEVEL TESTS:** For tests of 'B' and 'C' level as indicated in Annexure-I, the contractor shall provide all facilities for testing of materials at Zonal laboratory/NABL Accredited laboratories /IITs at his own cost. The lump sum/rates quoted by the contractor shall be deemed to be inclusive of these tests. The rates of various tests conducted in the laboratory of MES are indicated in Annexure –I. The contractor shall bear the actual charges of 'C' level tests (to be done in labs other than MES labs) irrespective of rates indicated in Annexure-I. Wherever it is convenient to get 'B' level tests done at NABL Accredited laboratory, the same can be done at the cost of the contractor and no separate recoveries will be made by the Department for the same.

(d) The recoveries on account of testing charges wherever applicable shall be effected from the Running Account payments due to the contractor payable after completion of the respective tests or whenever the test is due whichever is earlier.

31.1 INSPECTION OF SITE BY THE TENDERER:

31.1.1 The tenderer is advised to contact the Garrison Engineer for the purpose of inspection of site(s) and relevant documents other than those sent herewith, who will give reasonable facilities for the purpose. The tenderer shall also make himself familiar with the working conditions, accessibility of site(s), availability of materials and other cogent conditions which may affect the entire completion of work under this contract. The tenderers shall be deemed to have inspected the site(s) and made themselves familiar with the working conditions, whether they have actually inspected the site(s) or not.

ANNEXURE -A- TO SPECIAL CONDITION (CONTD../-

MATERIALS AND THEIR TESTS

Sl No	Materials	Test	Method of testing	Frequency of tests			Level of test	Rate per test Rs Ps	Remarks
1	2	3	4	5			6	7	8
1	Bricks	[I] Compressive strength	IS-3495 [Part II]	As per IS-5454 as given under			'A'	180/-	Check for visual and dimensional characteristics shall also be carried out as per IS :5454
		[ii]Water absorption	-do-	Lot size	Sample Size	Permissible no of defective bricks	'A'	150/-	
		[iii] Efflorescence	-do- Part I	1001 to 10000	5	0	'A'	180/-	
				10001 to 35000	10	0			
				35001 to 50000	150	1			
2	Coarse aggregate	[I] Sieve analysis	IS-2386 [Part I]	One test for every 15 Cum of aggregate or part thereof brought to sit			'A'	120/-	
		[ii] Flakiness index	-do-	-do-			'A'	90/-	
		[iii] Estimation of deleterious materials	-do-	One test for every 100 Cum of aggregate or part thereof brought to site			'A'	120/-	
		(iv) Organic impurities	-do-	One test per source of supply			'C'	120/-	
		(v) Moisture content	-do- Part II	Regularly as required			'A'	120/-	
		(vi) Specific gravity	-do-	One test for each source of supply			'B'	120/-	
3.	Fine aggregate	(i) Sieve analysis	IS-2386 (Part I)	One test for every 15 Cum of aggregate or part when brought to site.			'A'	180/-	
		(ii) Test for clay, silt and Organic impurities	IS-2386 (Part I)	-do-			'A'	90/-	
		(iii) Specific gravity	-do- (Part ii)	One for each source of supply			'B'	180/-	
		(iv) Moisture content	-do- (Part II)	Regularly as required subject to 2 tests/day when being used			'A'	180/-	
		(v) Test for organic impurities	-do-	One test for each source of supply			'C'	180/-	

ANNEXURE -A- TO SPECIAL CONDITION (CONTD../-

MATERIALS AND THEIR TESTS

Sl No	Materials	Test	Method of testing	Frequency of tests		Level of test	Rate per test Rs Ps	Remarks
1	2	3	4	5		6	7	8
4.	Cement	(i) Setting time	IS:4031-63 Affirmed 1980	Once for each consignment or as and when required		'B'	180/-	
		(ii)Soundness				'B'	120/-	
		(iii)Compressive strength				'B'	360/-	
		(iv) Fineness				'B'	120/-	
5.	Structural concrete (M-15 grade and above)	(i) Slump test or compacting factor test or Vee-bee time	IS-1199	The minimum frequency of sampling of concrete of each grade shall be as under:-		'A'	180/-	Random sampling shall be carried to cover all mix units.
		(ii) Compressive strength	IS-516	Qty of concrete in the work (Cum)	No of samples	'A'	120/-	As per IS-456 Clause 15.2.2 for frequency of sampling.
				1-5	1			
				6-15	2			
				16-30	3			
				31-50	4			
				51 and above	4+1 for each addl 50 cum of part there of			
6.	[a] PCC block for walling [hollow block]	[i] Compressive strength	IS-2156-1984 [Appx 'B']	08 blocks out of 14		'A'	60/-	Sample: 14 blocks from consignment of every 5000 blocks or part thereof
		[ii] Water absorption	-do- [Appx 'B']	03 blocks out of 14		'B'	120/-	
		[iii] Density	-do- [Appx 'C']	03 blocks out of 14		'B'	90/-	

ANNEXURE -A- TO SPECIAL CONDITION (CONTD../-

MATERIALS AND THEIR TESTS

Sl No	Materials	Test	Method of testing	Frequency of tests	Level of test	Rate per test Rs Ps	Remarks
1	2	3	4	5	6	7	8
6.	[b] PCC solid block for walling	[i] Compressive strength	IS-2185	12 blocks out of 18	'A'	60/-	Sample: 18 blocks from consignment of every 1000 blocks or part thereof. These blocks to be checked for dimension and weight.
		(ii) Water absorption	-do-	03 blocks out of 18	'B'	120/-	
		[iii] Density	-do-	03 blocks out of 18	'B'	120/-	
7.	Cement flooring tiles/terrazzo tiles	[i] Water absorption	IS 1237 [Apppx'D'	06 tiles out of 18	'B'	180/-	Sample of 18 tiles from each source of supply selected at random
		[ii] Wet transverse strength	-do- [Appx'E']	-do-	'B'	144/-	
		[iii] Resistance to wear	-do- [Appex'F']	-do-	'C'	540/-	
8.	Burnt clay roofing tiles [machine made] As per IS-2690 [part I]	[i] Water absorption	IS-3495 [Part II]	06blocks out of 12	'B'	216/-	Sample: 12 tiles from each source of supply selected at random.
	Length 150 mm to 250 mm, width 100 mm to 200 mm, thickness 35 mm to 50 mm.	[ii] Compressive strength	-do- [Part I]	-do-	'A'	180/-	
9.	Mangalore pattern roofing tiles	[i] Water absorption	IS-654 [Appx 'A']	06 tiles out of 32	'B'	180/-	Sample: 32 tiles from each consignment of 3000 tiles or part thereof.
		[ii] Breaking load	-do- [Appx'C']	-do-	'B'	120/-	These tiles shall be checked for dimension.

ANNEXURE -A- TO SPECIAL CONDITION (CONTD../-

MATERIALS AND THEIR TESTS

SL No	Materials	Test	Method of testing	Frequency of tests	Level of test	Rate per test Rs Ps	Remarks
1	2	3	4	5	6	7	8
10.	Timber	[i] Specific gravity and weight	IS-1708-1960	Minimum '3' samples from a lot of 4 cum or 250 pieces of seasoned timber	'B'	120/-	
		[ii] Moisture content	-do-		'A'	120/-	
11	Water for construction purposes, flooring tiles/terrazzo tiles.	[i] Test for acidity	IS -456 & 3015	One at the stage of approval of source of water.	'B'	240/-	Also refer clause 6.4 OF IS-456 and its subsequent sub clauses regarding suitability of water.
		[ii] Test for alkalinity	-do-	-do-	'B'	240/-	
		[iii] Test for solid content	-do-	-do-	'C'	300/-	
12.	Welding of steel work	Visual inspection test	IS-822-1970 Clause 7.1	100% by visual inspection	Work site	360/-	Specialized tests, their method and frequency to be decided on consideration of their importance by the Accepting Officer.
13	Timber paneled & glazed door/window shutter [including factory made shutter]	[a] Dimensions, sizes, workmanship and finish	IS-1003-1977 [Pt I]	Frequency of sampling from each lot shall be as under:-	'A'	180/-	

Lot size	Sample size
26 to 50	5
51 to 100	8
101 to 150	13
151 to 300	20
301 to 500	32
501 to 1000	50
1001 and above	80

ANNEXURE -A- TO SPECIAL CONDITION (CONTD../-

MATERIALS AND THEIR TESTS

Sl No	Materials	Test	Method of testing	Frequency of tests	Level of test	Rate per test Rs Ps	Remarks
1	2	3	4	5	6	7	8
		(b) Strength test (i) Slamming	IS-1303-1990	From the each lot 5% of the factory made shutters shall be tested by manufacturers for strength			
		[ii] Impact identification	-do-				
		[iii] Shock resistance	-do-				
		[iv] Edge loading	-do-				
14.	Plywood [IS-303-1989]	Moisture content	IS-1734-1983 [Pt I]	Six test pieces cut from each of the boards selected as per table shall be subjected to test.	'C'	240/-	Sampling shall be as per IS-7835-1975 Tables.
15	Wood particles board[Medium density] IS-3097-1985	[a] Density	IS-2360 [Part III]	Three test specimen from each sample [size 150 mm x 75 mm]	'A'	60/-	Sampling shall be as per IS-3487-1983.
		[b] Moisture content	-do-	-do-	'A & B'	60/-	
		[c] Water absorption	-do-[Part 16]	-do- [Size 300mm x 300mm]	'A'	60/-	
		[d] Swelling due to surface absorption	-do-[Part 17]	-do- [Size 125mm x 100mm]	'A'	60/-	
		[e] Swelling in water	-do-	-do- [Size 200mm x 100mm]	'A'	60/-	
		[f] Modulus of rupture	-do- [Part 4]	Three test specimen as per IS-2380-77	'B'	90/-	
		[g] Screw withdrawal strength	-do- [Part 4]	-do- as per Is-2385	'C'	120/-	
16	Reinforcement Steel	(i)Physical test up to 16mm dia (normal mass, tensile elongation, bend and rebend)	AS per IS	AS per IS	'B'	2500/-	
		(ii) –do- more than 16mm dia	AS per IS	AS per IS	'B'	2750/-	

LEGEND : A-SITE LAB, B- ZONAL LAB, C-NATIONAL TEST HOUSE/SEMT WING/ NABL APPROVED LAB/ENGG COLLEGE.
NOTE : List of tests given above is not final. Other due tests required as per SSR/BIS to satisfy the quality requirement will also be got done by the GE and necessary expenditure for the same shall be borne by the contractor.

SIGNATURE OF CONTRACTOR
DATED:

AGE (CONTRACTS)
FOR ACCEPTING OFFICER

PARTICULAR SPECIFICATIONS

1. GENERAL

1.1 Work under this contract shall be carried out as detailed in different parts of schedule 'A' and in accordance with particular specifications, special conditions, General specifications/ provisions given in MES Standard Schedule of Rates 2009 Part I and 2020 Part II including amendments and errata thereto (hereinafter referred to SSR-2009 Part-I and SSR- 2020 Part-II).

1.2 Term 'General specifications' referred to in these documents as well as referred in IAFW–2249 (General Conditions of Contracts) shall mean the specifications contained in the MES Schedule.

1.3 General Rules, specifications, Special conditions and all preambles in the MES Standard Schedule of Rates shall be deemed to be applicable to the work under this contract, unless stated otherwise in these documents. In case of variance between provisions in these documents and those given in MES schedule, the provisions in these documents shall take precedence over the aforesaid provisions in the MES Schedule.

(a) Rates quoted by the tenderer shall be deemed to include for any minor details/items of work and/or construction, which may not specifically mentioned/ specified but are obviously and fairly intended and which are essential for the execution and entire completion of the work.

(b) Decision of the Accepting Officer as to whether any minor detail of work and/or construction is obviously and fairly intended to be included in the contract shall be final and binding.

1.4 Reference to IS code means the latest edition including amendments thereto upto and including the date of receipt of the tender even if mentioned otherwise.

2. SCOPE OF WORK:

2.1 This contract includes for the full, final and entire completion of following described in Schedule 'A' and particular specifications forming part of this contract.

(a)	**SPECIAL REPAIRS OF ROAD FROM PERIPHERY ROAD TO HRDF SITE AT INS BAAZ, CAMPBELL BAY** **JOB NO : ISH/11/2023-24 (SPL REP ROADS)**
(b)	**SPECIAL REPAIRS TO SECURITY FENCE AROUND BAAZ HOUSE AT INS BAAZ, CAMPBELL BAY** **JOB NO : ISH/01/2023-24 (SPL REP BUILDINGS)**
(c)	**SPECIAL REPAIRS TO PASSAGE OF AGE(E/M) OFFICE OF GE(I) CAMPBELL BAY UNDER GE(I) CAMPBELL BAY** **JOB NO : ISH/05/2023-24 (SPL REP BUILDINGS)**

2.2 Any damage during the execution shall be made good by the contractor at his own cost. The contractor is advised to make maximum precaution to carryout work without disturbing the occupants and to avoid any possible damage to the structure.

PARTICULAR SPECIFICATIONS

2.3 In case of any damage, the same shall be made good with materials and workmanship, which will match with the existing at Contractor's cost. Decision of Accepting Officer regarding who is responsible for damage, what extent the compensation has to be levied in case contractor does not rectify the damages etc is final and binding and the provision for such precautionary work shall be deemed to be considered in the rates quoted against the respective items in Schedule 'A'

2.4 to 2.5 BLANK

2.6 MES SCHEDULE

2.6.1 General rules, specification for materials and workmanship preambles and special conditions in MES schedule 2020 shall apply to the works under this contract, where the provisions in the particular specifications of the MES schedule, the particular specification shall take precedence. The term `General Specification' referred in the tender documents shall mean the specification contained in the MES Standard Schedule of Rates-2009.

2.7 ACCEPTANCE OF MATERIALS

2.7.1 The order of precedence for acceptance of materials shall be as under:-

(a) Materials with ISI markings.
(b) In case of non manufacture of ISI marked materials, materials conforming to relevant ISI.

(c) In case of non availability of materials at (a) & (b) above, the best available materials in the market as defined in clause 1.3 (b) of SSR 2009 Part-I and approved by the GE.

3. EARTH WORK AND EXCAVATION

3.1 SUFACE EXCAVATION

3.1.1 Before staring excavation work, the contractor shall carry out surface excavation in hard/dense soil not exceeding 30 cm deep and averaging depth as specified in the Schedule “A”. The site shall be dressed to slope away from the structure. Surface excavation shall be carried out before the excavation for foundation is started.

3.2 EXCAVATION-GENERAL

3.2.1 Excavation shall be carried out as specified in Clause3.12 & 3.13 of SSR 2009 Part-I and as described in Schedule ‘A’.

3.2.2 All water which may accumulate during the progress of excavation from springs, rain or other causes, shall be bailed/pumped or otherwise removed at no extra cost to Government.

3.2.3 Excavation shall be done without timbering and the rates quoted shall be deemed to be for excavation without timbering. Timbering if required shall be provided only on prior written order from GE and shall be paid separately through deviation order.

PARTICULAR SPECIFICATIONS

3.3 EXCAVATION IN TRENCHES AND OVER AREA

3.3.1 Excavation in trenches and over areas shall be carried out in any type of hard soil/soft disintegrated rock/hard rock. Bailing and pumping of water, if required, will be done as described in para 3.17 of MES Schedule Part I. No extra payment shall be admissible for the above mentioned operation. If however, an inflow of water into excavations is caused by powerful springs, tidal or river seepage, broken water mains or drains and the like (other than those broken through the contractor's negligence), the contractor shall immediately inform the GE. If in the opinion of the GE the inflow is due to any of these causes, he shall issue instructions, in writing as to the method to be employed in clearing the excavation in water and additional payment shall be allowed as a Deviation Order.

3.4 DISPOSAL OF EXCAVATED MATERIAL

3.4.1 These shall be carried out as specified in schedule 'A' and as directed by the Engineer-in-Charge and as specified in SSR Part-I Clause No3.20.1 & 3.20.2. Surplus spoil shall be removed to a distance not exceeding 50M.

3.5 RETURNING FILLING IN AND DISPOSAL OF EXCAVATED MATERIALS

3.5.1 These shall be carried out as specified in Schedule 'A' and as directed by the Engineer-in-Charge and as specified in SSR Part-I Clause No 3.20.1 & 3.20.2. Surplus spoil shall be removed to a distance not exceeding 50M.

3.6 FILLING IN TRENCHES/UNDER FLOORS

3.6.1 The approved soil obtained from excavation (except earth/ spoil obtained from surface excavation) shall be used for returning, filling in trenches, under floors or any other situation after removing big stones, grass, roots and vegetables and other organic matter. Earth mixed with small stones/pebbles (if approved by the GE) is permitted for use in filling around pipes after the pipes are laid and tested.

3.6.2 Filling under floors / sides of trenches shall be in layers not exceeding 250mm and each layer shall be watered, compacted and rammed as approved by Engineer-in-Charge.

3.7 EXCAVATION IN POST HOLES

3.7.1 Excavation in independent post holes (or similar holes) each not exceeding 0.5 cum shall be described as 'excavation as post holes' and shall include return, fill in, ram and removal of surplus spoil.

3.8 **ROUGH EXCAVATION** : Rough Excavation shall be as per MES SSR part-I Clause 3.3.1 .

3.9 **SURFACE DRESSING** : Surface Dressings shall be as per MES SSR part-I Clause 3.3.6 .

3.10 **HARD CORE.** Hard core shall be crushed or broken trap / basalt / granite stones of gauge not exceeding 63mm as specified in the Schedule "A". The finished consolidated thickness of hard core shall be 150mm unless otherwise specified. The crushed or broken stones shall be brought from the source(s) approved by the GE.

PARTICULAR SPECIFICATIONS

4. **CONCRETE:**

4.1. Contractor shall submit quality assurance plan for concrete works to the GE within one month of the acceptance of the tender for approval of GE. The GE shall approve the same in writing. GE and contractor shall ensure that every person involved in the concrete work shall establish and implement a quality Assurance plan. The responsibility and tasks of all persons involved in the work shall be defined. The following documents shall be maintained:

[a] Test Reports and Manufactures Certificate for Material.
[b] Concrete Mix Design Details.
[c] Pour Card for site organization indicating, location of concrete, type of concrete, water cement ratio, proposition of concrete, ingredient for day adopted, surface moisture content of aggregate, weather, temperature of concrete, cement consumed and test specimen detail etc.
[d] Non Conformance Report, Change Orders.
[e] Statistical Analysis.

4.2. **COARSE AGGREGATE:**

4.2.1. Coarse aggregate for all cement concrete works shall be graded machine crushed or broken granite stone / hard stone all as specified and conforming to specifications as given in clause 4.4.1 to 4.4.7 of MES Schedule Part – I and shall conform to the grading given in clause 4.4.7 [I] of MES Schedule Part – I, Stone for coarse aggregate shall be obtained from quarries approved by the Garrison Engineer. Mixture of two types shall however, not be used. Nominal sizes of graded stone aggregate in various situations shall be as indicated hereinafter. The exact source of coarse aggregate shall be approved by GE.

4.2.2. **GRADING OF COARSE AGGREGATE:** Coarse aggregate shall be graded conforming to IS – 383. Size and grading of aggregate for reinforced concrete shall be as specified in IS – 456 but in no case more than 20 mm graded aggregate. Graded Aggregate of nominal sizes given hereunder shall be used, unless specified otherwise, in the specifications hereinafter:

[a] Plain or reinforced cement concrete except in Foundation of Brick or PCC Solid Block Walls or Stone Walls / Pillars, Floors and Sub base to Floors.
[i] For structural elements of depth/thickness more than and including 80mm **:** 20mm
[ii] For structural elements of depth/thickness less than 80mm **:** 12.5mm

Note: However, in no case the nominal size of aggregate shall be greater than one–fourth the minimum thickness of the member.

[b] Plain Cement concrete in Foundation of Brick / PCC Solid Block Walls / Stone Walls, Pillars, Floors and Sub Base to Floors

[i] Under 30mm thickness : 12.5 mm
[ii] 30 to 80mm thickness : 20 mm
[iii] Exceeding 80mm thickness : 40 mm

PARTICULAR SPECIFICATIONS

4.3. **FINE AGGREGATE:**

4.3.1. Fine aggregate for concrete work shall conform to material specifications and grading within the limits of Grading Zone – II of IS – 383 except for finer finishes and all as specified in clause 4.4 of MES Schedule Part – I [Specifications]. Fine aggregate shall be approved quality of Natural River sand. Natural River sand shall be obtained from approved source / river beds from **Main Land of India Only**. The sand shall be washed with clean and fresh water if percentage of deleterious materials exceeds the limit of 5% specified in IS. Crushed sand is also permitted to be used in concrete work adhering to the laid down provisions in the relevant clauses of IS – 383. Crushed sand shall not be used in plastering or pointing works.

4.3.2. The sand shall be stored at site in dumps. The contractor shall take necessary precaution to avoid contamination or risk of shoveling of earth or other impurities by keeping sand over firm level ground as stipulated in IS – 4082.

4.3.3. **Note for Clause No. 4.2. & 4.3.:** Source of Coarse Aggregates and Fine Aggregate shall be as specified in Schedule "A" Notes.

4.4. **WATER:** Quality of water to be used for mixing and curing of concrete work shall conform to the requirement of clause No 5.4 of IS – 456 and as per Clause 4.9 of MES Schedule Part – I [Specifications]. GE and contractor will ensure that testing of water is carried out in any Government laboratory at three months intervals to comply with the requirement. Cost of testing of water will be borne by the contractor. The GE will monitor the quality of water through test results to ensure that the water used for making concrete or mortar and for curing confirm to IS – 456.

4.5 **CEMENT:** Cement will not be issued under Schedule B. Refer Appx to Particular Specifications in detail.

4.6 **PLAIN CEMENT CONCRETE**

4.6.1 All materials, workmanship, inspection, sampling and testing shall be as per the requirements given in IS :456-2000

4.6.2 The contractor shall provide all facilities and equipment for casting, curing and conveyance of test cubes of cement concrete to the laboratory approved by the GE. Test of concrete cubes during execution shall be carried out as laid down in IS:456 by the contractor in laboratory. Contractor shall supply free of cost concrete cubes for testing any number of times as demanded by GE. All incidental charges for taking out cubes, transporting to laboratory, testing charges etc., shall be borne by the contractor. The number of cubes made and tested shall be recorded to account the quantity of cement consumed.

4.6.3 **BATCHING, MIXING, DEPOSITING AND RAMMING**

4.6.3.1 Controlled concrete materials shall be batched by weight only. Combined batching with digital weighing system and mixing plant with auto cut off and computer print out facility shall be used for concreting. The capacity of mini batching plant provided at site shall be adequate enough to execute the work as per the CPM, or otherwise the contractor shall provide single batching plant of higher capacity or more number of batching plants of adequate capacities as required at site, as asked by the GE.

PARTICULAR SPECIFICATIONS

4.6.3.2 No claim what so ever arising out on this account is admissible. The decision of GE in this regard shall be final and binding. The plant/plants shall have the digital system of adding specified quantity of water into concrete mix as per the design mix requirement.

4.6.3.3 Water shall be measured either by volume in calibrated tanks or weighed. All measuring equipments shall be kept in a clean serviceable condition and their accuracy checked periodically.

4.6.3.4 Provisions as in clause 4.11.3.2 to 4.11.3.5 of MES Schedule Part-I shall be followed. All batching of concrete and accuracy of batching shall be as per Clause 10.2 of IS-456: 2000.

4.6.3.5 The mixing shall be done for at least 2 minutes and until a uniform colour and consistency is achieved.

4.6.3.6 Quantity of concrete mixed in any one batch shall not exceed the rated capacity of the mixer. The whole of the mixed batch shall be removed before materials for fresh batch enter the drum. Concrete mix as approved shall not be modified by addition of water or otherwise in order to facilitate handling for any other purpose.

4.6.3.7 On ceasing of work and other stoppage exceeding 20 minutes, the mixer and other plants used for handling wet mix shall be thoroughly washed with clean water. Pickup and throw over blades in the drum of the mixer which are worn down 20mm or more in depth shall be replaced with new blades. All cement concrete, both plain and reinforced shall be mixed in mechanical mixer as specified in para 4.11.5 and 4.11.5.1 of MES Schedule Part-I. However for small quantity of concreting (other than RCC works) i.e., the quantity of concrete required being less than one batch of mix, the contractor may after obtaining written approval of Engineer in charge which shall be exceptional, adopt hand mix subject to addition of 10% extra cement without price adjustment where hand mixing permitted, it shall be carried out on a concrete platform and care shall be taken to ensure that mixing is continued until the concrete is uniform in colour and consistency.

4.6.3.8 All cement concrete both plain and reinforced concrete, shall be deposited and compacted all as specified in Clause 4.11.10 and 4.11.11 of MES Schedule Part-I. However, RCC work in columns, foundation, beams walls, chajjas and slabs etc., shall be compacted using mechanical vibrator, compaction of lean concrete shall be carried out by ramming and consolidated by tamping and rodding as specified. In the event of break down of mechanical mixer and vibrator, the contractor must have arrangements for standby mechanical mixer and vibrator.

4.7 TO 4.8.2 BLANK

4.8.3. **DESIGN MIX CONCRETE [CONTROLLED CONCRETE]:**

4.8.3.1. Grade of design mix concrete shall be as specified here-in-before and shall be as referred to in IS – 456 and as specified here-in-after. Design mix concrete may also be referred to as controlled concrete. Mix design shall be done as per IS – 10262 [Recommended Guidelines for Design Mix Concrete] and as described in SP – 23 [An IS Publication].

PARTICULAR SPECIFICATIONS

4.8.3.2. The requirement of cement per cubic meter of controlled concrete of **Grade M30** shall be as per IS - 456. The actual requirement of cement for the controlled concrete shall be ascertained by the tests as specified here-in-after. The design mix shall be carried out for SEVERE environment conditions and good quality control. The tenderer shall ascertain the quantity of cement required and quote accordingly. No claim whatsoever arising on account of quantity of actual cement incorporated in the work on account of design mix is admissible.

4.8.3.2.1 [a] Contractor shall use liquid admixtures [Super plasticizers] to achieve the work-ability and to reduce the water content in design mix. Admixtures shall conform to IS – 9103 and shall be from approved manufacturers as given below:

[a] M/s. Fosroc Chemicals [India] Pvt. Ltd
[b] M/s. Sika India Pvt. Ltd.
[c] M/s. Choksey Chemicals Pvt. Ltd.
[d] M/s. STP Ltd. / M/s. Bostik India Pvt. Ltd.
[e] M/s. Thermax Ltd.
[f] M/s. Hindcon Chemical Ltd.
[g] M/s. Asian Paints Limited

[b] Para 5.5 of IS – 456 be also referred for quality of admixtures.
[c] For maximum dose of admixtures, please refer Para 10.3.3 of IS – 456.
[d] Various tests as specified in IS – 9103 shall be carried out for each batch of Admixtures at contractor's cost.
[e] Contractor shall submit original purchase voucher and test certificate of manufacturer for complete quantity of admixtures used in the work before claiming payment for the same.
[f] Complete quantity of admixtures including name of manufacturer, its brand name, date of manufacturing, date of expiry, voucher No. and details of test certificates shall be entered in MB as "Not to be Abstracted" duly signed by JE, Engineer-in-Charge, GE and representative of contractor before making payment in RAR.

4.8.3.2. As soon as possible after receiving the work order to commence the work, the contractor shall submit samples of the materials required for preparing design mix concrete viz. cement, coarse aggregate, fine aggregate and admixtures for approval of GE and intimate the place out of the following where they propose to carryout the Design Mix and preliminary tests of Design Mix concrete:

[i] National Test House
[ii] SEMT Wing
[iii] Government Approved Laboratories.
[iv] Regional Research Laboratory
[v] IIT / National Institute of Technology
[vi] NABL Accredited Laboratory
[vii] Command Testing Lab

4.8.3.3. As soon as possible after receiving the work order to commence the work, the contractor shall submit samples of the materials required for preparing design mix concrete viz. cement, coarse aggregate, fine aggregate and admixtures for approval of GE and intimate the place out of the National Test House / SEMT Wing / Government Approved Laboratories / NABL Accredited Laboratories / Regional Research Laboratories / IIT / National Institute of Technology / Command Testing Lab where they propose to carryout the Design Mix and preliminary tests of Design Mix concrete.

PARTICULAR SPECIFICATIONS

4.8.3.4. The cement used in the work shall be as specified here in before. Coarse aggregate shall be crushed stone aggregate. The gradation shall be followed as per clause 4.2 table 2 of IS – 383 to obtain maximum density.

4.8.3.5. After the samples of all the materials are approved by GE in writing, sufficient quantities of these materials shall be forwarded by GE at contractor's expense for carrying out design mixes.

4.8.4. **PRELIMINARY TESTS:**

4.8.4.1. Preliminary tests are tests conducted on the trial mixes of concrete produced in the laboratory with the object of:

[i] Designing concrete mixes before the actual concreting operation starts.

[ii] Determining the adjustments required in the design, when there is change in the materials used during execution of work.

[iii] Verifying the strength of concrete mix at 28 day

4.8.4.2. The preliminary tests shall consist of 3 separate sets of tests covering possible variation of gradation of aggregates and each set of test using a minimum 9 Cubes of size 150 mm x 150 mm x 150 mm and one slump test. Three cubes shall be tested at 7 days to get indication of minimum strength of 28 days. Other 3 Cubes shall be tested at 28 days and 3 Cubes shall be preserved for Government use for subsequent testing. The compressive strength tests of cubes shall be performed as per IS – 516. Casting of cubes and testing of these cubes shall be carried out in the presence of contractor's representative, GE / GE's representative and representative of Accepting Officer. It will be contractor's responsibility to ensure that design mix is carried out at the earliest. Contractor shall ensure that design mix calculations, supporting trail mix [03 Nos] details and test results of trial mixes along with recommended trial mix are submitted to GE at the earliest for his further action.

4.8.4.3 Based on test results, the GE shall approve the design mix in writing. Copy of approved design mix shall be submitted to Accepting Officer within 10 days of approval by GE. The testing charges for the design mix and the tests conducted shall be borne by the contractor. The cost of materials, labour and transport shall also be borne by the contractor.

4.8.5. **WORK TEST:**

4.8.5.1. The work tests shall be carried out at Site Lab / Command Testing Lab situated in the premises of Chief Engineer [A & N] Zone, Port Blair.

4.8.5.2. Work test shall be conducted as per Clause 15 of IS – 456. At the commencement of the concreting, samples of concrete shall be taken on each day as specified in Clause 15 of IS – 456 and specimens made at the work site out of the concrete being used in the works, for the purpose of testing compressive strength.

4.8.5.3. From each of these samples, 9 Test Cubes of size 150 X 150 X 150 mm shall be taken to test 3 Specimens at 7 days and 3 Specimens at 28 days in Command Testing Lab. C. A. No., date of casting and location where concrete is being used shall be marked on each concrete cube. 3 Test Cubes of preliminary and work test shall be preserved duly marking the date of casting and C.A. No. for verification / subsequent testing, if required. The cube shall be preserved by the GE / Engineer-in-Charge until the defects liability period of the work is over.

PARTICULAR SPECIFICATIONS

4.8.5.4. The testing charges for the work tests conducted in the Command Testing Lab shall be at the rate mentioned here-in-after in Particular specifications and the same shall be effected from the payments due to the contractor in RAR / Final bill whichever is earlier. The cost of materials, labour and transport shall be borne by the contractor. The rate quoted shall include the cost of testing the concrete cubes both for Design Mix / Volumetric Mix.

4.8.5.5. In the event of contractor setting up the laboratory at site as specified here in before in Special Conditions, the contractor shall carry out cube testing in site lab, in presence of Engineer-in-Charge and as specified here in before. However, random testing up to 5 percent of total tests to check the compressive strength of cube shall be carried out in Command Testing Lab for which testing charges shall be recovered from the contractor at the rate mentioned here-in-after. Contractor shall include this aspect while quoting his rates.

4.8.5.6. The Engineer-in-Charge shall maintain the record for all the tests carried out in Site Lab / Command Testing Lab separately. The cost of testing including material, labour etc., incurred shall be borne by the contractor and the rate quoted shall be deemed to include this.

4.8.6. **MIXING:**

4.8.6.1. The mix design and also execution of work shall be carried out by weigh batching. The quantum of cement for execution of work by weigh batching shall be as per mix design.

4.8.6.2. It shall be ensured that the grading characteristics as adopted in the mix design are followed throughout. Wherever the type and/or batch of cement/aggregate is changed, a fresh mix design shall be carried out. Nothing extra is payable on this account.

4.8.6.3. The contractor during the progress of work shall not change the mix design without the prior approval of the GE.

4.8.6.4. Engineer-in-Charge shall maintain a record of actual consumption of cement in proper register [other than the cement register mentioned in special conditions] and shall initial the entries of quantity of materials issued to contractor every day. The register shall be got checked and signed by GE. In case the consumption of cement as per cement consumption register is found to be more than the estimated quantity of cement due to whatsoever reason, the contractor shall not have any claim, whatsoever for such excess consumption of cement.

4.8.7. **WATER CEMENT RATIO:**

4.8.7.1. It is most important to maintain the water cement ratio constant and to it's correct value. To this effect, determination of moisture content in both fine and coarse aggregate should be made as frequently as possible.

4.8.7.2. The frequency for a given job shall be determined by the Engineer-in-Charge. According to weather conditions, the amount of water to be added shall be adjusted to compensate any variations in the aggregate, IS – 2390 [Part – III] method of test for aggregate, for concrete, Part – II specific gravity, density, voids, absorption and bulking of aggregates due to variation in their moisture contents shall apply. The maximum quantity of water to be added shall be determined by mix design to be carried out as specified here-in-before. Workability of concrete shall be checked at frequent intervals. The slump test or where facilities exists the compacting factor test in accordance with IS – 1199 may be adopted for this purpose.

PARTICULAR SPECIFICATIONS

4.8.7.3. The slump for M30 Grade Concrete [except for piles] 50mm to 100mm for medium degree workability as given in Clause – 7 of IS – 456.

4.8.7.4. Curing shall be carried out all as specified in MES Schedule Part – I.

4.8.8. **ACCEPTANCE CRITERIA:** As per Clause 16 of IS – 456.

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6. **STONE MASONRY**

6.1 **STONES:**

6.1.1 Stones shall be of granite and shall be obtained from the approved quarries. Stone shall be hard, sound, durable and free from weathering, decay and defects like cavities, crack flaw, sand holes injurious veins, patches of loose or soft materials and other similar defects that may adversely affect its strength and appearance.

6.1.2 As far as possible, stones shall be of uniform colour and texture. Generally stone shall not contain crypto crystalline silica or chert, mica and other deleterious materials like iron oxide, organic impurities, etc. The compressive strength and water absorption of common types of stones should be as indicated in clause 6.2.3 of MES Schedule part-I. Test for compressive strength shall be carried out as laid down in IS: 111 Part-I, 1974. Test for water absorption shall be carried out as laid down in IS: 1124 1974.

6.2 **LATERITE STONE MASONRY**

6.2.1 Stone shall be of granite obtained from the approved quarries and shall be sound hard, durable and free from weathering, decay and defect that may adversely affect its strength and appearance all as specified in clause 6.2 of SSR Part I. The stone masonry shall be random rubble masonry uncoursed of brought to courses.

6.2.3 The masonry shall be built in CM 1:6 using stones after dressing of all side / faces as per local practice. The workmanship shall be as specified in Clause No 6 .11 of MES Sch Part-I.

6.3 **HAMMER DRESSING**

6.3.1 Hammer dressing shall be carried out as specified in clause 6.9 of SSR Part I, 2009 and shall be as specified in Sch ‘A’.

6.4 **BOND/ QUOINS STONE**

6.4.1 Bond/ Quoin Stones shall be provided as specified in MES Schedule. The contractor may, however, at his option provide cast-in-situ or Precast PCC (1:3:6) Type C-1 block in one piece in lieu of stone bond/ quion stones. The size and spacing of PCC bond stones shall be as specified for stone bond stone and shall be laid on the full section of the walling in one piece.

6.4.2 Unless otherwise specified all stone masonry shall be built in cement mortar (1:6).

PARTICULAR SPECIFICATIONS

6.5 KEYED POINTING

6.5.1 Keyed or ruled pointing shall be done by first finishing the joints as for flush pointing and then ruling off the horizontal and vertical joints with a round edged tool to form narrow semicircular grooves in proper alignment and all as per SSR Part I relevant clauses.

6.6 PRECAST PCC SOLID BLOCK MASONRY.

6.6.1 All masonry work shall be with the precast PCC solid block of grade (C-5) having block density of not less than 1800 Kg/Cum. The concrete shall be mixed as laid down in IS-2185 (Part-I).

6.6.2 AGGREGATES

(a) Coarse aggregate shall be crushed or natural aggregate 20mm and below size, conforming to IS-383.

(b) Fine aggregates (Sand/Crushed rock fines) shall be free from dust and well graded having fineness modulus 3.5 to 4 of combined aggregate conforming to IS-383.

6.6.3 The size of blocks shall be one of the following:-

Size designation	Nominal Length in CM	Breadth in CM	Height in CM	Actual Length in CM	Breadth in CM	Height in CM
A	40	30	20	39	30	19
B	40	20	20	39	20	19
C	40	10	20	39	10	19
D	20	20	20	19	20	19
E	20	20	10	19	10	19

Sizes other than these specified above may also be used with the approval of Engineer-in-Charge.

6.6.4 MANUFACTURE OF BLOCKS

6.6.4.1 The blocks shall be machined made. The process of concrete mixing, fabricating/ casting of blocks, curing and drying shall be in accordance with IS-2185 and all as specified and in MES Schedule. Faces of the blocks shall be flat and rectangular.

6.6.5 PHYSICAL REQUIREMENT

(a) **GENERAL :** All blocks shall be sound and free of cracks or other defects which interfere with proper placing of the block or impair the strength or performance of the construction. Minor chopping resulting from the customary methods of handling during delivery shall not be deemed ground for rejection.

(b) **TOLERANCE :** The maximum variation in the dimension shall be not more than ± 3.0 mm in height and breadth and ± 5mm for length.

(c) **BLOCK DENSITY :** The block density shall be as per IS-2185, (Part-I, 2005) for solid PCC block. It will not be less than 1800 Kg/cum.

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(d) **COMPRESSIVE STRENGTH :**The average compressive strength of eight blocks when determined in the manner described in IS-2185 shall not less than 5.0 N/Sq mm of the gross area. The strength of lowest individual block shall not less than 80 percent of the average compressive strength of eight (8) blocks.

(e) **WATER ABSORPTION:** The water absorption shall be as per IS-2185.

(f) **DRYING SHRINKAGE :**The drying shrinkage on the blocks shall be as per IS-2185.

(g) **MOISTURE MOVEMENT :** The moisture movement shall be as per IS-2185.

6.6.6 **TESTS** :Tests as described in ANNEX 'B' to 'G' of IS-2185 shall be conducted on samples of Blocks selected according to the sampling procedure as per Para 11 of IS-2185 to ensure conformity with the physical requirement laid in IS-2185.

6.6.7 **SAMPLING & CRITERIAL FOR CONFORMITY**

6.6.7.1 The blocks required for carrying out the tests laid down in the standard shall be taken by one of the methods given in Para 10 of IS-2185 and shall be considered as Conforming to the requirements of the specifications if the conditions mentioned in IS-2185 are satisfied,

6.6.8 **MORTAR FOR WALLING** : 20cm thick PCC solid block shall be in cement mortar 1:6 and 10cm thick PCC solid Block walling shall be built in cement mortar 1:4.

6.6.9 **LAYING** : The blocks shall be slightly wetted before and / or during laying in the wall. The blocks shall be laid with joints complete filled without any void left in the masonry. The thickness of the horizontal and vertical joints shall not exceed 1cm. The ½, 1/3 and 2/3, block shall be used for breaking the joints, The face joints shall be raked to a depth of 1cm by raking tool during the progress of the work when the mortar is still green, so as to provide proper key for plaster or to facilitate pointing to be done later. Where plaster or pointing is not required, the joints shall be struck flush and finished side by side.

6.6.10 **CURING OF WALLING :**Masonry work shall be kept constantly moist on all the faces for a minimum period of 7 days.

6.6.11 **SCAFFOLDING FOR WALLING :** Only double scaffolding shall be used. The scaffolding shall be strong and sound. No holes in the masonry for supporting scaffolding shall be allowed.

6.6.12 All 10cm thick partition walls on ground floor shall rest on sub base of the floor and on first floor shall rest on RCC slab, 10cm thick partition walls/columns shall be properly bonded at ends into adjoining walls/columns. Irrespective of whether shown in drawings or not, 10cm thick partition walls shall be provided with the following:-

(i) 2 No 6mm dia MS round bars at every fourth course or 2 No 8mm dia TMT bars at every sixth course.

6.6.13 **CURING OR WATERING** – Masonry work shall be kept constantly moist for a minimum period of 7 days.

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9. **STEEL AND IRON WORK**

9.1 **GENERAL:** All steel required for the work under the contract shall be procured, supplied and incorporated in the works by the contractor under his own arrangement

9.2 **GRADE AND QUALITY:** Steel supplied by the contractor shall conform to the following grades and quality:

(a) **REINFORCEMENT STEEL**

(i) High strength deformed steel bars produced by Thermo Mechanical Treatment Process (TMT Steel bars of grades Fe 500/ Fe 500D and Fe 550/ Fe 550 D) meeting all other requirements of IS : 1786

(b) **STRUCTURAL STEEL** (Refer clause 10.4 of MES Schedule Part I (2009)

(i) Standard Quality- Conforming to IS : 2062

(ii) Ordinary Quality- Conforming to IS : 1977

(i) Structural Steel of Standard Quality shall conform to IS-2062 and Grade Fe-410-S. This type of steel shall be provided for all structural steel works in the locations specified and in clause 10.4.1 of MES Schedule Part I.

(ii) Structural Steel "Ordinary Quality" shall be used in frames of doors and windows, guard bars, grills and like and shall conform to IS-1977 Grade Fe-310-0. This type of steel shall be provided in the locations specified and in clause 10.4.3 of MES Schedule Part- I.

(c) Steel tubes for structural purposes shall conform to IS-1161 and shall be of Grade YST-240.

(d) **GALVANISED STEEL SHEETS (PLAIN & CORRUGATED)** - Conforming to IS:277

(e) **FABRIC REINFORCEMENT FOR CONCRETE** – Conforming to IS :1566

9.3 **MATERIALS:**

(i) The Galvanised Iron sheets & fabric reinforcement for concrete to be supplied by the contractors, shall be ISI marked and shall be procured from main manufacturers/ authorized dealer of main producer in case of small quantity for GE works.

(ii) The structural steel supplied by the contractor will be procured from main producers like SAIL (Steel Authority of India Limited), RINL (Rashtriya Ispat Nigam Limited, also known as Vizag Steel), Jindal Steels and Powers Ltd (JINDAL),TISCO (Tate Iron and Steel Company Limited) or secondary producers who manufacture structural steel out of ISI marked billets and are having BIS certification of ISI marking on their products and approved by HQ E-in-C's Branch, New Delhi.

(iii) TMT steel supplied by the contractor will be procured form main producers of steel as approved by HQ E-in-C's Branch, New Delhi

9.4 **SOURCE OF PROCUREMENT –TMT STEEL BARS**

(i) TMT steel bars of all sizes supplied by the contractor shall be procured directly from any one of the following main producers of steel/ authorized dealer of main producer in case of small quantity for GE works.

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Sl No	Company Name & Brand	Address	Type of Steel
1.	Rashtriya Ispat Nigam Limited (RINL) Brand : RINL	Visakhapatnam steel plant, Visakhapatnam- 530 031(India) Tel : (91 891) 518226, 518376 Fax : (91891) 518316 E:Mail- cmdvsp@itpvis.ap.nic.in	All
2.	Tata Iron & Steel Company (TISCO or Tata Steel) Brand : TATA	Bombay House ,2 ,4 Homi Modi Street, Mumbai – 400 001 (India) Tel: (91 22) 204 9131 Fax: (91 22) 204 9522, 287 0840 E-mail: corpcomm@jsr.tatasteel.com (Br Office for North : Jeevan Tara Bldg, Patel Chowk, New Delhi)	All
3.	Steel Authority of India Ltd (SAIL) Brand: SAIL	Central Marketing Organization, Northern Region, 17th floor, Scope Minar, Laxmi Nagar Distt centre, Delhi - 110092	All
4.	M/s Rashmi Metaliks Limited, WB Brand : RASHMI TMT	Premlata Building 39, Shakespeare Sarani 6th Floor, Kolkata – 700 017 Tele :033-22894255/56 Fax:033-22894254 E-mail mkt.domesticdip@rashmigroup.com Website: www.rashmigroup.com	TMT Bars of grade Fe 500 (size 8-32mm), Fe 500D & Fe 550D (size 8-25mm)
5.	M/s Incredible Industries Limited (Formerly M/s Adhunik Industries Limited) Brand : ADHUNIK Fe 500 SD	Lansdowne towers, 2/1 A, Sarat Bose Road, Kolkata-700 020 Tel: 033 66384700 Fax: 91 33 2289 0285 Email: www.adhunikindutries.com	(TMT Bars of grade Fe 500D (Size 8-32mm)
6.	M/s Real Ispat & Power Ltd, Chhattishgarh Brand : GK TMT	M/s Real Ispat & Power Limited (CG) Vrindavan, Near IDBI Bank, Civil Lines, Raipur -492 001, Chhattishgarh,Tele : 0771-4224000, Fax: 0771-4224010 E-Mail : real@realispat.com, www.realispat.com	(TMT Bars of grade Fe 500D)
7.	M/s Electrotherm (India) Ltd Brand : ET TMT	Survey No.72, Palodia, (Via-Thaltej, Ahmedabad), Gujarat-382115 Tele : +91-2717-234553/660550 Fax: +91-2717-234866 Website: www.electrotherm.com	(TMT Bars of grade Fe 500, Fe 500D &CRS(Size 8-36mm)
8.	M/s Shyam Metalics & Energy Ltd Brand : SEL	Viswakarma, 1st Floor, 86 C, Topsia Road, Kolkata-700 046 Tel : +91 33 2285 2212 Website: www.shyamgroup.com	(TMT Bars of grade Fe 500 & Fe 500D (Size 8-32mm)

PARTICULAR SPECIFICATIONS

Sl No	Company Name & Brand	Address	Type of Steel
9.	M/s Bajrang Power & Ispat Ltd, Raipur Brand : GOEL TMT	M/s Bajrang Power & Ispat Ltd, Raipur Vill -Borjhara, Urla Industrial Area Raipur – 493 221 (CG) Tele -0771-4288019/29/39 Fax : 0771-2323601/602, 4288123 E-mail : infobir@giekgriyo.co.in Commercial.bir@goelgroup.co.in	(TMT Bars of grade Fe 500 & Fe 500D)(Size 8-32mm)
10.	M/s Gallantt Ispat Ltd, Gujarat Brand : GALLANTT TMX	Corp Office :Ward 10BC, Plot No 123,Ground floor, Gandhi Dham, Kutch, Gujarat – 370201 Tel : + 91-2836-228164 Fax : + 91-2836-235787 E-mail: gml@gallantt.com Website: www.gallantt.com	(TMT Bars of grade Fe 500 ,Fe 500D and CRS (Size 8-32mm)
11.	M/s SPS Steel Rolling Mills Ltd (WB) Brand : ELEGANT TMT	M/s SPS Steel Rolling Mills Ltd (WB) Regd Office : 'Diamond Heritage' , 16 Strand Road, Room No H523 A, 5th Floor, Kolkata – 700001 Corporate Office : 'Diamond Heritage', 41 A, A.J.C Bose Road, 8th Floor, Room No 801, Kolkata – 700017, Mob :9831055568/ 9830281273 E-mail : info@spsgroup.co.in	(TMT Bars of grade Fe 500D)
12.	M/s Steel Exchange India Ltd (AP) Brand : SIMHADRI TMT	M/s Steel Exchange India Ltd (AP) D.No 1-65/K/60, Plot No 60, Abhis Hiranya, 1st Floor, Kavuri Hills, Hyderabad, Telangana-500 081 Tel :040-23403725/23413267 E-Mail : info@seil.co.in	(TMT Bars of grade Fe 500D & Fe 500D (HSCRM)
13.	M/s Super Smelters Limited, Kolkata Brand : SUPER SHAKTI	M/s Super Smelters Limited, Kolkata Premlata, 39, Shakespeare Sarani 2nd Floor, Kolkata -700 017 Tele/ Fax :91-33-2289-2734/36 E-mail : info@supershakti.in Website : www.supershakti.in	(TMT Bars of grade Fe 500D & Fe 550 (Size 8-32mm)
14.	M/s Shyam Steel Industries Ltd (WB) Brand : SHYAM	M/s Shyam Steel Industries Ltd (WB) Shyam Towers, EN-32, Sector-V, Salt Lake, Kolkata – 700 091 (WB) Tele : 033-40074007 Fax :033-40074010 E-mail : marketing@shyamsteel.com Website : www.shyamsteel.com	(TMT Bars of grade Fe 500D & CRS
15.	M/s Jai Balaji Industries Ltd (WB) Brand : BALAJI SHAKTI	M/s Jai Balaji Industries Ltd (WB) 5, Bentinck Street, 1st Floor, Kolkata-700 001 Tele : 033-22489808/22488173 Fax :033-22430021/22107893/ 22426263 E-mail : info@jaibalajigroup.com	(TMT Bars of grade Fe 500D Fe- 500D CRS) (Size 8-32mm)

Sl No	Company Name & Brand	Address	Type of Steel
16.	M/s MSP Steel & Power Limited, Chattisgarh Brand : MSP TMT 500D	M/s MSP Steel & Power Limited, Chattisgarh 16/S, Block-A, New Alipore, Kolkata-700 053 Tele :033 4005 7777/2398 2239 Mob : 7381028976 E-mail : contactus@mspsteel.com Website : www.mspsteel.com	(TMT Bars of grade Fe 500D) (Size 8-32mm)

(ii) The steel procured from main manufacturer, secondary manufacturers shall conform to the requirements as stated in clause 9.2 above. The steel items supplied by the producer/manufacturer, through their authorised conversion agents or authorized dealer/agent shall also conform to the requirements as stated in Clause 9.2 and producer shall stand guarantee for quality of steel and shall countersign the test certificates issued by conversion Agent.

(iii) The contractor shall produce original vouchers from main manufacturer's / secondary producers for the total quantity of steel supplied, under each consignment and to be incorporated in the work. The original vouchers shall be kept on record of the GE duly defaced by Engineer-in-Charge and authenticated.

(iv) **THERMEX/ TEMPCORE/ EVCON TURBO/ SIEMENS'S HYQST/ TMT** : Process of manufacturing TMT steel is being installed by some more manufacturers. The cases for their approval will be progressed in due course as and when the installation of Thermex, Tempcore, EVCON Turbo or SIEMENS's HYQST/ TMT process of manufacturing TMT steel is completely installed.

9.5 **SOURCE OF PROCUREMENT –STRUCTURAL STEEL**

(a) **STRUCTURAL STEEL:** The contractor shall procure structural steel sections directly from main producers like SAIL/RINL/TISCO/JINDAL or approved primary producers viz M/s Jindal Steel & Power Ltd, Gurgaon. In case of non-availability of structural steel with any of the main/primary producers the same can be procured from approved secondary producers as listed below with necessary minus price adjustment with a reduction of 5% (Five Percent) of the accepted rate of structural steel. In case desired section of structural steel is not rolled / manufactured by primary producers, there shall be NO price adjustment in use of structural steel procured from approved secondary producers. Galvanised sheets and Fabric Reinforcement for concrete shall be procured directly from Main producers and shall be ISI marked.

LIST OF APPROVED SECONDARY PRODUCERS FOR PROCUREMENT OF STRUCTURAL STEEL

(i) Shree Parashnath Rolling Mills Ltd, 3511 Part, Dr. BC Roy Avenue, Durgapur, West Bengal - 713201

(ii) M/s Amba Shakti Ispat Ltd , Plot No. 6 Phase II Industrial Area , Kala Amp, Dist Sirmour, HP – 173030

(iii) M/s Karam Steel Corporation, Nasrail Road, PO Box No. 56, Mandi, Gobinandgargh, Punjab - 147301.

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(iv) M/s K L Steel Pvt Ltd, PO Box No. 61,Lal Kuah buland Shahar Road, Ghaziabad-(UP)

(v) M/s Pushpak Steel Industries Pvt Ltd, Gate No. 119, Alandi Market Road, Dhabore, Tah Khed, Pune

(b) However, non structural steel as for guard bars, hold fast, railing and the structural steel where total requirement under the contract is less than 2.00 tonnes may be procured locally after obtaining permission of the Accepting Officer in writing.

(c) The contractor shall produce original vouchers from suppliers for the total quantity of steel supplied, under each consignment and to be incorporated in the work.The original vouchers shall be kept on record of the GE duly defaced by Engineer-in-Charge and authenticated.

(d) All finished steel shall be well and clearly rolled to the dimensions, section and weights specified. The finished material shall be reasonably free from cracks, surface flaws, laminations, rough jagged and imperfect edges and any other harmful defects and shall be finished in a proper manner. Tolerance on size and weight of reinforcement bars shall not be more than as specified in clause10.17.4 and 10.17.5 of SSR Part-I and as specified in IS-1786 and IS- 2062 and as per relevant IS codes.

(e) Contractor will give to GE manufacturer's test certificate (IN ORIGINAL) along with the test sheet giving result of each mechanical test and the chemical composition of steel (as per IS-1786) for reinforcement steel or authenticated copy thereof duly singed by manufacturer with each consignment. The documents such as original purchase vouchers and test certificates in support of the purchases of steel shall be produced by the contractor to the site staff & GE for verification and record.

9.6 **APPROVAL OF STEEL BROUGHT BY THE CONTRACTOR**

9.6.1 Following action shall be taken by GE before incorporating steel procured by the contractor for the work:-

(a) Physical verification of steel received to confirm the actual quantity of steel as well as to verify aspects brought out in foregoing.

(b) GE will obtain original purchase vouchers of manufacturer from contractor.

(c) No consignment or part thereof will be allowed to be incorporated in the work until and unless the test results of independent testing are obtained and the consignment is passed by GE. Schedule of procurement will be prepared keeping in view the time lost for testing etc.

9.7 **TESTING OF STEEL**

(a) The manufacturers of steel are to carry out inspection and testing of steel in accordance with the relevant BIS provisions.

(b) The contractor shall submit manufacturer's test certificate in original or authenticated attested true copy by the manufacturers only along with the test sheet giving the result of each mechanical test as applicable in accordance with relevant IS provision and the chemical composition of the steel or authenticated copy with each consignment.

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(c) The Engineer-in-Charge shall record these details in a steel acceptance register which will be signed by the Junior Engineer, Engineer-in-Charge, GE and Contractor after due verification and Engineer-in-Charge shall send a certified true copy of test sheet to GE for his records.

(d) Independent testing of structural steel by the GE shall be optional at the discretion of the GE in case of procurement of steel from main producers and testing charges shall be borne in accordance with Condition 10A of IAFW 2249 i.e. testing charges shall be borne by the Department if the test results are found in order otherwise these shall be borne by the Contractor.

(e) Independent testing of structural steel by the GE shall be mandatory in case of procurement of steel from secondary producers and testing charges shall be borne by the Contractor irrespective of the outcome of test results.

(f) For independent testing ,random samples of steel drawn from various lots and shall be got tested from a National Test House, SEMT CME, Regional Research Labs, CTL Chennai, IITs, NITs, any Govt Engg Colleges, Government approved Labs, NABL approved labs, Zonal Labs, etc as per the minimum frequency given below. Samples from each lot shall be also tested for quality and elongation. Samples from each lot should be tested for quality and elongation. The elongation shall not be less than 18 %.

(g) In all cases mentioned above contractor at his cost shall provide all facilities required for the testing. Cost of samples, transportation and testing shall be borne by the Contractor. The records of such checks would be maintained in the steel test register.

(h) Ultimate tensile strength elongation, bend and re bend test for reinforcement steel bars shall be carried out as per clause 9 and test specimen shall be as per clause 11 and delivery inspection shall be as per clause 12 of IS-1786. Bend tests and tensile tests for structural steel shall be carried out as per IS-2062 & recorded.

9.7 **FREQUENCY OF SAMPLING FOR INDEPENDENT TESTING BY GE**

9.7.1 Frequency for nominal mass, tensile strength, bend and re-bend tests of steel for checking nominal mass, tensile strength, bend, re-bend test, test specimen at random shall be selected by the GE at following frequency :-

(a) **STEEL FOR CONCRETE**

(i)	Bars size less than 10mm	Sample (3 specimens) for each test for every 25 tonnes or part thereof.
(ii)	Bars size 10mm to 16mm	Sample (3 specimens) for each test for every 35 tonnes or part thereof.
(iii)	Bars size over 16mm	Sample (3 specimens) for each test for every 45 tonnes or part thereof.

(b) **STRUCTURAL STEEL**

(i)	Tensile Test	Test for every 25 tonnes of steel or part thereof
(ii)	Bend Test	Test for every 25 tonnes of steel or part thereof

9.7.2 The testing by GE as per above frequency is mandatory before payment is released to the contractor in case of structural steel from secondary producers. The GE may also increase the frequency and number of samples / tests for his satisfaction. The cost of these additional tests shall be governed as per condition 10 (A) of IAFW-2249. However cost of samples, transportation and other overheads shall be borne by the contractor irrespective of test results.

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9.7.3 Test shall not be insisted upon for the steel required for guard bars, holdfasts, grills and such other allied items.

9.7.4 In case test results of testing pursuant to clause 9.7.1 are not within the acceptable limits, then that consignment of steel shall stand rejected and contractor shall remove the same from site at his own cost. The rejected material shall not be incorporated in the work. The contractor shall have no claim on this account.

9.7.5 Cost of test samples as per frequency given in clause 9.7.1 above shall be borne by the contractor irrespective of test results.

9.8 DOCUMENTATION

9.8.1 Original purchase vouchers from the manufacturer, and original or authenticated test certificates of the manufacturers for the total quantity of steel supplied under each consignment to be incorporated in the work shall be produced to the Engineer-in-Charge of the work by the contractor.

9.8.2 All consignments received at the work site shall be inspected by the GE along with the relevant documents before acceptance. The original vouchers and the test certificates shall be defaced and signed by the Engineer-in-Charge and kept on record in the office of the GE duly authenticated and with cross reference to the control number recorded in the steel acceptance register.

9.8.3 The steel acceptance register shall be signed by JE, Engineer-in-Charge, GE and contractor. The entire quantity of all consignments shall also be suitably recorded in the measurements book for record purposes as 'NOT TO BE ABSTRACTED' before incorporation in the work and shall be signed by the Engineer-in-Charge and contractor.

9.8.4 The following provisions shall also be complied :-

(a) All original vouchers will be kept in a file serially numbered and to be kept in GE's office
(b) Test certificates of each steel consignment will be kept in a file, serially numbered and to kept in GE's office.
(c) Steel Acceptance Register will be maintained by the GE
(d) In/Out Register for details of receipt, acceptance/rejection and consumption of steel will be maintained.
(e) Register containing results of independent and additional testing by GE
(f) Inspection registers

9.8.5 GE will check the documents personally, connected with the steel, at least once a month and record of these check will be kept in the Inspection Register (Para 9.8.4(f) above).

9.9 STORAGE ACCEPTANCE / PRESERVATION OF STEEL

9.9.1 The steel procured by the contractor shall be stored in the site of work as directed by Engineer-in-Charge / GE neatly in separate stacks at least 15 cm above GL for various grades/quality / sizes / consignments with distinct paint marks for identification. The steel so stacked shall be removed for incorporation in the work only in the presence of departmental representative.

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9.9.2 The quantity of steel of various sizes received at site and recommended for incorporation in the work shall be entered in a separate register and signed by the contractor and the Engineer-in-Charge daily

9.9.3 Steel will be stored in a manner so as to prevent distortion and corrosion till it is consumed in the work. It will be preserved either by giving a coat of red oxide primer or by cement wash and its costs will be deemed to be included in rate quoted by the contractor. For deviation purpose coat of red oxide primer will be considered, if steel not preserved properly. Any section that has deteriorated and corroded or if considered defective for any other reason, the same shall be removed from site by contractor at his cost.

9.9.4 The contractor will keep a separate stack of steel brought at site for inspection, away from the accepted stack of steel. In case, the consignment does not meet any of the requirements of the relevant IS codes, the steel will be rejected by the GE and it will be removed from the site within 24 hours at the cost of the contractor.

9.10 CONVERSION WEIGHT OF STEEL

9.10.1 The weight of steel shall be calculated as per the conversion factors specified in the SSR. For sections not listed in SSR, ISI conversion table shall be followed or manufacturer's certificate if the weights are not available in SSR/ISI tables.

9.10.2 Normal waste and off-cuts shall be stacked neatly which shall be the property of contractor. Contractor shall be allowed to remove such cut pieces after inspection and certification by the Engineer-in-Charge.

9.10.3 Advance on account of payment made towards these cut pieces shall be adjusted from advance on account of payment immediately falling due and before removal of such cut pieces from site.

9.11 PAYMENT IN RAR

9.11.1 Payment of the steel brought by the contractor should only be released by the GE after taking action on points enumerated in para here-in-before and after completing the documentation mentioned here-in-before in this regard.

9.11.2 Before procurement of steel, contract and structural drawing shall be read thoroughly and various grades/types of steel to be incorporated in the work shall be identified by contractor and got approved by the GE. Steel shall be procured sufficiently in advance as mentioned here-in-after under Clause 9.13

9.12 SAFETY OF STEEL

9.12.1 It will be responsibility of contractor to make sure that all possible arrangement are made for safe custody of the steel. In case of any loss of steel, only contractor will be responsible and the loss will be made good by contractor without any delay or claim what so ever.

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9.13 **SCHEDULE OF SUPPLY**

9.13.1 Contractor shall work out complete requirement of steel size wise and phase the same as per the activities planned to be executed in terms of CPM networking. The contractor shall procure all the steel sections in accordance with this CPM chart.

9.13.2 Schedule of supply of steel will be finalized by GE in consultation with contractor and same will be incorporated in CPM chart so that supply of steel is monitored in a way to avoid any delay in completion of the work. The schedule of supply of steel will be vetted by CE from time to time.

10 TO 12. BLANK

13. **FLOORING:**

13.1. **GENERAL:** Provision contained in clause 13.25, 13.27, 13.28, 13.32, 13.34, 13.38, 13.39, 13.40, 13.43, 13.44, 13.45 and 13.47 of MES Schedule Part – I are to be adopted for laying, jointing of floors. Floors shall be laid to levels or to falls as shown on drawings and as directed by Engineer-in-Charge. Floor finish shall be extended over dwarf walls, door openings and other openings. The dividing lines between the floors of different types wherever they so meet between adjoining rooms shall be determined of the basis of the finish visible when the doors are closed and the applicable finish shall accordingly be provided. Floor finish over RCC slabs shall be laid all as specified in clause 13.32.5 of MES Schedule Part – I. Sub floor may not be laid in panels.

13.2. Floors of types and composition as indicated in the Schedule "A" and shall be laid as specified in Section – 13of MES Schedule Part – I and as directed by Engineer-in-Charge.

13.3. **CEMENT CONCRETE SUB FLOORS / BASE / SUB BASE:** Cement concrete of specified mix in sub floor / base / sub base shall be mixed all as specified in Section - 4 [Concrete] of MES Schedule Part – I and shall be laid, compacted and finished as specified in clauses 13.27, 13.32 to 13.32.10 of Section 13 of MES Schedule Part – I. The top surface of sub base shall be broomed to have adequate bond with topping. Flooring shall be commenced within 48 hours of laying the concrete sub base, failing which the surface of sub base shall be roughened with steel wire brushes, wetted by sprinkling water and smeared with a coat of cement slurry at 3 Kg / Sq.m.

13.4. **PCC FLOOR**: PCC floors [except floors in sub base] shall be laid in bays adopting panel dimension of 1200 mm X 1200 mm. The bays shall be so arranged that the end bays shall be of equal width not exceeding 1200 mm. The floor shall be laid in alternate bays.

13.4.1 Floor topping shall be finished even and smooth using extra cement. 3mm thick glass dividing strips shall be provided to form joint in all cement concrete floors. The glass dividing strips shall be 3mm less than the thickness of PCC floor finish. The glass dividing strip shall be inserted in the PCC while the concrete is still plastic and the strips shall be left in position. PCC floors upto 40mm thickness only shall be provided with glass strips. PCC floors to be provided with glass strip shall not be laid in alternative bays and this will not involve any price adjustment.

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13.4.2 The surfaces of PCC floor shall be finished with a steel trowel to even and smooth surface using extra cement. Floors finishes of Ramps shall be worked up to an even and smooth surface without using extra cement and making of expanded metal impressions while the concrete is green. Wherever indicated in schedule of finishes and on treads of steps floor shall be provided with chequered finish. PCC Floor finish shall be provided at locations as specified in Schedule “A” and shall be laid over cement concrete sub floor to the thickness as indicated on drawings. PCC floor finish shall be in 1:2:4 mix, Type B1, using 20mm graded stone aggregate. The surface shall be cured efficiently by water ponding as directed by Engineer-in-Charge.

13.5. **NONSKID CERAMIC TILE FLOORING:** The size of the tile shall be as specified in Schedule "A". Non-skid ceramic tiles shall be first quality, flat true to shape, sound and free from flaws and other manufacturing defects. Ceramic tiles shall conform to IS – 15622, Grade B–II[a] for Toilets and B–II[b] for other locations for floor tiles and Grade B–III for wall tiles. The shade / design of tiles shall be as approved by GE. The screed for laying the tiles shall be 15 mm thick in cement mortar 1:4. The tiles shall be laid as per pattern described by GE, set, jointed in cement slurry. Joints shall be pointed flush using polymer based cementitious tiling joint filler to match the shade of the tile. The workmanship shall be all as specified in clause No. 13.40.2 of MES Schedule Part – I. The top surface of RCC slab shall be cleaned with hard wire brush and cement slurry @ 1.2 KG / Sq.m shall be applied before laying the floor.

13.6. **POLISHED CHEQUERED TILES:** The size shall be as per Schedule "A" and the tiles shall comply the requirement of clause 13.17 of MES Schedule Part – I. Workmanship shall be all as specified in Clause 13.25 & 13.39 of MES Schedule Part – I. Tiles shall be laid over screed bed in cement sand mortar as specified in Schedule "A". The tiles shall confirm to IS – 13801, Specification for Chequered Cement Concrete Tiles except for the wear quality and when tested in the manner specified in clause 11.6 of the same IS. The wear for tiles shall not exceed the following values:

[a] Average wear **:** 2mm
[b] Wear on individual specimen **:** 2.5mm

13.7 TO 13.8 BLANK

13.9. **SKIRTING:**

13.9.1. **CEMENT PLASTER SKIRTING:** Skirting shall be of 5 mm thick in cement mortar [1:3] over 10 mm thick rendering in cement mortar [1:6], 100mm high. Surface shall be finished even and smooth with steel trowel using extra cement.

13.9.2. **NONSKID CERAMIC TILE SKIRTING:** Provide Non-Skid Ceramic Tile skirting 100 mm high over 10 mm thick screed in cement mortar [1:3]. Tiles shall be all as specified here-in-before for nonskid ceramic tile flooring. Tiles shall be set and jointed in cement slurry. Joints shall be pointed flush using polymer based cementitious tiling joint filler to match the shade of the tile. The skirting thickness projecting from the rendered wall surface shall be provided with triangular coving in CM 1:3, finished even and smooth. The height of coving shall be equal to the projection of skirting from the wall surface.

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13.10. **GLAZED CERAMIC TILE DADO:** The size of tile shall be as specified in Schedule "A". The tile shall be plain, coloured / design, Type B-III, Grade-I quality, shall conform to IS – 15622 and shall be got approved by the GE. The tile shall be set over cement screed 10 mm thick in cement mortar [1:3].Joints shall be pointed flush using polymer based cementitious tiling joint filler to match the shade of the tile. The height of dado shall be as specified. The height of dado shall be 1.50 M for W/C, and shall be 2.0 M for bathrooms. The dado thickness projecting from the rendered wall surface shall be provided with triangular coving in CM 1:3, finished even and smooth. The height of coving shall be equal to the projection of skirting from the wall surface.

13.11. **COVING TO SKIRTING:** Wherever Skirting / Dado is provided, the top portion of Skirting / Dado shall be provided with triangular coving in CM 1:3, finished even and smooth. The height of coving shall be equal to the projection of skirting from the wall surface.

14. **PLASTERING:**

14.1. **GENERAL:**

14.1.1. Prepare surface to be plastered and wet the surface thoroughly.

14.1.2. Curing of plasterwork shall be properly done. The contractor may use a spray pump or similar device without any price adjustment taking precaution against damage to plaster due to impact of splicing water.

14.1.3. The contractor shall take every precaution right from the commencement of plaster work to prevent any craziness that may appear on the surface of plaster and shall be responsible to make good any portion of plaster work which in the opinion of the GE requires removal and redoing.

14.1.4. External plaster surfaces shall be finished to fair and even surfaces and internal plastered surfaces shall be finished to even and smooth surface without using extra cement.

14.1.5. The thickness of plaster/dado/skirting is the finished thickness [exclusive of dubbing] over proudest portion.

14.1.6. Plastering at the junction of walls and floors and to all internal and external angles shall meet at right angle as directed by Engineer-in-Charge. Plastering shall be returned in openings involved.

14.1.7. Internal plastering shall be carried out to the full width of jambs for the door, windows and openings i.e. up to external edge of the wall.

15. **WHITE WASHING / COLOUR WASHING / OIL EMULSION DISTEMPER / CEMENT BASED PAINT:**

15.1. **ACRYLIC WASHABLE DISTEMPER:** Provide Acrylic Washable Distemper two coats over a coat of alkali resistant primer over wall care putty where specified. Acrylic Washable Distemper shall conform to IS – 428. Preparation of surfaces application of primer and Acrylic Washable Distemper shall be in accordance with clause 15.14 of MES Schedule Part – I. Acrylic Washable Distemper shall be of any of the makes specified here-in-after. Tint of Acrylic Washable Distemper shall be as approved by GE.

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15.2. **EXTERNAL WEATHER PROOF PAINT:**

15.2.1. **GENERAL:** Unless samples of all materials are approved, the contractor will not be allowed to commence the work. Brushing for painting etc., shall be got approved by Engineer-in-Charge based on manufacturer's recommendations. No improvised brushes or substandard brushes shall be brought to site and used.

15.2.2. **MATERIALS:**

15.2.2.1. Paint shall be weather proof Acrylic emulsion, exterior grade [100% Acrylic] premium quality. Paint shall be procured from any of the makes listed here-in-after

15.2.2.2. Shade of the paint shall be as approved by GE.

15.2.2.3. Primer shall be water based acrylic suitable for exteriors as per manufacturer's instructions. Primer shall be of same make as of paint.

15.2.2.4. The paint and primer shall be brought in manufacturer's sealed containers only by the contractor duly marked with batch number from the manufacturer.

15.2.2.5. The contractor shall produce manufacturers test certificate along with purchase voucher in original for the paint and primer brought to site before claiming payment for the same, Purchase voucher of paint and primer shall contain the complete description of material, batch No., net weight, test certification No., quantity in each package, No. of packages etc., The quantity of material brought at site indicating No. of packages, quantity in each package, batch No., purchase voucher number, test certification number, date of manufacturing, date of expiry etc., shall be entered in MB as "Not to be Abstracted" and shall be signed by the JE, Engineer-in-Charge, GE and contractor.

15.2.2.6. Each container of paint and primer shall bear the following particulars:

[a] Manufacturer's Trade Mark
[b] Reference to Indian Standard to which they comply
[c] Name of Product
[d] Net Weight
[e] Date of Manufacturing
[f] Batch No.
[g] Storage Requirement
[h] Storage Life
[i] Date of Expiry

15.2.2.7. Each lot of paint and primer shall be checked by Engineer-in-Charge and approved by him after verifying from invoices, package, batch No. and test certificate. Materials shall be incorporated in the work only after written approval from Engineer-in-Charge.

15.2.3. **WORKMANSHIP:**

15.2.3.1. All brushes, tools, pots, kettles etc., used in carrying out the work shall be free from foreign matter and shall be thoroughly cleaned with hot water and solutions at the end of a day's work or before use for a different type of material. No finish shall be executed until a sample of the finish to the required colour and shade has been approved by the GE. Where more than one finish is indicated, each coat shall be approved by the GE before the subsequent coat is applied.

15.2.3.2 The colour shall be even shade over the whole surface, if it is patchy or otherwise bad, the work shall be redone by the contractor at his own expense.

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15.2.4. **PREPARATION OF SURFACES:** The surface shall be thoroughly cleaned of loose particles, dust, dirt, efflorescence, chalking, grease, mortar drops and other foreign matter. The surface shall be sand papered with Grade – I abrasive paper and dusted off to achieve an even and smooth surface free from all dust particles. The contractor will use electric blowers for this purpose. If surface so obtained is uneven, it shall be brought to a perfectly even surface by applying putty and allowing it to dry completely and then it shall be rubbed with the abrasive paper and dusted off and finally area cleaned by use of electric blower.

15.2.5. **PRIMING COAT:** After preparing the surface as approved by GE, one coat of exterior water based acrylic primer as approved by manufacturer, thinned with water in 1:1 ratio shall be applied with brush as per manufacturer's instructions and as directed by Engineer – in – Charge.

15.2.6. **APPLICATION OF WEATHER PROOF PAINT:**

15.2.6.1. The acrylic emulsion weather proof paint shall be applied by brush or roller. No stainer or colorants shall be used. The paint shall be stirred well before use. The primer coat shall not be left without application of top coats for a long period time.

15.2.6.2. Two coats of 100% Acrylic emulsion weatherproof paint thinned with 400 ml water per litre of paint shall be applied. The drying period between two coats shall be minimum 4 hours or as per Manufacturer's instructions. The shade shall be as approved by GE. The finish of Acrylic emulsion weather paint shall be smooth matt finish.

15.2.6.3. The paint shall be as per Manufacturer's original colour as available or shade card. No mix of tint shall be made into original shade.

15.2.7. **GUARANTEE:**

15.2.7.1. The work of applying primer and paint shall be got carried out under the supervision and guidance of accredited representative of the manufacturer. A certificate from their representative shall be obtained by the contractor to the effect that work of painting and primer has been carried out under their strict supervision and as per manufacturer's instructions. The same shall be submitted by the contractor to GE.

15.2.7.2 The contractor shall also obtain a written guarantee for effectiveness of paint against fading out, peeling off, cracking, dust / algae accumulation etc., for a period of **5 [Five] Years** from the certified date of completion of entire work from the manufacturer and submit the same to GE before completion of work.

15.2.7.3. Should the GE at any time during construction or reconstruction or prior to the expiry of the Guarantee period, finds defective performance of the paint, the contractor shall, on demand in writing from the GE specifying the location complained of notwithstanding that the same may have been inadvertently passed / certified and paid for, undertake to carry out such treatment as may be necessary forthwith to rectify the defects to the full satisfaction of GE and render complaint, free from any type of defects.

15.2.7.4. In the event of his failure to do so, within the period as specified by the GE in his aforesaid demand, the GE may undertake such defective work through other agency at the risk and cost of the contractor in all respects.

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15.2.7.5 The liability of the contractor under this condition shall not extend beyond the period of **5 Years** from the certified date of completion unless the GE had previously given notice to the contractor. Security Deposit against the Guarantee Period for effectiveness of Acrylic Emulsion Paint for **5 Years** shall be calculated **@ 3%** of the Amount of Acrylic Emulsion Paint at contract rates [Subject to a minimum of ₹ 5,000.00] shall be retained by the Government from the contractor's dues. If contractor fails to rectify the defects noticed in the treatment or found in the material, the aforesaid amount so retained shall be utilised for rectification of defects and contractor shall have no claim whatsoever on this account. This Security Deposit shall be released after successful expiry of the **5 Years** guarantee period from the certified date of completion of the entire work by GE, provided always that the contractor shall first have been paid the Final Bill and have rendered "No Demand Certificate [IAFA – 451]". Alternatively, the contractor may furnish Fixed Deposit Receipt in lieu of Security Deposit from a Schedule Bank in favour of GE for the above said Guarantee Period. Condition 10, 46 & 68 of General Conditions of Contracts [IAFW – 2249] shall be deemed to be amended to the extent mentioned above.

15.2.8. **SCAFFOLDING:**

15.2.8.1. The exterior painting work shall be carried out by using scaffolding. No zoola is permitted for the work under any circumstances. Suitable scaffolds shall be provided for workmen.

15.2.8.2. Scaffolding or staging more than 3.5 Metres above the ground or floor, swung or suspended from any overhead support or erected with stationary support shall have a guard rail properly attached, braced and otherwise secured at least 1 Metre high above the floor or platform of such scaffolding or staging and extending along the entire length of the outside and ends thereof with only such openings as may be necessary for the delivery of materials. Such scaffolding or staging shall be so fastened as to prevent it from swaying from the building or structure.

15.2.8.3. Every opening in the floor of a building or in a working platform be provided with suitable means to prevent the fall of persons or materials by providing suitable fencing or railing whose minimum height shall be 1 Metre.

15.2.8.4. Safe means of access shall be provided to all working platforms and other working places.

15.2.8.5. The rates quoted are deemed to include the above provision of scaffolding and no extra will be payable to contractor on this account. The scaffoldings shall be removed only after obtaining clearance of Engineer-in-Charge / Garrison Engineer after considering the quality of the work undertaken on completion of the painting.

15.2.9. **SAFETY PRECAUTIONS:**

15.2.9.1. Contractor shall provide all safety precautions for the labour engaged for this work. All the labours shall be provided with safety belts, helmets, Nose screens etc. and the contractor shall adhere to all safety precautions as per Labour Welfare Act.

15.2.9.2. It is also advised that contractor shall have insurance cover for his workmen working at heights against any eventuality from any reputed insurance agencies. Department will not be responsible for any untoward incident happening due to lack of safety precautions taken by contractor.

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15.2.10. **STAGE PASSING:**

15.2.10.1. The work shall be executed in a workman like manner and to the entire satisfaction of the GE. Contractor shall obtain the approval of GE stage wise as indicated below. The contractor shall give due notice in writing with sufficient time in advance to the Engineer-in-Charge and the GE, when each stage is about to be completed. The contractor shall start the work of subsequent stage only after obtaining written approval of the GE for previous stage. In default of such notice being received from the contractor, if he commences the work of subsequent stage without approval of the GE, then the GE shall have all the rights reserved to reject the work and ask the contractor to demolish the said portion executed. The contractor shall have no claim on this account or otherwise. The decision of the GE in this regard shall be final and binding.

STAGES:

[i] Extent of area to be painted.
[ii] Sample of painting work [Area to be decided by GE]
[iii] After preparation of surface and before applying priming coat.
[iv] After applying priming coat.
[v] After applying first coat of Acrylic emulsion paint.
[vi] After applying second coat of Acrylic emulsion paint.

15.3 **CEMENT BASE PAINT**

15.3.1 Refer Clause No.15.7 of MES SSR 2009 Part-I (Specifications). Cement base paint finishes shall be two coats of cement base paints over a coat of alkali resistant primer shall be applied. Cement base paint shall be of superior quality manufactured by standard firms given in here-in-after.

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17. **PAINTING:**

17.1. Members specified to be painted shall be first be passed by the Engineer-in-Charge and marked as such before commencement of painting work. Each coat of paint shall be passed by the Engineer-in Charge before successive coat is applied.

17.2. Surfaces which become inaccessible for applying further coats shall be painted before fixing.

17.3. The synthetic enamel paint shall be of 1stquality as approved by GE. The exact tint of paint shall be as approved by the GE.

17.4. The primer and paint shall be of the same manufacturer. Pink primer shall be used for wooden surfaces and zinc chrome primer for steel surfaces.

17.5. The contractor shall inform the GE, within three weeks of the acceptance of the tender, the brand names of the manufacturer of paint proposed to be used in the works and submit samples thereof and obtain prior written approval of the GE before their use in the work.

17.6. If the undercoat of paint is not executed within six months after applying the priming coat of paint, the priming coat shall be redone by the contractor at no extra cost to the Government.

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17.7. Surfaces which are specified to be treated with synthetic enamel paint shall be prepared and painted with two coats of approved synthetic enamel paint over a coat of primer all as specified. Primer for wood surfaces shall be pink primer and for that of steel surfaces shall be zinc chrome primer. Colour and shade for under coat and finishing coat shall be decided by Garrison Engineer.

17.8. **PAINTING TIMBER SURFACES/STEEL SURFACES:** Where painting to timber / steel surfaces is indicated, prepare surfaces and apply two coats of synthetic enamel paint over one coat of primer as specified here-in-before. Painting works shall be executed as per Schedule "A", MES Schedule Part – I and as directed by Engineer-in-Charge. Preparation of surfaces for all locations except gantry girders shall be all as specified in MES Schedule Part – I.

17.8.1 The steel surfaces of gantry girder shall be prepared by sand blasting. The prepared surface shall be got passed by the GE before applying paint, gantry girders shall be painted with primer and under coat before erection. Painting shall be carried out by spray painting.

17.9. **WORKMANSHIP:**

17.9.1. All wood work required to be painted shall be smoothened, sized and knotted and then applied with priming coat. Stopping and filling[filler coat] shall be done after priming coat and surfaces rubbed down to a level and smooth surface and thereafter under coat and finishing coat applied, all as specified in clause 17.6 of MES Schedule Part – I.

17.9.2. The steel surfaces which are required to be painted shall be given two coats of paint, priming coat and undercoat after fabrication but before assembly and erection and finishing coat after assembly and erection.

17.10. Unless otherwise indicated and/or specified in these particular specifications, all surfaces of iron and steel work shall be prepared and treated with two coats [one undercoat and one finishing coat] of synthetic enamel paint over a coat of primer all as specified here-in-before. Reinforcement bars, tinned or galvanized iron surfaces and steel-work embedded in concrete/plaster shall however not to be painted.

17.11. Bottom of door shutters shall be given one coat of primer only.

17.12. Irrespective of what is indicated on drawings and specified elsewhere, finishing coat and undercoat shall be with the same paint.

17.13. **CLEANING:** Before commencement of plastering / painting work all doors / windows / ventilators, Balcony railings etc. shall be covered properly for protecting from the drips of paint / primer while painting, to the entire satisfaction of Engineer-in-Charge. Covering shall be done with tarpaulin or ply wood. After completion, the surrounding area, glass panes, flooring shall be cleaned and all the paint marks on it shall be removed. Contractor's quoted rates shall be deemed to be included of the above said provision and nothing extra shall be admissible on this ground.

17.14. **TARRING:** Prepare and apply two coats of tar to the hold fasts, backs of wooden frame in contract with brick work/plaster, etc. Hold fast shall be given two coats of tar and sanded.

PARTICULAR SPECIFICATIONS

18. SECURITY FENCING / GATE

18.1. GENERAL:-

18.1.1. Refer relevant Clauses 4.22 of MES Schedule Part I 2009 (Specifications)

18.2 RCC FENCING POSTS

18.2.1 Fencing post may be rectangular or square, of uniform section throughout their length or tapering on two sides or tapering on all four sides as indicated. The tolerance on cross section shall be + 3 mm.

18.2.2 Cover :The longitudinal reinforcement shall have a minimum concrete cover of 20 mm or twice the diameter of the main reinforcement whichever is greater. For posts and struts to be used in coastal areas, the minimum concrete cover for the longitudinal reinforcement shall be increased to 25 mm.

18.2.3 RCC posts shall be cast with holes as indicated through which the fencing wires would pass. The holes may be formed by inserting steel rods, slightly greased, horizontally through the holes in the divisions of the forms/mould and withdrawing them before the concrete sets too hard. Holes shall have a uniform diameter 10 mm or as directed. They shall present a reasonably smooth surface.

18.2.4 Where fencing wire is to be tied to the fence post using cast in metal projections or clips, these shall be embedded in the fence post along the centre line of the post at right angle to one face of post during casting. Inner ends of the hook or clip shall be bent or hooked to prevent extraction. The slips shall be sufficiently robust and strong, to withstand service conditions and repeated fixing and unfixing.

18.3 PVC COATED CHAIN LINK FENCING

18.3.1 PVC Coated Chain Link fencing shall be as per Sch A BOQ and as per manufacturer’s instruction/ latest IS Codes.

19. SOLID PVC WALL LINING, PARTITION, CEILING

19.1 PVC SHEET CLADDING

19.1.1 PVC sheet shall be stuck using rubber based adhesive for cladding. PVC sheet may be heat bent. i.e. lipping to cover the edges of plywood/flush/flush door shutter

19.2 SOLID PVC WALL LINING

19.2.1 Solid PVC Wall Lining shall consist of PVC sheet stuck with solvent cement on PVC sheet paneling fillets as per direction of Engr-in-Charge & manufacturer's instruction. Rectangular or circular pillars or other insets in the wall shall be covered using heat bent PVC sheet to ensure heat bending of the PVC sheet at the required angle: suitable groove shall be cut in the PVC sheet if required. Two PVC lining sheets shall be stuck adjacent to each other after leaving a gap of 2mm. If so desired strips of 5 mm thick PVC sheet of width 75 mm feathered at the edges can be stuck at the places where two PVC sheets are joined, using solvent cement.

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20. **ROAD WORK:**

20.1. **GENERAL:** Road work shall be executed all as specified in MES Schedule Part – I. Road shall be constructed to the widths, alignments, cambers, super elevations and gradients etc. as specified / shown on drawings and as directed by Engineer-in-charge. Alignment, profile and surface evenness etc. shall be provided all as specified in clause 20.A.19 of MES Schedule Part – I. The aggregate / materials shall be stacked all as specified in clause 20.A.9 of MES Schedule Part – I.

20.2. **SCARIFYING:** The work of scarifying shall be carried out as described in Schedule "A" and specified in clause 20.A.34.1 of MES Schedule Part – I.

20.3. **ROAD SIDE FILLING:** Before commencement of earth filling, representative of the GE and the contractor will be required to take the levels jointly of existing ground surface at intervals decided by the GE [The decision of the GE shall be final and binding in this respect] shall be taken for calculating the reduced levels of the existing ground and recorded in level register.

20.3.1 After completion of the filling of earth including compaction/ramming, finished levels shall also be taken at decided internal and the reduced levels shall be entered in the level register. The quantity of filling shall be arrived out based on the Prismoidal Formula [Simpson's rule]. Payment for earthwork in the above-mentioned situations shall be allowed as per the above method subject to ten percent deduction for voids.

20.4. **FORMATION SURFACES:** Before laying soling the formation shall be prepared all as specified in MES Schedule Part – I clause 20.A.21.2 and rolled by 8/10 capacity power roller.

20.5. **BLOCK LEVELS:**

20.5.1. Before commencement of road-work, the block levels of the area [after clearing the shrubs and vegetation etc.] shall be taken jointly by the contractor and Engineer-in-Charge. Block levels shall be taken at 3 meter intervals.

20.5.2. Drawing showing the block levels on the basis of the levels taken as mentioned above shall be prepared and signed by both parties in token of their acceptance. Formation levels to which the cutting/filling is required to be carried out shall then be marked on these drawings and cutting/filling shall be carried out strictly according to these signed drawings.

20.5.3. Levels of the finished formation [after consolidation] shall be taken to ensure that the correct levels as indicated in the aforesaid drawings have been, in fact achieved. A certificate to this effect shall be endorsed by the Engineer-in-Charge in the works passing register in addition to the entries made in the measurement book.

20.5.4. From the above mentioned signed drawing and final levels, volume of cutting/filling shall be computed by using standard formula of Simpson's rule and Prismoidal formula. 10% [TEN] deduction shall be done from the volume of filling.

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20.6. **MOORUM / RED BAJRI:** Moorum/Red Bajri shall be all as per cause No 3.21.1 of MES Schedule Part – I. Moorum / Red Bajri shall be obtained from approved pits and quarries of disintegrated rocks containing siliceous material and natural mixture of clay of calcareous origin. These shall not contain any admixtures of ordinary earth. Red Bajri shall be dark red in colour consisting of coarse grains, free from mica and other foreign matter. Size of Moorum/Bajri shall vary from dust to 40 mm gauge.

20.6.1 Anything over in size shall be rejected or shall be broken down to bring within 40 mm size. It shall conform to or be superior to the samples kept in Garrison Engineer office. No price adjustment shall be made on this account. Sources of good quality Moorum shall be approved by GE. After site clearance, the levels of filling shall be marked by fixing pegs on both sides at regular intervals as a guide before commencement of earthwork. The filling materials shall be spread in layers of uniform thickness not exceeding 15 cm [Spread thickness] over entire width. Successive layers shall not be placed until the layer under construction has thoroughly compacted.

20.7. **SOLING:** Stones for soling shall be crushed or broken hard stone obtained from approved quarries and shall be broken to size ranging from 100mm to 50mm and conform to the samples kept in GE's office and approved by the GE before incorporation in the work. Thickness of broken stone soling shall be as indicated in Schedule "A", laid in layers as indicated, levelled, watered and rolled to required surface and camber all as specified in clause 20.A.20.1 of MES Schedule Part – I.

20.8. **WATER BOUND MACADAM:** Stone for water bound macadam shall be hard broken graded granite aggregate of grading 2, 63 to 40mm size obtained from approved quarries. Screenings shall be of granite and shall be of grading Type B. WBM work shall be completed six months before laying bituminous carpet. The surface of WBM shall be inspected before laying carpet by GE and representative of contractor for rectifying the defective construction if any, specified in MES Schedule Part – I clause 20.A.21.11, without extra cost to the Government. Bituminous carpet work shall not be carried out unless the 'stage passing' is given by GE.

20.9. **MATERIALS:**

20.9.1. **BITUMEN**: Bitumen shall be paving bitumen of Grade **VG–10 & VG-30** as per IS – 73, Specification of Paving Bitumen and shall be Contractor's supply and shall be procured by the contractor directly from **HPCL / IOCL / BPCL**. Contractor shall make his own arrangements to store the same. The contractor shall produce paid vouchers and test certificates for bitumen used in the work immediately on receipt of materials. Contractor shall make his own arrangements to store the bitumen. After the bitumen has been brought to site, independent tests shall also be carried out by the GE, to ascertain the quality of the bitumen. Testing of bitumen shall be in accordance with IS – 73. For sampling the numbers of containers to be selected from the lot of bitumen depend upon size of the lot and shall be in accordance with Table – III of IS – 73. Number of tests and types of tests to be carried out shall be as per relevant IS. In case the bitumen is not of requisite standard as verified by the GE through independent testing as mentioned here-in-before, the Contractor shall remove the total consignment from the site at his own cost after written rejection order of the consignment by the GE despite manufacturer's test certificate. The cost of test shall be borne by the Contractor irrespective of the results of tests.

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20.9.1A The bitumen containers shall be stacked at site before use and the Engineer-in-Charge shall verify the grade and manufacturer's monogram before the container is opened. The stack measurement of the materials shall be entered in the measurement book under the heading "NOT TO BE ABSTRACTED" and shall be signed by both the contractor and the department representatives to ensure that required quantity of materials have been brought to site for incorporation in the work.

20.9.2. **COARSE AGGREGATE:** Coarse aggregate shall be of broken/crushed hard stone all as specified in Clause 20.B.2 and shall conform to the samples and grading kept in the office of the GE.

20.9.3. **FINE AGGREGATE:** Fine aggregate shall be crushed stone dust all as specified in Clause 20.B.2 and shall conform to the samples and grading kept in the office of GE.

20.9.4. All the aggregates brought to site shall be stacked and stack measurements of the same shall be recorded in the MB as "NOT TO BE ABSTRACTED" to check that the requisite quantities of aggregates have been brought and incorporated in the work.

20.10 **BITUMINOUS MACADAM**

20.10.1 The combined grading of coarse aggregate and fine aggregate shall be as specified in clause 20.B.2.5.4 (b) of SSR-2009 Part-I for bituminous macadam.

20.10.2 The contractor shall get the mix design done through approved Government laboratory and the cost shall be deemed to be included in the rate quoted by the contractor for relevant item of Schedule "A".

20.10.3 The design requirement shall be as under for bituminous macadam: -

Sl No	Properties	Specified value of bituminous macadam
(a)	Number of compaction blows each and of Marshall specimen.	75
(b)	Marshall stability (75 blows) at 60ºC minimum in kg	545(min)
(c)	Marshall Flow at 60ºC,MM	2-4
(d)	% void in compacted Mix	10-15
(e)	Type of binder	VG-30 Grade
(f)	Binder content % by weight total mix	As per design mix

Bituminous macadam shall be prepared, spread and compacted as specified in clause 20.B.3 to 20.B.3.10.1 of SSR-2009 Part-I.

20.10.4 **TRANSPORTATION & LAYING OF BITUMINOUS MACADAM.**

20.10.4.1 Mixed material as per approved design mix shall be transported from Mechanical Mixer to the site in a covered/ properly protected clean tipper/ dumpers. Temperature of mix delivered at site shall be maintained & checked prior to discharge into the spreader. If temperature of any batch is below the laying temperature mix shall be rejected.

20.10.4.2 Mix transported from the Mechanical Mixer for Macadam shall be laid, spread, tampered & finished to the specified width & profile. Temp of the mix at the time of laying shall be in the range of 120^0C to 160^0C.

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20.10.5 **QUALITY CONTROL FOR BITUMINOUS MACADAM.** At least one sample for every 100 tons or part thereof of bituminous mix or one sample per day shall be collected and following test shall be done.

(a) Three marshal specimens for each sample shall be conducted and tested for average stability, flow, void ratio and density and shall be as per designed value.

(b) Binder shall be extracted from about 1000 grams of mix for each sample and binder content shall be determined. Contents of other ingredients shall also be determined.

(c) The permissible variation of various ingredients and binder shall be as per designated value as given in to Section 20-B of SSR Part I.

(d) Field density of Bituminous layer shall be checked for every 100 square meters. The field density shall not be less than 98 % lab density.

(e) The contractor shall arrange the material, labour and machinery required for testing and the cost shall be deemed to be included in the rate quoted by him for relevant item of Schedule "A".

(f) The record of all test results shall be kept in record duly defaced by Engineer-in-Charge and the contractor.

20.10.6. **CONTROL ON LAID THICKNESS:** After the mix is compacted, the thickness shall be checked by noting the depth of penetration of hot scale. Except cross the crown off camber or change with the finished surface of the wearing course is to such regularity that when tested with 3 meters straight edge placed anywhere in any direction there shall not be a gap greater than 3 mm between the bottom edge and surface anywhere. Any depression or protrusion shall be rectified immediately.

20.10.7 **MIXING**

20.10.7.1 The temperature of binder at the time of mixing shall be in the range of 150^0C to 163^0 C and that of the aggregates in the range of 155^0 C to 163^0 C provided that the difference in temperature between the binder and aggregates at no time exceeds 14^0 C. Mixing shall be thorough to ensure that a homogeneous mixture is obtained in which all particles of the aggregates are coated uniformly and discharge temperature of mix shall be between 130^0 C to 160^0 C. The temperature of mix at the time of laying shall be in the range of 120^0 C to 160^0 C.

20.10.7.2 Mixing of the aggregates with binder shall be carried out in Mechanical Mixer.

20.10.8 **ROLLING**.

20.10.8.1 The rolling shall start longitudinally along the edges and proceed towards the centre of the road, overlapping on successive trips of the roller by at least half the width of the rear wheel on the super elevated portion of the road rolling shall commence from the lower edge and progress gradually towards the upper edge of the pavement. Rolling shall be carried out all as specified here-in-below

20.10.8.2 **WBM ROLLING**: WBM surface to be rolled with 8 to 12 tonne power roller all as specified in Clause No. 20.A.21.4 of SSR Part-I.

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20.10.8.3 **BITUMINOUS MACADAM ROLLING**: Rolling shall be with power driven tandem roller of weight 8 to 12 tonne. Static Weight Bituminous Macadam shall be initially rolled in vibratory mode & finally in static mode all as specified in clause 20.B.3.8 of SSR Part-I.

20.10.8.3.1 The speed of the roller shall not exceed 5 KM per hour to avoid displacement of the mix.
20.10.8.3.2 To prevent adhesion of the mix to the roller wheels, the wheels shall be kept damp with water.

20.10.8.3.3 Excess water shall be avoided. Fuel oil, lubricating oil or kerosene shall not be used for the purpose.
20.10.8.3.4 Rolling operation shall be completed in very respect before the temperature falls down below 100ºC

20.10.7 **BITUMINOUS CARPET (30 MM THICK OR MORE)**

20.10.7.1 **PREPARATION OF BASE :** Before the carpet is applied to the existing base, the surface shall be cleaned by removing dirt, caked earth and other foreign matter with wire brushes, sweeping with broom and finally dusting with sacks. Where the existing base is potholed or rutted, these irregularities shall be corrected with premixed chippings or coated macadam, as indicated or ordered depending upon the depth of the pothole, laid after applying a tack coat or binder and well rammed thereafter.

20.10.7.2 **TACK COAT :**The binder shall be heated, wherever required, to the temperature appropriate to the grade of binder used and applied uniformly to the base at the rate specified by means of a sprayer. The binder shall be evenly brushed, if need be. The tack coat should be applied just ahead of the spreading of the premix.

20.10.7.3 **PREPARATION OF PREMIX :**Hot mix plant of suitable capacity or Mechanical mixers of approved type shall be employed for mixing as indicated. For small quantities for work, improvised hand mixing drums may also be used as approved by the EIC. The binder shall be heated to the temperature appropriate to the grade of bitumen used. Where paving bitumen is used, the aggregate shall be suitably heated before these are placed in the mixer. After about 15 seconds of dry mixing the heated binder shall be distributed over the aggregates at the rate specified. The mixing of the binder with chippings shall be continued until the chippings are thoroughly coated with the binder.

20.10.7.4 **SPREADING OF PREMIX:** Immediately after the application of the tack coat, the premix shall be spread evenly with mechanical paver or rakes or distributed evenly by means of a drag spreader as indicated to the desired thickness and to the correct camber. The surface shall then be checked for camber by means of camber board and all inequalities corrected. For premix carpet of thickness 25 mm or more, Mechanical Pavers shall only be used unless otherwise indicated.

20.10.7.5 **ROLLING :**As soon as sufficient length has been spread with the premix, the surface shall be rolled with 8 to 10 tonnes smooth wheeled power rollers or pneumatic tyre rollers. Rolling shall commence at the edge and progress towards the center longitudinally, except in the case of super-elevated curves, where rolling shall progress from the inside towards the outside of the curve.

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20.10.7.6 When the roller has passed once over the whole area any high spots or depressions which become apparent, shall be corrected by removing or adding premixed chippings. When this has been done, the surface shall be rolled to compaction and all roller marks eliminated. Excessive rolling shall be avoided as this serves no useful purpose and may spoil the carpet. In each pass of the roller and preceding track shall be overlapped uniformly by at least 1/3 width.

20.10.7.7 The roller wheels shall be kept damp to prevent the premix from adhering to wheel and being picked up. Fuel and lubricating oils shall not be used for this purpose. Longitudinal and transverse edges of the carpet laid and compacted earlier shall be cut to their full depth so as to expose fresh surface, which shall be painted with a thin surface coat of binder before new mix is placed against it.

20.10.7.8 **SEAL COAT :**A premixed sand seal coat or liquid seal coat, as indicated, shall be applied to the surface immediately after laying the carpet. No traffic shall be allowed on the road till the seal coat has been placed.

20.10.7.9 **FINISHED SURFACE :**The finished surface shall be uniform and conform to the lines, grade and cross-sections specified. The surface evenness of the compacted carpet in the longitudinal and transverse directions shall be within the tolerances specified.

20.10.7.10 **Opening to Traffic :**Traffic may be allowed on the road after providing the seal coat.

20.10.7.11 **QUANTITIES OF MATERIALS**

20.10.7.11.1 Quantities of materials required for 10 Sqm(30mmthick) of road surface shall be as under :-

Sl No	Treatment	Stone Chippings Cum	Binder Kg
1.	Tack Coat : (a) On water bound macadam surface (b) On existing black top surface	— —	7.5 to 10 5 to 7.5
2.	Carpet : (a) 12.5 mm size chippings (b) 10 mm size chippings	 0.27 0.135	 14.25 7.65
	Total	**0.405**	**21.90**

20.10.7.12 **LIQUID SEAL COAT**

20.10.7.12.1 **BINDER :**Binder shall be paving bitumen of suitable penetration grade or a cut back bitumen of suitable viscosity, as indicated. The binder shall be heated to the temperature appropriate to the grade of bitumen being used and applied to the cleaned surface, blinded with chippings and rolled. The construction operation shall be as described under surface dressing.

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20.10.7.12.2 **STONE CHIPPINGS FOR LIQUID SEAL COAT** Stone chippings shall be of 6.3 mm size defined as 100 percent passing through 10mm size IS Sieve and retained on 2.36mm size IS sieve. The chippings shall satisfy the quality requirements specified for coarse aggregate for bituminous carpets, except that the upper limit for flakiness index shall be 30.

20.10.7.12.3 **QUANTITIES OF MATERIALS**

The quantities of materials required per 10 Sqm of road surface shall be as under

Type	Stone Chippings Cum	Sand Cum	Binder Kg
Premixed sand seal coat	-	0.06	6.8
Liquid seal coat	0.09	-	9.8

20.11 **SEMI DENSE ASPHALTIC CONCRETE [SDAC]:**

20.11.1 **MATERIALS:**

20.11.1.1 The Semi Dense Asphaltic Concrete shall consist of coarse aggregate, fine aggregate and filler in suitableproportions and mixed with sufficient binder. The combined grading of fine and coarse aggregates shall conform to the following:

Layer Thickness	**35–40 mm**	**25–30 mm**
IS Sieve [mm]	**Cumulative % by weight of total aggregate passing**	
19	100	–
13.2	90 – 100	100
9.5	70 – 90	90 – 100
4.75	35 – 51	35 – 51
2.36	24 – 39	24 – 39
1.18	15 – 30	15 – 30
0.6	–	–
0.3	09 – 19	09 – 19
0.15	–	–
0.075	3 – 8	3 – 8

Note: The combined grading shall not vary from the low limit on one sieve to the high limit on theadjacent sieve.

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20.11.1A **COARSE AGGREGATE:**

20.11.A.1 Coarse aggregate shall be crushed stone retained on IS 2.36mm sieve. They shall be clean, hard, durable, of cubical shape, free from disintegrated pieces, free from dust and soft or friable matter, organic or other deleterious substances and adherent coatings.

20.11.A.2 The aggregates shall not contain more than 6 percent by weight of flat/longed pieces [Flat piece is one having ratio of "width/thickness" of more than 4, elongated piece is where the ratio "length / width" is more than 4].

20.11.A.3 The aggregates shall preferably be hydrophobic and of low porosity. If hydrophanous aggregates are to be used, which in normal circumstances shall be avoided, bitumen shall be treated with anti-stripping agents of appropriate quality in suitable doses as per the manufacturer's recommendations, without additional payment. Before approval of the source of aggregate, the aggregates shall be tested for stripping.

The aggregate shall satisfy the following physical requirements:

Property	Test	Specification
Cleanliness [Dust]	Grain Size Analysis [IS – 2386, Part – I]	Max 5% Passing 0.075mm Sieve
Particle shape	Flakiness and Elongation Index [Combined][IS – 2386, Part – I]	Max 30%
Strength	*Los Angeles Abrasion Value [IS – 2386, Part – IV]	Max 35%
	*Aggregate Impact Value [IS – 2386, Part – IV]	Max 27%
Polishing	Polished Stone Value [BS:812 Part – 114]	Max 24%
Durability	Soundness [IS – 2386, Part – V] Sodium Sulphate Magnesium Sulphate	Max 12% Max 18%
Water Absorption	Water absorption [IS – 2386, Part – III]	Max 2%
Water Sensitivity	Retained Tensile Strength [AASHTO T 283]	Min 80%
Stripping	Coating & Stripping of Bitumen Aggregate [Mixtures] [IS – 6241]	Minimum Retained Coating 95%

Note:

1. *Aggregate may satisfy requirements of either of these two tests.
2. Water sensitivity test is only required for the minimum retained coating in the striping test is less than 95%.
3. If minimum retained coating is less than 95% and it is required to use anti-stripping agent, the same shall be provided as per manufacturer's instructions or as advised by testing lab as per approved job mix formula.

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20.11.1B **FINE AGGREGATE:** Fine aggregate shall be the fraction passing IS – 2.36 mm sieve and retained on 75micron sieve, consisting of Natural River Sand conforming to IS – 383. It shall be clean, hard, durable, dry and free from dust, and soft or friable matter, organic or other deleterious matter.

20.11.1C **FILLER:** Filler shall consist of Ordinary Portland Cement [43 Grade] as approved by the GE. The filler shall be graded within the limits indicated in Table below.

IS Sieve [mm]	Cumulative per cent passing by weight of total aggregate
0.6	100
0.3	95 – 100
0.075	85 – 100

The rate quoted for the Semi Dense Asphaltic Concrete works shall be including the cost of the filler to be incorporated in the mix as per approved mix design.

20.12 **DESIGN CRITERIA FOR SEMI DENSE ASPHALTIC CONCRETE:**

20.12.1 Semi dense asphaltic concrete mixes should be properly designed so as to satisfy certain criteria needed to assure adequate stability and durability. The mix design shall be done by Marshall Method of mix design [ASMD 1559 – 1979]. The mix as designed and laid should satisfy the requirements as given under:

Criteria		Specified Value
		Semi Dense Asphaltic Concrete
[a]	Number of compaction blows each end of Marshall Specimen	75
[b]	Marshall Stability in KGs	1200 [Min]
[c]	Flow Value [mm]	2 – 4
[d]	Percentage Air Void in Mix [to prevent bleeding]	3 – 5
[e]	Percentage Air Voids filled with Bitumen [VFB]	65 – 78
[f]	Loss of Stability on immersion in Wear at 60°C [ASTMD 1075]	–
[g]	Binder content % by weight of Total Mix.	5.5%**
[h]	Percentage Voids in Mineral Aggregates [VMA]	See table below

Note : ** Minimum Bitumen content shall be 5.5% as specified hereinabove & Schedule "A". However, the binder content to be used in the works shall be as per approved Mix Design as per Job-Mix formula obtained from approved Institutions / laboratories. Please note that if the actual quantity of bitumen to be incorporated on approval of Mix-Design is found more than the minimum bitumen content specified above, No price adjustment shall be made for actual bitumen content of Mix-Design.

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MINIMUM PERCENT OF VOIDS IN MINERAL AGGREGATES:

Nominal Maximum Particle Size*	**Minimum VMA, Percent related to Mix Design Air Voids, Percent****		
	3.0	**4.0**	**5.0**
9.5	14.0	15.0	16.0
12.5	13.0	14.0	15.0
19.0	12.0	13.0	14.0
25.0	11.0	12.0	13.0
37.5	10.0	11.0	12.0

* The nominal maximum particle size is one size large than the first sieve to retain more than 10%.

** Interpolate maximum voids in the mineral aggregate [VMA] for design air voids values between those listed.

20.13 **JOB-MIX FORMULA FOR SEMI DENSE ASPHALTIC CONCRETE:**

20.13.1 The Design Mix / Job-Mix formula for Semi Dense Asphaltic Concrete shall be carried out from National Test House / SEMT CME / Regional Research Labs / CRRI, New Delhi / NABL Accredited Laboratories / Zonal Labs / Government Engineering Colleges, NITs & IITs etc. The mix designation shall be all as specified in table under clause 20.B.2 of MES Schedule Part – I. The cost of transportation of material and testing charges shall be borne by the contractor.

20.13.2 The proportions of coarse aggregate, fine aggregate and mineral filler shall be indicated as percentage by weight of total aggregate including mineral filler. Contractor shall submit these mix proportions to Garrison Engineer for approval on meeting the specifications mentioned here-in-before.

20.13.3 The tenderer shall submit their job mix formula to the Garrison Engineer within 30 days of the acceptance of contract also giving the details as mentioned in the preceding clauses for approval of the Job–Mix formula for actual use. A re-verification of mix proportion is essential for every change in source of aggregates and bitumen and by obtaining fresh job mix formula and mix design.

20.13.4 Material, quality, workmanship, mix design criteria for semi dense asphaltic concrete shall be all as described in clause 20.B.2 of MES Schedule Part – I. Anti stripping agent shall be added to the aggregate to reduce the stripping time of aggregate all as per manufacturer's instructions without any extra cost to the Government. The binder shall be paving bitumen of Grade **VG–30** as per IS – 73.

20.13.5 Preparation of mix, laying, spreading, compaction and control for semi dense asphaltic concrete shall be done all as specified in clause 20.B.4 of MES Schedule Part – I.

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20.14 **PLANT TRAILS:** Permissible variation in Job mix formula.

20.14.1 Once the laboratory job mix formula is approved the contractor shall carry out plant trails at the mixer to establish that the plant can be set up to produce a uniform mix conforming to the approved job mix formula. The permissible variations of the individual percentages of the various ingredients in the actual mix from the job mix formula to be used shall be within the limits as specified in table below.

Description	Permissible Variation
	Semi Dense Asphaltic Concrete
Aggregate Passing 19mm Sieve or Larger	± 7%
Aggregate Passing 13.2mm, 9.5 mm Sieve	± 6%
Aggregate Passing 4.75 mm Sieve	± 5%
Aggregate Passing 2.36mm, 1.18 mm, 0.6 mm Sieve	± 4%
Aggregate Passing 0.3mm, 0.15 mm Sieve	± 3%
Aggregate Passing 0.075mm Sieve	± 1.5%
Mixing Temperature	± 10°C

20.14.2 Once the plant trials have demonstrated the capability of the plant, and the trials are approved, the laying operation may commence. GE may order additional testing of the product to establish the reliability and consistency of the plant. No claim whatsoever on this account will be admissible.

20.15 **LAYING TRIALS: [FOR SEMI DENSE ASPHALTIC CONCRETE]**

20.15.1 Once the plant trials have been successfully completed and approved, the contractor shall carry out laying trials at the locations as indicated by the Engineer-in-Charge, to demonstrate that the proposed mix can be successfully laid, and compacted all in accordance with contract conditions. The laying trial shall be carried out one suitable area, which is not to form part of the works, unless of 100 Sq.m of construction similar to that of the proposed work, and its shall be in all respects similar, particularly compaction shall be same as required in the proposed work, on which the bituminous material is to be laid.

20.15.2 The contractor shall previously inform the GE of the proposed method for laying and compacting the material. The plant trials shall then establish if the proposed laying plant, compaction plant, and methodology is capable of producing satisfactory results. The density of the finished paving layer shall be determined by taking cores, no sooner than 24 hours after laying or by other approved method

20.15.3 Once the laying trials have been approved, the same plant and methodology shall be applied to the laying of the material on the project, and no variation of either shall be acceptable, unless approved in writing by the GE, who may at his discretion require further laying trials without any extra cost to the Government.

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20.16 **MIXING AND TRANSPORTATION OF MIX:**

20.16.1 The temperature of binder at the time of mixing shall be in the range of 165°C to 170°C and of the aggregates in the range of 153°C to 163°C, provided also that at no time, the difference in temperature between the aggregates and binder shall exceed 14°C. The mixing shall be thorough to ensure that a homogeneous mixture is obtained in which all particles of the mineral aggregates are coated uniformly and temperature of mix shall not exceed 160°C.

20.16.2 Hot mix plant shall be used for mixing of aggregate and the binder. The binder shall be heated to the specified temperature. The aggregate shall be suitably warmed or heated before loading into hot mix plant. The correct quantity of each size of aggregate shall be fed into mixer with specified quantity of each size of aggregate shall be fed into mixer with specified quantity of binder. Each batch shall be mixed to ensure thorough coating.

20.16.3 The mix shall be transported from the mixing plant to the point of use in suitable tipper vehicles specified here in after. The vehicles employed for transport shall be clean and be covered using suitable covers in transit to ensure that the temperature of mix does not fall below 120°C at the time of laying.

20.17 **SPREADING OF THE MIX:** Spreading of asphaltic concrete shall be done by means of self–propelled mechanical Paver with a provision of electronic sensing device for automatic levelling and profile control within the specified tolerances and internal heating arrangement for the screed.

20.18 **ROLLING AND COMPACTION:** The rolling and compaction process for Semi Dense Asphaltic Concrete shall be in four stages as described here in after. The initial or breakdown rolling shall be done with 8 to 10 tone Tandem Vibratory Roller used on static mode. Intermediate rolling shall be with a smooth sheet pneumatic roller of 15 to 30 ton capacity having tyre pressure of 7 Kg/Sq.cm. There after the compaction will be carried with the help of Tandem roller with vibratory mode till compaction levels are achieved. Final compaction and surface finish shall be achieved with the help of pneumatic tyred roller.

20.19 **QUALITY ASSURANCE:** Adequate quality control at every stage of work is essential and the contractor is responsible in conducting day to day quality control tests as enumerated in succeeding clauses. This shall be in addition to any other tests which will be required by the Garrison Engineer and Engineer-in-Charge through approved laboratory/test house. Expenses on all test shall be deemed to be included in the rates quoted.

20.20 **PERMISSIBLE TOLERANCES FOR SEMI DENSE ASPHALTIC CONCRETE:**

20.20.1 The permissible variation in binder content with reference to Schedule "A" shall be within the following limits:

[i] Binder Content : ± 0.3%

20.20.2 When the road is opened to traffic, it should be ensured that a minimum density of 95% is achieved.

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20.21 **FINISHED SURFACE PAVEMENT UNIFORMITY [APPLICABLE FOR SEMI DENSE ASPHALTIC CONCRETE]:**

20.21.1 The finished pavement shall be of uniform thickness and thickness shall not vary more than 3mm from that specified. The surface tolerance of any paving course shall be such as not to exceed 3mm when tested transversely. The uniformity of finished and compacted surface shall be checked with Profilometer / road unevenness recorder. Any layer with deviation beyond this limit shall be corrected and or removed and replaced by contractor at his own expense. The rectification in such cases shall be carried out with fresh materials and compacted to specifications.

20.22 **MISSING DETAILS / SPECIFICATIONS:** In case of any missing specification details, relevant provisions contained in IRC [Indian Road Congress] shall be followed during execution of the work. In case the same are not available in IRC also, the same will be followed from latest version of relevant technical Instructions published by E–in–C's branch.

20.23 **MEASUREMENTS:** Refer special condition 20.A.1 on Page 459 of MES Schedule Part – II.

20.24 **OPENING TO TRAFFIC:** Traffic may be allowed immediately after completion of the final rolling when the mix has cooled down to the ambient temperature.

20.25 **USE OF ROAD:** During the progress of work on road the contractor has to provide suitable temporary screen barricades preferably of galvanization sheets duly painted on back side of intersections of road to avoid any untoward accident.

21. **CAT'S EYES (ROAD STUDS):**

21.1 The cat eyes shall conform to ASTM D 4280. Cat Eyes shall be designed for bonded and anchored within the road surface for lane marking and delineation for night time visibility to the traffic.

21.2 It shall be capable of reflecting the incident light in the direction close the direction from which it come. The Cat eyes shall be working exclusively on micro prismatic lens type technology and dual directional.

21.3 The Cat eyes/ Reflective pavement marker shall have both faces reflective in the red colour. It shall be assembled in a weather resistant polymer blend housing filled with an inert potting compound that adds strength to provide the increased durability, less damage by traffic ad breakage due to spot loading from uneven road surfaces. It shall have a tough layer of untempered glass, which protects acrylic lense from abrasion and makes it to get self cleaned with every contact of tyre. The body of Cat eye/road stud shall be moulded from HIP (Hi-impact Polystyrene).

21.4 The Cat eyes body shall be produced in neutral white or yellow colour. The colour of the retro reflective element shall be white, yellow, or red.

21.5 The cat eyes should have two shanks for fixing arrangements. The Cats eye (Road studs) shall support of a load of 13635 Kg tested in accordance with ASTMD 4280. The luminance in testing shall be as per the specifications and shll be tested as described in ASTM 1:809 as recommended in BS:873 Part 4:1073.

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21.6 The Cats eye (Road studs) shall be fixed to the road surface using adhesive conforming to IS as per procedure recommended by the manufacturer and as directed by the Engineer-in-Charge.

21.7 The contractor should submit the manufacturers test certificates consists of following test results of cat eyes before incorporating it in the work.

(a) Specific Intensity test
(b) Compressive strength test
(c) Abrasion test

22. **POLYCARBONATE CONVEX MIRROR :** Polycarbonate convex mirror shall be as provided in Sch 'A' and as directed by Engr-in-Charge

23. **THERMOPLASTIC ROAD MARKING PAINT :** Thermo plastic road marking paint shall be as provided in Sch 'A' and as directed by Engr-in-Charge

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33. **OUTPUT OF ROAD ROLLERS:**

33.1. [Refer Condition – 15 of IAFW – 2249]. Where road rollers are hired by the Department to the contractors, a logbook for each road roller shall be maintained by the Department recording hours of working of the road roller. In case, however, when the contractor procures road roller from, sources other than the Department, a logbook for each road roller shall be maintained by him for recording hours of working of the road roller. Entries in the log book shall be signed by the contractor or his authorised representative and by the Engineer-in-Charge.

33.2. To ensure proper consolidation roller must work for at least number of days assessed on the basis of output given hereinafter. If the roller has not worked for the number of days so assessed, recovery shall be effected from the contractor for the number of days falling short of the days assessed on the basis of output stipulated. The recovery shall be effected as under:

[a] Where Road Roller is hired out only by the Department to the contractor at rates given in Schedule "C".

[b] Where Road Roller is hired by the contractor only from sources other than the Department at the rate of **₹ 2,500.00** per working day of 8 hours for static power roller and at the rate of **₹ 4,000.00** per working day for tandem vibratory roller.

[c] Where Road Roller is hired by the contractor from the Department and also from sources other than the Department, at higher of the two rates given in Schedule "C" of the contract and Para [b] above.

33.3. The above provisions shall not, however, absolve the contractor of his responsibility of properly consolidating surface as required under the provisions of contract.

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33.4. **OUTPUT OF ROAD ROLLER PER DAY OF 8 HOURS WORK:**

33.4.1 Road rollers shall be arranged by the contractor, a logbook for each road roller shall be maintained by the Department recording hours of working of the road roller. In case, however, when the contractor procures road roller from, sources other than the Department, a logbook for each road roller shall be maintained by him for recording hours of work of the road roller. Entries in the log book shall be signed by the contractor or his authorised representative and by the Engineer-in-Charge.

33.4.2 To ensure proper consolidation roller must work for at least number of days assessed on the basis of output given hereunder. If the roller has not worked for the number of days so assessed, recovery shall be effected from the contractor for the number of days falling short of the days assessed on the basis of output stipulated.

33.4.3 The output of road roller required for the work for different items are as under:-

	Items of Work	**Output per day of 8 hours work**
(a)	Consolidation of formation Surface/Sub Grade	1850 Sq.m
(b)	Consolidation of Stone Soling/Hardcore	
	[i] 10 cm thick [Spread Thickness]	1000 Sq.m
	[ii] 15 cm thick [Spread Thickness]	800 Sq.m
	[iii] 23/20 cm thick [Spread Thickness]	518 Sq.m
(c)	Consolidation of Water Bound Macadam [Stone Metal] including Spreading and Consolidation with Binding Material:	
	[i] 7.5 cm [Compacted Thickness	372 Sq.m
	[ii]10 cm [Compacted Thickness]	175 Sq.m
	[iii] 11.5 cm [Compacted Thickness]	248 Sq.m
(d)	Consolidation of Premixed Carpet Including Seal Coat:	
	[i] 20 mm thick [Compacted Thickness]	744 Sq.m
	[ii] 25 mm thick [Compacted Thickness]	600 Sq.m
	[iii] 40 mm thick [Compacted Thickness]	500 Sq.m
(e)	Consolidation of Single Coat Surface Dressing	774 Sq.m
(f)	Consolidation of Two Coats of Surface Dressing	558 Sq.m
(g)	Consolidation of bituminous mixture 2 Parts broken stone metal and 1 Part of sand and bitumen:	
	[i] 4.0 cm [Compacted Thickness]	372 Sq.m
(h)	Consolidation of 15 cm thick [Spread Thickness] Earthen / Moorum Berms	1800 Sq.m
(j)	Premixed Bituminous Macadam	15 Cu.m
(k)	Semi Dense Asphaltic Concrete	18.40 Cu.m

Note: Regarding output of road roller in respect of other items catered in the CA, GE shall order a board of officers and ascertain the required output of road roller to achieve the desired / specified compaction over a trail area which will be the basis for the corresponding works.

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33.5. Road roller shall not be issued by the Department under Schedule "C" and shall be arranged by the contractor under his own arrangements. Provision of Condition hereinbefore shall be deemed amended accordingly.

33.6. The number of hours / days assessed for proper consolidation and number of hours / days for each surface of each stretch / piece of road consolidated, shall be submitted by the contractor to Engineer-in-Charge and GE for each stage and only after written approval of GE, contractor shall proceed to next stage of work.

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40 **PAID VOUCHERS/TEST CERTIFICATE**

40.1 The contractor shall produce original with the photo state copies of paid vouchers from the original manufacturers or from their authorised agents for all the materials/as soon as these are purchased and brought at site.

40.2 The contractor shall produce manufacturer's test certificate for the equipments as soon as the same is brought at site. The contractor shall produce necessary test certificates from manufacturer for the parts/spares provided. Contractor shall also produce necessary paid vouchers for the spares/parts procured and fixed duly defaced by the Engineer-in-Charge.

40.3 Contractor shall furnish guarantee for all the components provided by him or repaired by him against any manufacturing defects for minimum period of one year from the date of completion of respective item of work.

41 **RECORD OF MATERIALS AND PURCHASE VOUCHERS**

41.1 The quantity of materials as applicable such as cement, steel, paint, water proofing compound, and the like, as directed by the Engineer-in-charge (the quantity of which cannot be checked after incorporation in the works), shall be recorded in measurement books and signed by the contractor and the Engineer-in-charge as a check to ensure that the required quantity has been brought to site for incorporation in the work. Materials brought to site shall be stored as directed by the Engineer-in-charge and recorded in the Measurement Book shall be suitably marked for identification

41.2 It will be mandatory for the contractor to produce genuine receipted bills/cash memos, invoices from the manufacturer and/or their authorized agents for the full quantity of the following materials as applicable, as pre-requisite before submitting claim for payment of advances on account of work done and/or materials collected in accordance with condition 64 of General Conditions of Contracts (IAFW-2249).

41.3 As per E-in-C’s policy in vogue purchase vouchers are to be submitted by contractor for the following :

(a) Material lying at site for which contractor claims payment in RAR

(b)Cement, Steel, Major E/M equipment like transformers, DG sets, pumps, motors, AC & Lift equipment.

PARTICULAR SPECIFICATION

41.4 Original purchase vouchers shall be defaced by the Engineer-in-Charge and photocopies shall be kept in record. Genuineness of purchase voucher shall be verified by the Engineer in Charge before considering the same for making payment. Initially JE may verify the vouchers after taking confirmation on phone or through e-mail.

42. **SUPPLEMENTARY SPECIFICATIONS**

42.1 In case no Particular Specifications are given for any items/materials and workmanship in these tender documents for respective trade/item of Sch 'A' the Specifications and workmanship given in SSR 2009 and/or as per manufacturer's instructions shall be followed as approved by the Accepting Officer without any price adjustment.

43. **LIST OF DRAWINGS** : Nil

SIGNATURE OF CONTRACTOR

AGE (CONTRACTS)
FOR ACCEPTING OFFICER

PARTICULAR SPECIFICATION

44. **LIST OF MANUFACTURERS**

(i) The makes given in this Appendix shall be applicable to the extent required as per the Scope of this tender. The materials provided under this tender shall be of the approved brand/ manufacturer / Makes given here under except in Schedule 'A'/ or Particular Specifications.

(ii) In case makes are specifically mentioned in Schedule 'A', then makes specified in Schedule 'A' shall only be provided.

(iii) Makes specified in Particular Specifications shall be provided only when no makes are specified in Schedule 'A'.

(iv) In case where makes of any item are not given in Schedule 'A'/ specifications of relevant item in these Particular Specifications, the same shall be provided as per Appendix 'A'. However, makes specified in Sch 'A' shall take precedence over those specified in Appendix 'A'

(v) In case makes of any item(s) is/ are not specified in any of the aforesaid documents, such items shall be of reputed make as approved by GE in writing and shall bear ISI marking.

(vi) Irrespective of makes, manufacture in tender documents, material(s) should not be procured from/of the makes, manufacturers, brands, models etc, if as on date, there is ban due to technical defaults or not operating/ manufacturing for a long time or, non existent etc. No claim whatsoever on this account shall be admissible to the contractor later on.

(vii) In order to decide the equivalency of cat part of different makes, models and manufacturers; cost, specifications, and parameters etc. shall be taken into consideration by the GE. In case of difference in the opinion of GE and Contractor, decision of the CWE shall be final, conclusive and binding. No claim whatsoever on this account shall be admissible to the contractor later on.

(viii) The makes of the various items of materials are as under. The contractor has to obtain approval of the make by Garrison Engineer before procurement.

SL NO	DESCRIPTION OF ITEM	NAME OF MANUFACTURERS
1	CEMENT	REFER APPX TO PS HERE-IN-AFTER
2	STEEL	CLAUSE 9.4 & 9.5 REFERS
3	SAND	AVAILABLE QUARRY AS APPROVED BY GE OR LOCATION AS APPROVED BY STATE GOVT /AUTHORITY
4	AGGREGATE/ STONE	AVAILABLE QUARRY AS APPROVED BY GE OR LOCATION AS APPROVED BY STATE GOVT /AUTHORITY
5	READY MIX CONCRETE (RMC)	LAFARGE / ULTRA TECH / ACC / RMC READY MIX / RAMCO

PARTICULAR SPECIFICATION

SL NO	DESCRIPTION OF ITEM	NAME OF MANUFACTURERS
6	ADMIXTURE	M/S SIKA INDIA PVT LTD / M/S FOSROC CHEMICALS / M/S STP LTD / M/S CHOKSEY CHEMICALS PVT LTD / M/S MAPEI CONSTRUCTIONS PRODUCT INDIA PVT LTD, / M/S MARUTI BITUMEN PVT LTD / M/S HINDCON CHEMICALS LTD /M/S IWL INDIA LTD, / M/S THERMAX LTD
7	CURING COMPOUNDS	M/S SIKA INDIA PVT LTD. / M/S FOSROC CHEMICALS / M/S STP LTD / M/S CHOKSEY CHEMICALS PVT LTD / M/S MAPEI CONSTRUCTIONS PRODUCT INDIA PVT LTD, / M/S MARUTI BITUMEN PVT LTD / M/S BOSTIK INDIA PVT LTD
8	WATERPROOFING COMPUND/ CHEMICALS	M/S SIKA INDIA PVT LTD. / M/S FOSROC CHEMICALS / M/S STP LTD / M/S CHOKSEY CHEMICALS PVT LTD / M/S APURVA INDIA LTD / M/S BOSTIK INDIA PVT LTD / M/S STP LTD / M/S MARUTI BITUMEN PVT LTD / M/S KRYTON BUILDMAT CO PVT LTD / M/S BERGER PAINTS INDIA LTD / M/S MAPEI CONSTRUCTIONS PRODUCT INDIA PVT LTD
9	SYNTHETIC ENAMEL PAINT	M/S ASIAN PAINTS M/S NEROLAC PAINTS M/ AKZO NOBEL (DULUX) M/S JENSON & NICHOLSON PAINTS M/S JOTUM M/S SHALIMAR PAINTS
10	EXTERIOR EMULSION PAINT	M/S ASIAN PAINTS M/S BERGER PAINTS M/S NEROLAC M/S AKZO NOBEL (DULUX) M/S SHALIMAR PAINTS M/S JENSON & NICHOLSON M/S PIDILITE WEATHERPROOF M/S JOTUN
11	CEMENT BASE PAINT	M/S SUPER SNOWCEM/ M/S DURACEM/ M/S AQUACEM M/S SHALIMAR PAINTS/ M/S BERGER PAINTS M/S ASIAN PAINTS/ M/S ACCROCEM M/S NITCO MUMBAI/M/S JENSON & NICHOLSON
12	CEMENT PUTTY	M/S BIRLA CEMENT/ M/S JK WHITE CEMENT M/S ASIAN PAINTS/ M/S JENSON & NICHOLSON PAINTS M/S SHALIMAR PAINTS/ M/S GOLDEN MOHAR
13	ADHESIVES	M/S PIDILITE/ M/S FEVICOL/ M/S VERMICOL M/S ARMACELL INDIA PVT LTD
14	CHAIN LINK FENCING	M/S A1 FENCE PRODUCTS COMPANY LTD OR EQUIVALENT AS APPROVED BY GE
15	PVC SHEET AND TILE FLOORING	M/S ARMSTRONG/ M/S MARBLE TILES M/S POLYFIN TILES/ M/S SQUARE FOOT M/S KRISHNA VINYL TILES/ M/S NEELKAMAL M/S PREMEIR VINYL FLOORING NEW DELHI M/S RIKVIN FLOORS NEW DELHI/ M/S WONDER FLOOR

PARTICULAR SPECIFICATION

SL NO	DESCRIPTION OF ITEM	NAME OF MANUFACTURERS
16	BITUMEN VG- 30	IOCL, BPCL, HPCL
17	THERMOPLASTIC ROAD MARKING PAINT	M/S CMS BOMBAY, RITACTOR MEERUT, M/S SHIVALIK PRIME CHANDIGARH M/S SN INDUSTRIES DELHI M/S CMS TRAFFIC SYSTEMS LTD MUMBAI M/S RELIANCE THERMOPLAST : AD-131B, SHALIMARBAGH NEW DELHI- 110 088. ASIAN PAINTS (APCO MARK)/ STP SHALIMAR
18	SEALING COMPOUND	COLPOR 200 OF FOSROC, CIPY, POLYURETHANE PVT LTD, SIKA INDIA PVT LTD, PIDILITE INDUSTRIES LTD OR EQUIVALENT MAKE AS APPROVED BY GE.
19	WALL LINING, PVC FALSE CEILING & SOLID PVC PARTITION	M/S RAJSHRI PLASTIWOOD M/S KUMAR ARCH TECH PVT LTD(ECHON) M/S NAVRATNA SPECIALITY CHEMICALS (GIZA)
20	VITRIFIED TILES	M/S KAJARIA/ M/S SOMANY/ M/S ASIAN GRANITE (I) LTD M/S JOHNSON MARBONITE/M/S RAK CERAMICS LTD M/S ORIENT BELL LTD/M/S MURUDHESHWAR CERAMICS M/S BELL GRANITO/ M/S NAVEEN TILES M/S VARMORA GRANITO PVT LTD M/S SWASTIK CERACON LTD M/S SUNSHINE TILES CO PVT LTD M/S CENGRESS TILES LTD M/S OASIS VITRIFIED PVT LTD M/S QUTONE CERAMIC PVT LTD MS AMBANI VITRIFIED PVT LTD M/S APARNA LTD(VIETRO)

SIGNATURE OF CONTRACTOR
DATED:

AGE (CONTRACTS)
FOR ACCEPTING OFFICER

PARTICULAR SPECIFICATION

APPENDIX TO PARTICULAR SPECIFICATION

1. **CEMENT.**

1.1 **GENERAL**. Cement required for the work under the contract shall be procured, supplied and incorporated in the works by the contractor under his own arrangement. Cement shall be tested quality and comply with the requirements mentioned in the drawings, SSR, IS specifications (as amended) and particular specifications given hereinafter.

1.2 **TYPES OF CEMENT**. Type of cement for the subject work shall be Ordinary Portland Cement grade 43 (Forty Three) conforming to IS: 8112-1989 or grade 53 (fifty three) conforming to IS : 12669-1987.
(Note: -If OPC 53 grade cement is used curing must commence within 6 hours of completion of concreting work).

1.2.1 In case of non availability of OPC (Ordinary Portland Cement) in the market, the contractor can use PPC (Pozzollana Portland Cement) in accordance with IS 1489-1991 Part I with prior permission from the Accepting Officer and his decision shall be final and binding, that case PPC shall meet the strength criteria of 43 grade/53 grade OPC (Ordinary Portland Cement) as laid down is IS-8112-1989. Mixing of OPC and PPC is not allowed.

1.3 **PROCUREMENT/SUPPLY OF CEMENT BY CONTRACTOR**

(a) **PROCUREMENT**. The cement supplied by the contractor shall only be procured from main producers of cement. The particulars of the manufactures of cement along with the date of manufacture shall be obtained from the contractor for every lot of cement separately. The documents in support of the purchase of cement shall be verified by the Engineer-in-Charge/GE and site staff. Cement shall be procured by the contractor from any of the manufacturers listed below.

Sl No	Company Name	Brand	Cement Grades
1	The Associated Cement Companies Ltd	ACC	All
2	Ultra Tech Cement Ltd	Ultratech	All
3	The India Cement		All
4	Dalmia Cement(Bharat) Ltd	Dalmia Infra Pro	All
5	Century Cements	Century	All
6	Saurashtra Cement	Saurashtra	All
7	The Ramco Cements Ltd (Formerly Madras Cement)	Ramco	All
8	Mangalam Cement Ltd	Mangalam	All
9	Birla Corporation Ltd	Birla	All
10	Orient Cement	Birla A-1	All
11	Nuvoco Vistas Corporation Ltd (Formerly Lafarge Cement)	Nuvoco	All
12	Shree Cement	Shree	All
13	J K Cement	J K	All
14	J K Lakshmi Cement Ltd	J K Lakshmi	All
15	Jaypee Rewa Cement	Jaypee	All
16	Ambuja Cement Ltd	Ambuja	All
17	M/s Chettinad Cement Corporation Pvt Ltd, Chennai	Chettinad Cement	OPC 43 & PPC
18	M/s Kesoram Industries Ltd	Birla Shakti	OPC 43, OPC 53 PPC
19	M/s Sagar Cements Ltd, Hyderabad	Sagar	OPC 43,OPC 53,PPC, PSC

PARTICULAR SPECIFICATION

Note:-Any manufacturer not included above but already approved by E-n-C Branch and also approved during execution shall be deemed to be included without any price adjustment .

1.3.1 **USE OF PPC (POZZOLONA PORTLAND CEMENT)**

1.3.1.1 Portland Pozzollana cement shall not be used for the following:-

(i) In cold climate season.
(ii) Over head reservoir
(iii) Under ground sump
(iv) Prestigious Building
(v) Building with spans 10 meter or more
(vi) PPC and OPC shall not be mixed

1.3.1.2 While using PPC the following conditions shall generally be met :-

(i) Strength criterion for PPC cement shall be as per IS-8112-1989
(ii) Stripping time shall be 14 days
(iii) Both OPC and PPC shall not be permitted for use in the same building except for plaster and mortar.
(iv) Mandatory certificates of testing and quality assurance shall continue to be submitted as hither-to-fore, with fly ash content as per IS-1489-1991 (Part-I).

1.3.1.3 While procuring Portland Pozzolona Cement, the following requirement are to be ensured and certificate to be obtained from the manufacturer for each batch of PPC procured:

(i) The quality of fly ash is strictly as per IS-1489 (Part I)-2002
(ii) Fly ash is inter ground with clinker and not mixed with clinker.
(iii) Dry fly ash is transported in closed containers and stored in silos only pneumatic pumping is used.
(iv) The fly ash is received from thermal power plant using high temperature combustion above 1000ºC.
(v) The fly ash content in PPC is not more than 25%.

1.3.2 The contractor shall furnish the particulars of the manufacture/supplier of cement along with the date of manufacture to the Garrison Engineer for every lot of cement separately. The cement so brought shall be fresh and in no case manufactured before 60 days.

1.3.3 The document in support of purchase of cement shall be verified by the Garrison Engineer. Before placing the order for supply of cement by the contractor, he shall obtain written approval form the GE regarding name of manufacturer, quantity of cement etc. Cement shall be procured for minimum requirement of one month and not exceeding the requirement of the same for more than two months at a time. The cements shall be consumed in the work within three months after receipt. Cement shall conform to the requirement of Indian Standard Specification and each bag of cement shall bear relevant ISI mark. The weight of each consignment shall be verified by the GE and recorded. The content of cement shall be checked at random to verify the actual weight of cement per bag. However, the content per bag shall be 50 Kgs only subject to tolerance give in clause 9.2.1.1 and Annexure-‘B’ of IS-8112-1989.

1.3.4 However, when estimated requirements of cement for this work is less than 1200 bags, the contractor can procure cement from the authorized distributors/dealers of approved firms but the contractor will have to submit test certificates of the batch issued by the main producers. The particulars of the main manufacturers of cement along with the date of manufacturing shall be obtained from the contractor for every lot of cement separately. The documents in support of the purchases of cement shall be verified by the site staff and GE”.

PARTICULAR SPECIFICATION

1.4 **TESTING OF CEMENT**

1.4.1 The manufacturer shall carry out inspections and testing of cement in accordance with relevant BIS provisions. The Contractor shall submit the manufacturer's Test Certificate in original alongwith Test Sheets giving the result of each physical test as applicable and chemical composition of the cement in accordance with relevant IS provisions and the chemical composition of the cement or authenticated copy thereof duly signed by the manufacturer with each consignment, as per the following IS provisions:

(a) Method of Sampling Hydraulic Cement as per IS-3535-1986
(b) Method of Physical Tests for Hydraulic Cement as per IS-4031
(c) Method of Chemical Analysis of Hydraulic Cement as per IS-4032-1985

1.4.1.1 The test certificate and test sheet shall be furnished with each batch of the manufacture. The Engineer-in-Charge shall record these details in the Cement Acceptance Register as given in Appendix-'A' to be maintained by him which will be signed by Supdt B/R Gde-I/JE (Civil), Engineer-in-Charge, Garrison Engineer and the Contractor as given in the format hereinafter for verification.

1.4.2 (a) The Contractor shall however, organise setting time and a compressive strength test of cement through designated laboratory on samples collected from the lot brought at site before incorporation in work. The Contractor will be allowed to use the cement only after satisfactory compressive strength of seven days to meet this requirement, Contractor is required to keep minimum 10days stock before any new lot brought at site which can be used in the work. The Contractor shall be required to remove the cement not meeting the requirement from the site within 24 hours. Seven days strength test will be relied upon to accept the lot of cement to commence the work. 28days compressive strength test will be the final criteria to accept/reject the lot.

(b) The GE shall carry out independent testing as per the tests mentioned in the `CEMENT SUPPLY/ ACCEPTANCE FORM' of random samples of cement drawn from various lots, if sample fails in 7 days compressive strength.

The testing shall be carried out through NATIONAL TEST HOUSE, SEMT wing CME, REGIONAL RESEARCH LABORATORIES IITs, GOVT Engineering Colleges, MES ZONAL LABORATORIES, National Institute of Technology as per BIS and as per test methods referred hereinbefore and shall be recorded in respective portion of Appendix-'C-1'. The decision as to where the testing of cement is to be done shall be taken by the GE. In case the cement is not of requisite standard despite manufacturer's test certificate, the Contractor shall remove the total consignment from the site at his own cost after written rejection order of the consignment by the GE. The cost of test shall be borne by the Contractor irrespective of the results of testing.

(c) The random samples as per relevant IS shall be selected by the GE before carrying out testing. The random samples of cement to be tested shall be drawn as per MES Quality Assurance Manual. The record of such samples selected by the GE for testing shall be properly maintained in the `Cement Testing Register' giving cross reference to relevant consignment of cement and quantity received etc.

PARTICULAR SPECIFICATION

(d) Cost of transportation of samples to the approved laboratory / test house and all testing charges including cost of sample shall be borne by the Contractor.

1.4.2 The Contractor shall be required to set up adequate testing facilities at site to the entire satisfaction of the Garrison Engineer for conducting 'Setting Time Test' and 'Compressive Strength Test' as per IS codes referred to hereinbefore for the samples collected from the lot brought at site. These tests shall be carried out within 7days of receipt of cement at site. The tests can alternatively be carried out at the Zonal Laboratory so designated by the GE.

1.4.3 The Contractor shall submit original purchase vouchers for the total quantity of cement supplied under each consignment to be incorporated in the works. All consignments received at the work site shall be inspected by the GE along with the relevant documents to ensure the requirements as mentioned hereinbefore, before acceptance. The original purchase vouchers and the test certificates shall be verified for subject contract and defaced by the Engineer-in-Charge and kept on record in the office of GE duly authenticated and with cross reference to the consignment/ control number recorded in the Cement Acceptance Register. The 'Cement Acceptance Register'

1.4.4 The Accepting Officer may order a Board of Officers for random check of cement and verification of connected documents during the currency of contract.

1.4.5 For repair, maintenance and works where entire requirement of cement is less than 300 bags, cement can be procured from authorised dealers of the firm listed hereinbefore.

1.5 **STORAGE / ACCOUNTING / PRESERVATION OF CEMENT**

1.5.1 Cement shall be stored in covered godown over dry platform at least 20cm high in such a manner as to prevent deterioration due to moisture or intrusion of foreign matters. In case of store rooms, the stack should be at least 20cm away from floors and walls. The stacking of cement shall be not more than 10 bags and or done as specified in relevant IS. The storage, accounting and preservation of cement supplied by the Contractor shall be done as per standard engineer practice till the same is incorporated in the work and the cost of the same is deemed to be included in the unit rate/amount quoted by the tenderer.

1.5.1.1 The EIC shall inspect once a day to verify that cement lying at site is stored, accounted, preserved and maintained as per the norms. The cement shall be stored so as to differentiate each tested and untested consignment separately with distinct identification. If the GE is not satisfied with the storage/ preservation of cement, he may order for any test(s) of cement as applicable for that consignment to ensure its conformity to the quality mentioned in the manufacturer's test certificate. The Contractor shall bear the cost of necessary testing(s) in this regard and no claim whatsoever shall be entertained.

1.5.2.1 Stacking of cement shall be done as per relevant IS and as under:

(a) Each cement consignment shall be stacked separately and removal shall be made on the basis of `First in First out'.

(b) Adequate top cover will be provided

PARTICULAR SPECIFICATION

(c) Stacks in no case shall be higher than 10 bags. The maximum width of each stack shall be 3.00m. If the stack is to be more than 7 or 8 bags high, the bags shall be arranged in header and stretcher fashion, alternatively lengthwise and crosswise so as to tie the piles together and avoid danger of topping over.

(d) Adequate space shall be kept between two stacks.

1.5.3 Cement godown shall be provided with two locks on each door. The key of one lock at each door shall remain with the EIC or his representative and that of the other lock with the Contractor's authorised agent at site of works so that cement is removed from the godown only according to daily requirement with the knowledge of both the parties. During the period of storage, if any cement bag(s) found to be in damaged condition due to whatsoever reason, the same shall be removed from the cement godown on written orders of the GE and suitable replacement for the cement bag(s) so removed shall be made and no claim whatsoever shall be admissible on this account.

1.5.4 Cement shall be removed from the store only according to daily requirement with the knowledge of both the parties and daily consumption of cement shall be recorded in cement consumption register which shall be signed by the Engineer-in-Charge and the Contractor. Cement constants given in Appendix `A' to E-in-C's branch letter No.19280/E8 dated 03 May 1976 shall form the basis of consumption of cement for various items of works unless indicated otherwise.

1.5.5 In case of consumption of cement as per cement consumption register is found to be more than the estimated quantity of cement due to whatsoever reason, the Contractor shall not have any claim whatsoever for such excess consumption of cement.

1.6 **SCHEDULE OF SUPPLY**. The Contractor shall procure the cement timely as required in accordance with CPM chart agreed between GE and himself. The Contractor will forfeit his right to demand extension of time if the supplies of cement get delayed due to his failure in placing order in time to the manufacturer.

1.7 **DOCUMENTATION, MEASUREMENT AND PAYMENT OF CEMENT**

1.7.1 The Contractor shall submit original vouchers from the supplier for the total quantity of cement supplied under each consignment to be incorporated in the work, all consignment received at the work site shall be inspected by the GE alongwith relevant documents received at the work site. The original vouchers and the test certificate shall be defaced by the Engineer in Charge and kept on record in the office of GE duly authenticated with cross reference to the control number recorded in cement acceptance register. This register shall be signed by the JE, Engineer in Charge, GE and the Contractor.

1.7.2 The Accepting Officer may order a Board of Officer for random check of cement and verification of connected documents. The entire quantity of cement shall also be suitably recorded in the Measurement Book for record purposes as `NOT TO BE ABSTRACTED' before incorporation in the work and shall be signed by the Engineer-in-Charge and the Contractor.

PARTICULAR SPECIFICATION

1.7.3 The payment shall only be allowed after production of original purchase voucher, certified copies of test certificates from manufacturer for each consignment and results of testing carried out in laboratory on receipt of cement (7 days compressive test) are found satisfactory after testing as specified hereinbefore. Cement shall be paid as material lying at site under Condition 64 of IAFW-2249. Rate of cement given in SSR shall be applicable for cement irrespective of grade of cement specified for use in the work.

SIGNATURE OF CONTRACTOR
DATED:

AGE (CONTRACTS)
FOR ACCEPTING OFFICER

PARTICULAR SPECIFICATION

Annexure 'A'

Cement supply & Acceptance Register

1. CA No & Name of work

2. Control No

3. Name of Manufacturer / Brand Name / Gde of Cement

   (A) Manufacturer _____

   (B) Brand _____

   (c) Grade _____

4. Qty. of cement & Lot No. / week No (in Bags) : (Qty _____)
   (a) Lot No/Week No._____

5. Manufacturers's test certificate No. _____

6. Random Test Details (a) Physical test report from vide their letter No.

   (b) Chennai test report form _____ vide their letter No.

7. Details if Physical & Chemical properties : Refer Annexure 'B'

PARTICULAR SPECIFICATION

Annexure 'B'

	Physical Requirements (As per IS 4031)								Chemical Requirements (As per IS 4032)								
						Compressive Strength											
	Specific Surface (M2 G)	Soundness by Chancellor	Soundness by Auto Clave Expansion	Initial setting time (Minutes)	Final setting time (Minutes)	03 Days (Mpa)	07 Days (Mpa)	27 Days (Mpa)	Temp during testing OC	Standard consistency (%)	Lime saturation Factor (Ratio)	Almina Ion Ratio (Ratio)	Insoluble Resdue (%)	Sulphuric Anhydride (%)	Loss opn ignition (%)	Alkales (%)	Chlorides (%)
As per Relevant IS																	
As per manufacturer's test certificate																	
As per random test certificate																	

PARTICULAR SPECIFICATION

Annexure 'C'

IN/OUT CEMENT REGISTER

Sl No	Date	Cement IN		Cement OUT			Qty Balance in Bags	Signature Remarks		
		Qty in Bags	Control No	Qty in Bags	Reasons	Age of Cement		Contractor AGE/GE		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(j)	(k)	(l)

*** NOTE: The following reasons may be mentioned for taking out cement from store:-**

For Testing purpose

For use in work

Rejected cement taken out of site.

PARTICULAR SPECIFICATION

Annexure 'D'

STEEL SUPPLY/ACCEPTANCE FORM

Contract No.

Name of Work:

Control No :

Details of purchase
Particulars of Manufacturer
Details of Supplier (if any)
Details of Test Certificate
No and date
Particulars of issuing authority

Sl No.	Nomenclature and size		Quantity Actual	(t) Conversion	UTS T S	(N/Sq mm) Test Sheet	Random Test	Physical properties Elongation (%) I Test Random Bend S sheet test* test				Remarks
1	2	3	4	5	6	7	8	9	10	11	12	13

* As per frequency of testing given vide clause 10.3.6 above

Remarks with signatures

Junior Engineer Engineer-in-charge Contractor

Accepted/Rejected

Garrison Engineer

Remarks of BOO/Inspecting Officer/CWE

PARTICULAR SPECIFICATION

Annexure 'E'

TEST CERTIFICATE

No. of Test

Name of Articles (s)

Samples submitted by

Ref. to the letter No/Date

Test No.

Sl No	Sample (s)	Carbon C %	Sulphur S%	Phosphors%	MangnMn%	SilconSi%	ChromiumCr%	NiKel %	Copper Cu %	TinSn %	LeadPb %	Ferrous Fe %	Zinc Za%	Aluminum Al%	Imp %	WtofZa	UTS	% E1ong	P/SMPA
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

Remarks with signature

JE (civ)

Engineer-in-Charge

Contractor

Accepted/Rejected

Remarks of BOO/CWE/CE Zone

Garrison Engineer

PARTICULAR SPECIFICATION

Annexure 'F'

TEST REPORT

Name of article material :

Particulars identification Mark:

Name of the firm manufacturer :

This is to certify than the article, the particulars of which are given has been tested in the centre another test results are as follows:-

Chemical tests on metal (Percentage by Weight)

FERROUS	C	S	P	Mn
Si	Ni	Cu	Mo	
NON FERROUS	Sn	Ph	Cu	Fe
Zn	Others			

Ser No	Weight/ Metre	Wt/Metre	Mechanical/physical tests		Elongation	Bend test	Rebend test	Misc.	Remarks
			Yield/Proof stress (N/Sq mm)	Tensile Strength (N/Sq mm)					
1	2	3	4	5	6	7	8	9	10

Remarks with signature

JE (Civ)

Engineer-in-Charge

Contractor
Accepted/Rejected

Garrison Engineer
Remarks of BOO inspecting officer/CWE

PARTICULAR SPECIFICATION

Annexure 'G'

IN/OUT STEEL REGISTER

Sl No	Date	Steel IN			Steel OUT			Qty Balance
		Qty (Tons)	Section	Control No	Control No	Section	Reasons*	

*** NOTE: The following reasons may be mentioned for taking out steel from storage:-**

For testing purpose
For use in work
Rejected steel taken out of size.